NEW APPLICATIONS OF THE TIME DOMAIN ELECTROMAGNETIC METHOD (TDEM) IN ISRAEL: FEASIBILITY STUDY

February 1988 X5/1715/87

המכון למחקרי נפט וגיאופיסיקה
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by

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A B S T R A C T

A series of tests on TDEM equipment and working techniques was carried out on selected geological and hydrogeological problems in Israel in order to evaluate:

1. the applicability of the equipment to different working and geological conditions; and
2. the types of problems which may be solved by the TDEM method.

Two types of equipment were used in the survey: the Geonics EM-37 was used for targets no deeper than 300-400m, while the Geonics EM-42 was used for much greater depths from 1 to 1.5km.

A total of 40 soundings was carried out in five areas throughout Israel. Of these, 34 measurements were of very high quality which permitted quantitative interpretation. Five soundings did not show the existence of the target and only an estimation of the minimum depth to a possible target was made. One measurement station was located so close to a noisy power line that the field data collected here were completely distorted.

This survey has demonstrated that the geology of Israel presents a variety of attractive targets which can be explored by the TDEM method. Arranged in the order of applicability of the TDEM method, the geological objectives and the main results obtained are listed below.

- The most appropriate targets for the TDEM method using the equipment mentioned above are seawater intrusions into the coastal aquifer of Israel. Experience (although not yet statistically significant) has shown that the

freshwater/seawater interface can be determined with almost the same degree of accuracy as in observation wells.

- A boundary between the Mesozoic sediments and the quartz-syenite intrusions in southern Ramon is detected very confidently.
- Fairly accurate detection of the seawater intrusion into the carbonate aquifer of Mount Carmel seems to be possible, at least in the depth range of up to a few hundred meters.
- The contact between the Neogene basaltic sequence and the clastic rock units in the Golan Heights area can be clearly detected by the TDEM measurements. This is also true for the contact between the clastic rocks and the limestone/dolomite sequence. It is not possible, however, to distinguish the clastic rocks/basaltic contact from the clastic rocks/limestone (or dolomite) contact, since the resistivity contrast across the contact is approximately the same in both cases.
- Very deep targets, such as the Precambrian basement and/or hypothetical magmatic intrusions in the southern Negev, could not be detected by the TDEM measurements.

I. INTRODUCTION

A geophysical survey using the Time Domain Electromagnetic (TDEM) method was conducted during the period April-May, 1987 in a number of areas throughout Israel (Fig. 1). The survey was carried out by the Institute for Petroleum Research and Geophysics, in close collaboration with the Geological Survey and Hydrological Service of Israel and the Soreq Nuclear Research Center.

The aim of this project was to define the applicability of the TDEM method for a variety of geological problems encountered in Israel.

Two types of equipment were used in this survey, depending on the required depth to target. The Geonics EM-37 equipment was used for relatively shallow targets (up to 300-400m), while the Geonics EM-42 was used for much greater depths in the range between 1 to 1.5km. This survey was the first to experiment with the EM-42 equipment in Israel and one of the first in the world. Both types of equipment were rented from the Earth Technology Corporation (ERTEC) of Denver, Colorado and were handled in the field by Mr. Mark Blohm, a specialist from the company.

The survey test areas were:

1. Zenifim-Ramon
2. Southern Makhtesh Ramon
3. Eastern Makhtesh Ramon
4. Ramat Korazim and Central Golan
5. Mount Carmel
6. Nitzanim

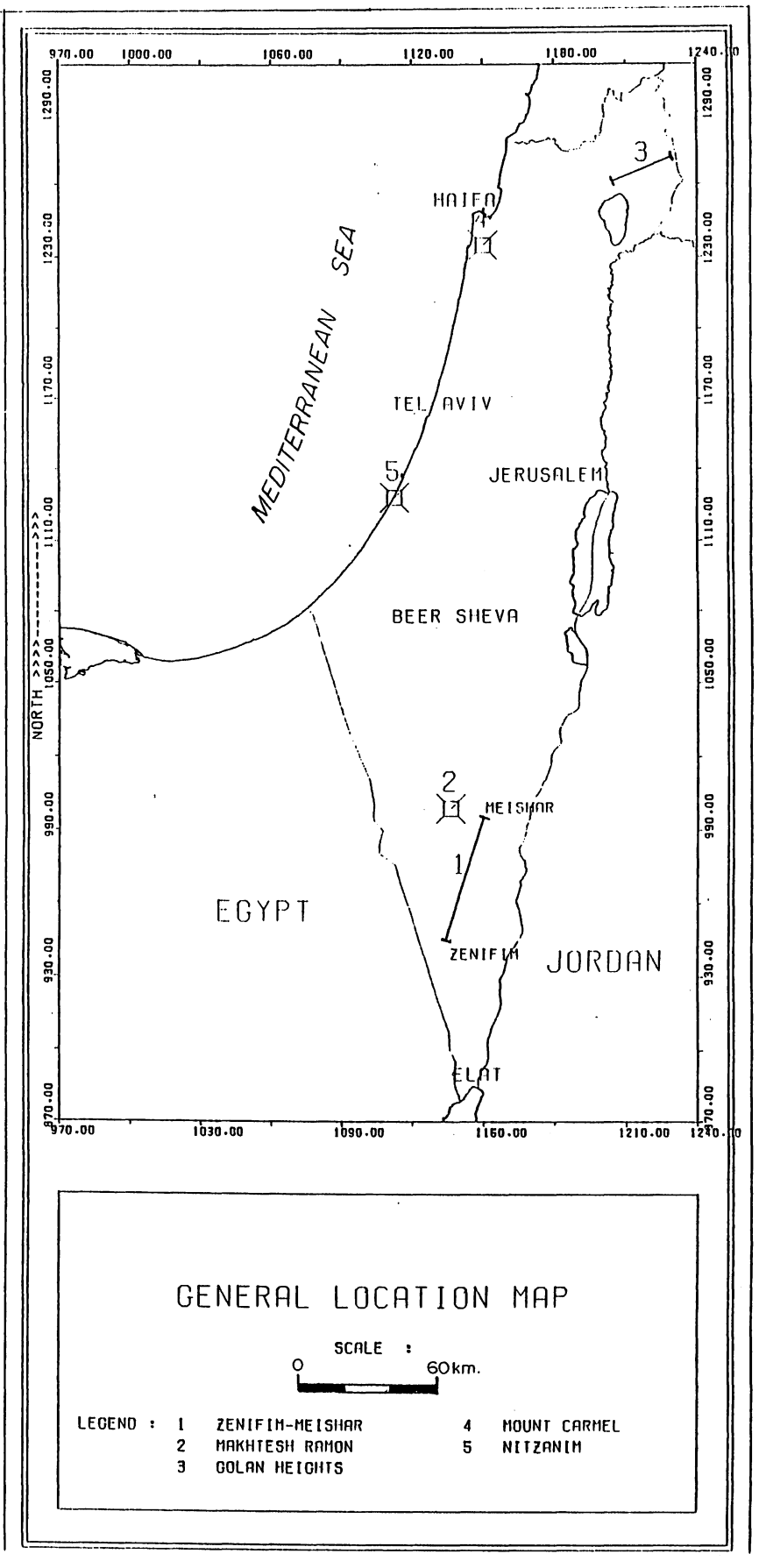

970.00
1000.00
1060.00
1120.00
1180.00
1240.00
1290.00
1230.00
1170.00
1110.00
1050.00
990.00
930.00
870.00
NORTH
MEDITERRANEAN SEA
HAIFA
TEL AVIV
JERUSALEM
BEER SHEVA
MEISHAR
ZENIFIM
EGYPT
JORDAN
ELAT
1
2
3
5
970.00
1030.00
1090.00
1160.00
1210.00
1240.00
GENERAL LOCATION MAP
SCALE :
0
60km.
LEGEND : 1 ZENIFIM-MEISHAR
2 MAKHTESH RAMON
3 GOLAN HEIGHTS
4 MOUNT CARMEL
5 NITZANIM

Figure I

1. ZENIFIM-RAMON AREA

The deep TDEM sounding technique using the Geonics EM-42 equipment was tested in the area between the Zenifim structure, including the Sinaf-1 site and the Ramon structure. The target was the contact between the Precambrian Zenifim Fm. and the overlying Paleozoic and Mesozoic sediments. According to the electric log data from the only borehole in the area (Sinaf-1), the resistivity contrast between the Zenifim Fm. and the Paleozoic sediments seems to be sufficiently large. A similar contrast was expected at the location of several magnetic anomalies which were interpreted by Domzalski (1967) as shallow magmatic intrusions. Five deep soundings (EM-42) were carried out near these targets along a profile located between Zenifim and Ramon regions.

2. SOUTHERN MAKHTESH RAMON AREA

Sixteen TDEM soundings were carried out in the area. The geological objective was to trace a possible extension of a large, probably mineralized, quartz-syenite intrusion buried beneath the sediments. Two soundings were carried out using the EM-42 and the remainder using EM-37 equipment.

3. EASTERN MAKHTESH RAMON AREA

One sounding was carried out using the EM-37 equipment in the Red Valley in order to detect a possible volcanic intrusion below the Red Valley Cauldron.

4. RAMAT KORAZIM AND CENTRAL GOLAN AREA

The Geonics EM-37 system was tried out in order to detect the contact between the exposed plateau basalts and the underlying sedimentary sequence. A total of ten soundings was carried out in the area.

5. MOUNT CARMEL AREA

Four measurement stations were established in the Mount Carmel region in order to detect the freshwater/seawater interface within the Judea Group aquifer. Three soundings were carried out using the Geonics EM-37 and one using the EM-42 equipment.

6. NITZANIM AREA

Four TDEM soundings were carried out at various distances from the sea in order to detect the seawater intrusion into the coastal plain aquifer. All measurements were carried out using the EM-37 equipment.

II. METHOD

1. BASIC PRINCIPLES OF THE TDEM METHOD

A procedure commonly utilized for ground exploration using time domain or transient electromagnetic techniques is to lay a large loop (as shown in Fig. 1) in the vicinity of the area to be examined. A steady current is generated in the loop for a period of time long enough to allow turn-on trasients in the ground to dissipate. The quiescent current is then sharply terminated in a controlled manner: for example, the current turn-off may be a linear ramp waveform (Fig. 2a). In accordance with Faraday's Law, rapid reduction of the transmitter current, and thus also of the transmitter primary magnetic field, induces an electromotive force (emf) in nearby conductors. The magnitude of this emf is proportional to the time rate-of-change of the primary magnetic field at the conductor. Therefore, it is desirable to reduce a large transmitter current to zero within a short time so as to achieve a large emf of short duration (Fig. 2b). This emf causes eddy currents to flow in the conductor with a characteristic decay, which is a function of the geoelectrical parameters of the conductor (target). The decaying currents generate a proportional (secondary) magnetic field (Fig. 2c), the time rate-of-change of which is measured by a receiver coil. In order to obtain the geoelectrical parameters of the target, either automatic inversion or various types of manual interpretation techniques are applied to the measured data.

EM - 37 SYSTEM

FIELD LAYOUT DIAGRAM

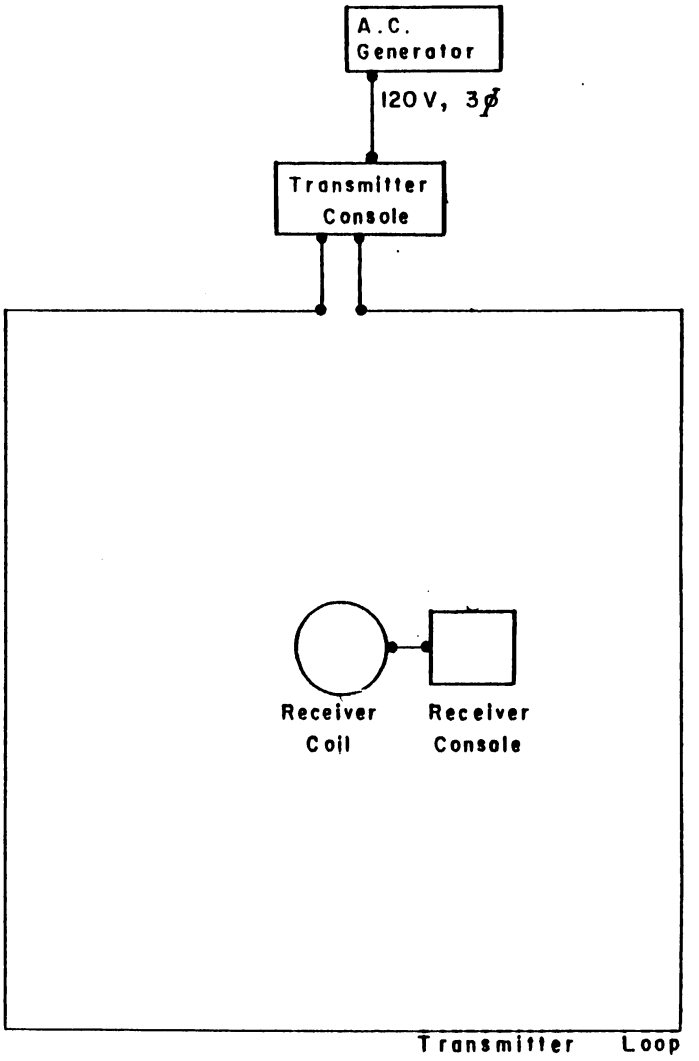
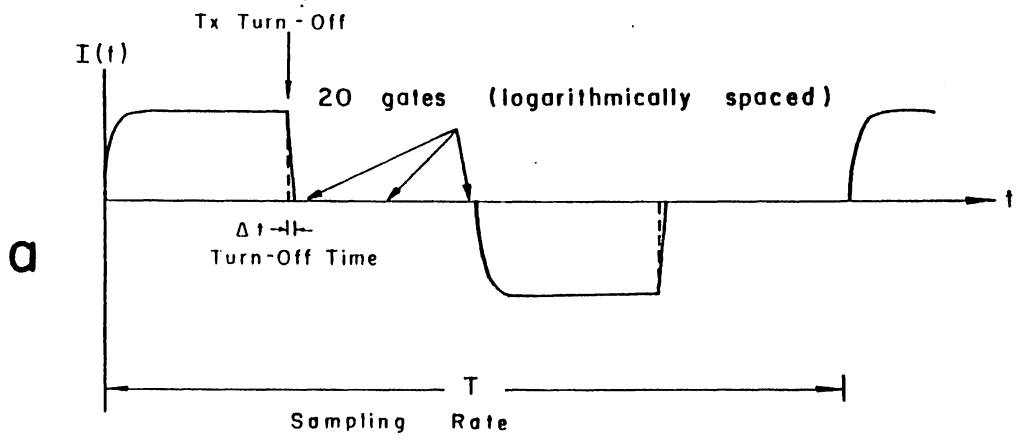
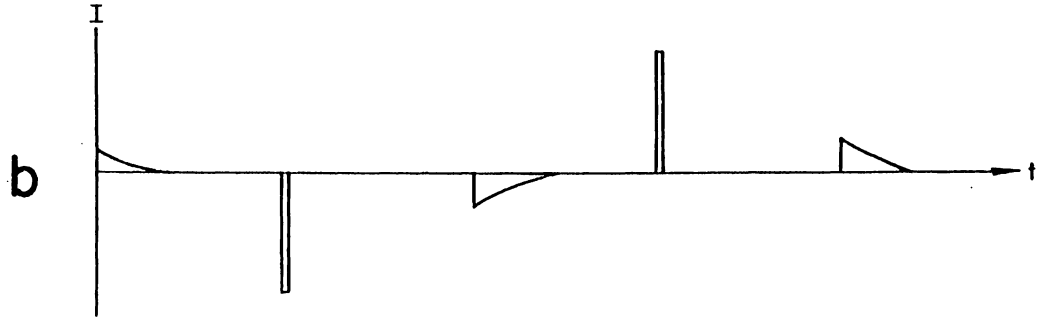


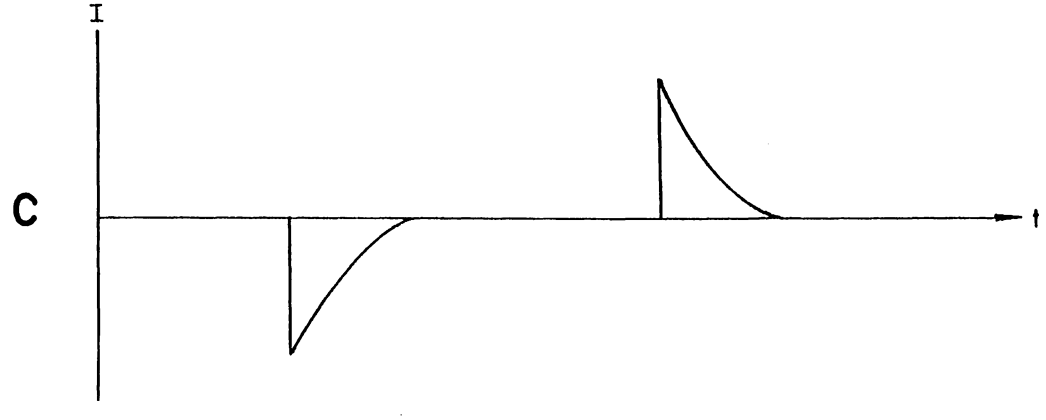
Figure 1



Transmitter Current Waveform



Induced E.M.F.



Secondary (EDDY) Currents

Figure 2

At this point it is important to emphasize that the TDEM method is the only technique which allows measurements to be carried out in the absence of the primary field. As a result, transmitter/receiver separation no longer affects investigation depth (as it is in all other electrical and EM methods) and the receiver may be located wherever convenient in the surveying area or it can even be combined with the transmitter (one loop configuration). This feature, amongst others, considerably improves lateral resolution of the method.

Historically, the time domain electromagnetic (TDEM) method developed in two different directions. On the one hand as a sounding technique for solving structural geologic problems in oil and gas prospecting as well as in groundwater and general reconnaissance explorations. On the other, as a horizontal profiling method of searching for good conductors applicable for locating massive sulfide ore bodies.

TDEM Soundings

To some extent the idea of TDEM soundings is similar to drilling, being the only method in electrical prospecting which does not require that large separations be used for deep penetration. In fact one can measure the signal inside the transmitter loop or even use the latter as a receiver.

The vertical resolution depends on the target to be detected. If the target is essentially conductive, then the resolution is excellent since the transient response is extremely sensitive to the so-called longitudinal conductance of a layer (the product of

the layer's thickness and conductivity). Thus, it is possible, for example, to detect a conductor such as a brine saturated layer 30 meters thick at a depth of more than 2 km. On the other hand, a thin resistive layer is transparent to the transient field allowing to probe the underlying strata. However, if the high resistivity layer is thick enough, it is still a good target since the transient signal is very sensitive to the longitudinal conductance of the overlying strata and, as a result, to the depth to the top of the thick resistive layer.

The most impressive advantage of the TDEM method is closely associated with the so-called equivalence problem, which is a measure of the nonuniqueness of the inverse problem solution. Experience shows that ambiguity in the interpretation of the TDEM results is significantly less than in other electrical prospecting methods. As an example, a case is presented from the east Nethanya area where 168 meters of Neogene to Quarternary shaly sediments (supposed to have a low resistivity) overlie high resistivity Mesozoic carbonates (borehole data). The results of the interpretation of the TDEM data at this location are shown in Fig. 3. The best fit is achieved with a 3-layer, H-type, apparent resistivity curve (curve #II). In order to check the equivalence, the resistivity and thickness of the second layer were simultaneously perturbed in such a manner that the longitudinal conductance of the layer would remain constant (S - equivalence). Comparing all three synthetic curves, it is concluded that there is no equivalence in the case under consideration. An essential increase in resitivity is determined uniquely at a depth of about

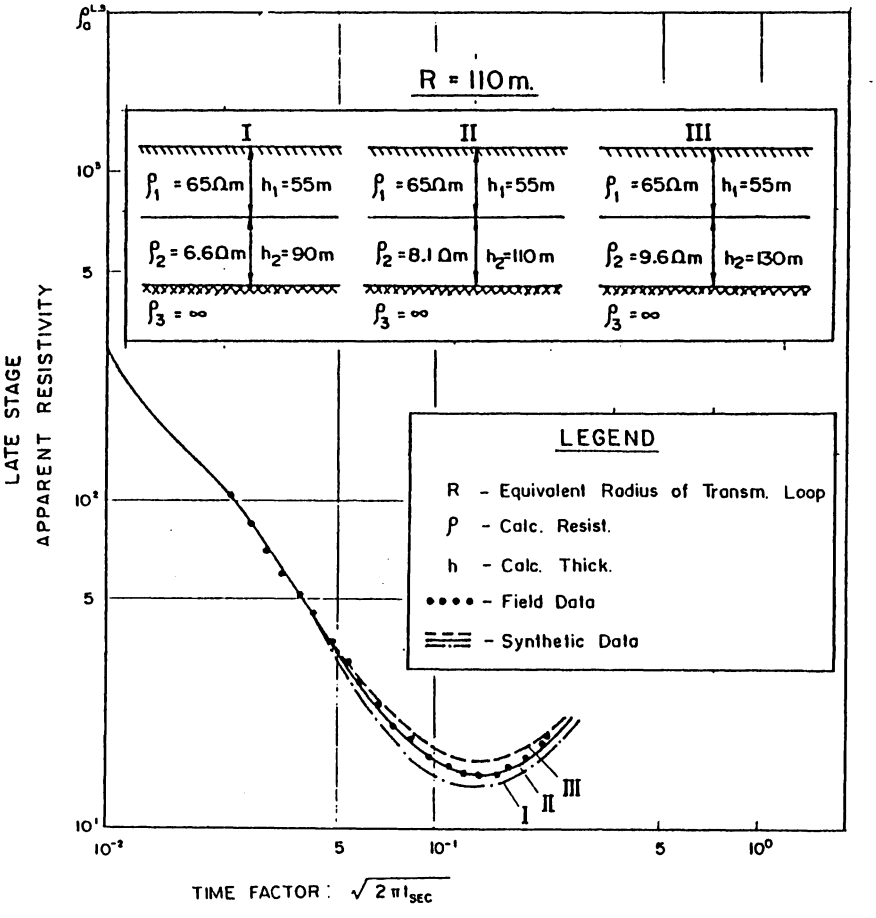

R = 110m.
I
II
III
ρ1 = 65Ωm h1 = 55m
ρ2 = 6.6Ωm h2 = 90m
ρ3 = ∞
ρ1 = 65Ωm h1 = 55m
ρ2 = 8.1 Ωm h2 = 110m
ρ3 = ∞
ρ1 = 65Ωm h1 = 55m
ρ2 = 9.6Ωm h2 = 130m
ρ3 = ∞
LEGEND
R - Equivalent Radius of Transm. Loop
ρ - Calc. Resist.
h - Calc. Thick.
•••• - Field Data
- Synthetic Data
LATE STAGE APPARENT RESISTIVITY
TIME FACTOR : √2πt_sec

Figure 3

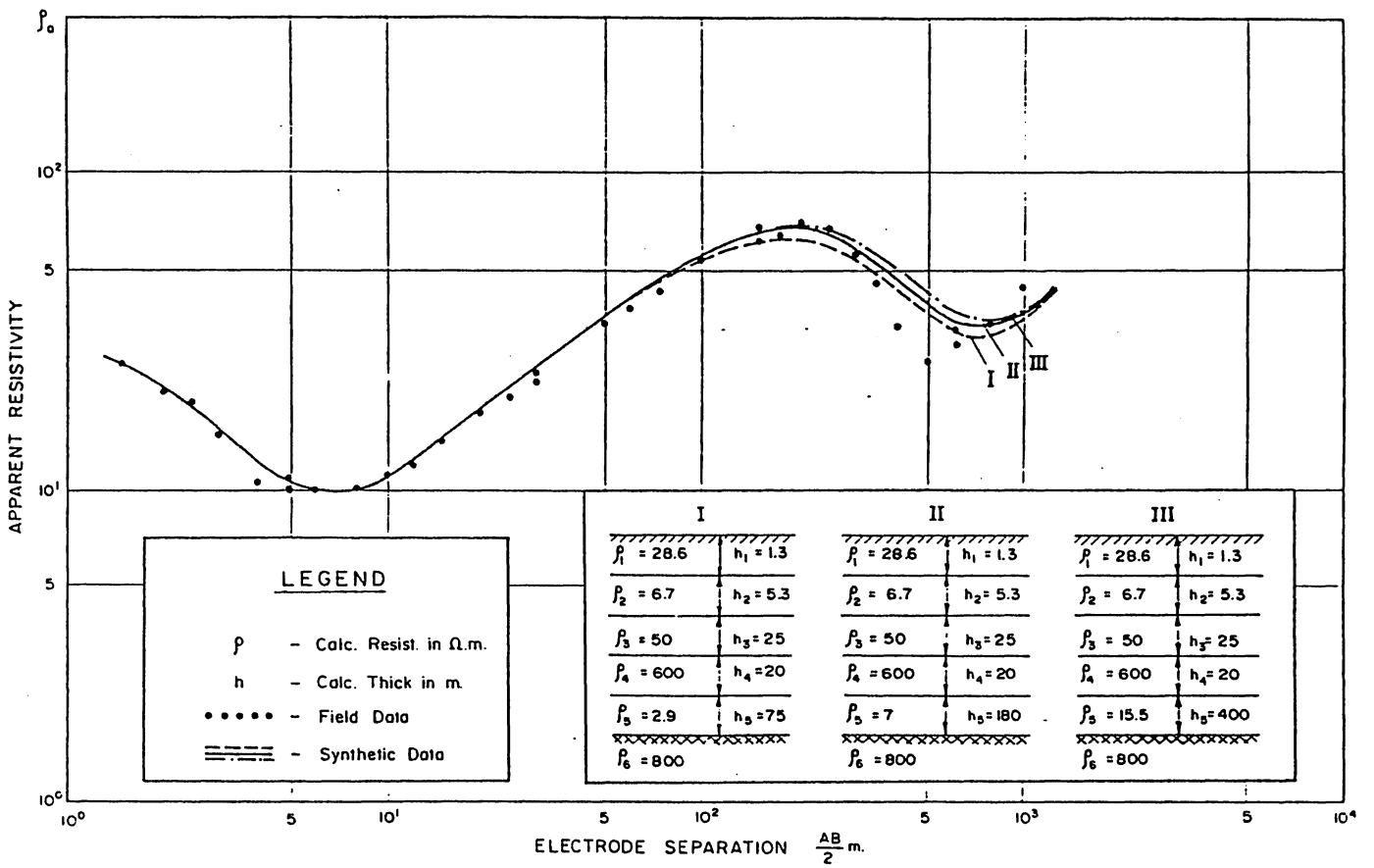

ρ_a
APPARENT RESISTIVITY
10^2
5
10^1
5
10^0
10^0
5
10^1
5
10^2
5
10^3
5
10^4
ELECTRODE SEPARATION $\frac{AB}{2}$ m.
I
II
III
LEGEND
ρ - Calc. Resist. in Ω.m.
h - Calc. Thick in m.
- Field Data
- Synthetic Data
I
ρ_1 = 28.6 h_1 = 1.3
ρ_2 = 6.7 h_2 = 5.3
ρ_3 = 50 h_3 = 25
ρ_4 = 600 h_4 = 20
ρ_5 = 2.9 h_5 = 75
ρ_6 = 800
II
ρ_1 = 28.6 h_1 = 1.3
ρ_2 = 6.7 h_2 = 5.3
ρ_3 = 50 h_3 = 25
ρ_4 = 600 h_4 = 20
ρ_5 = 7 h_5 = 180
ρ_6 = 800
III
ρ_1 = 28.6 h_1 = 1.3
ρ_2 = 6.7 h_2 = 5.3
ρ_3 = 50 h_3 = 25
ρ_4 = 600 h_4 = 20
ρ_5 = 15.5 h_5 = 400
ρ_6 = 800

Figure 4

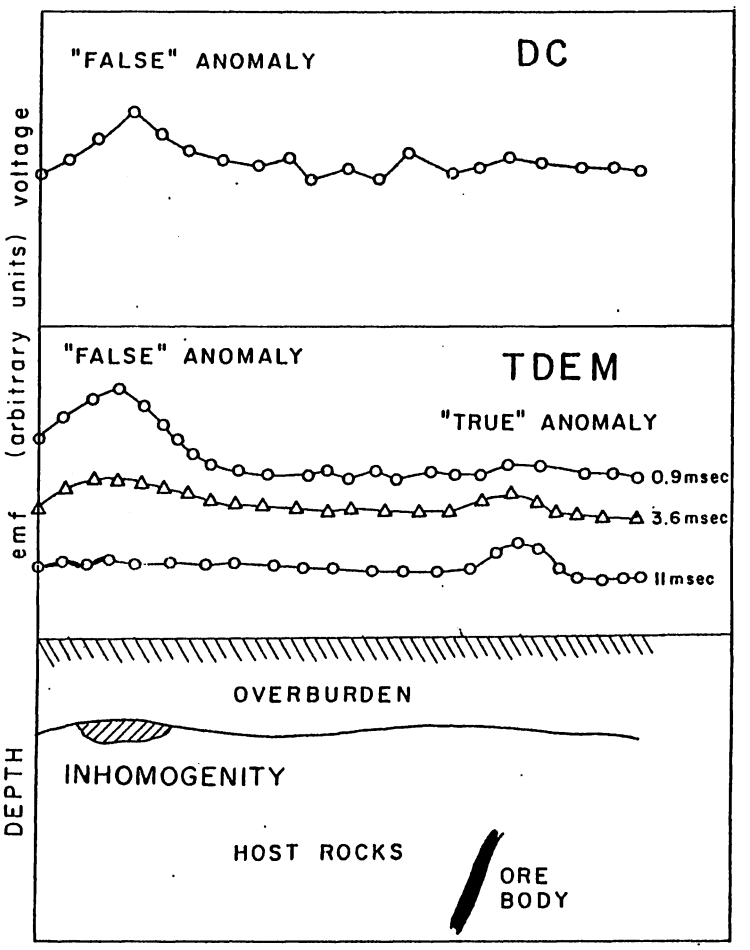

"FALSE" ANOMALY
DC
voltage (arbitrary units) emf
"FALSE" ANOMALY
TDEM
"TRUE" ANOMALY
0.9msec
3.6msec
11msec
OVERBURDEN
INHOMOGENITY
DEPTH
HOST ROCKS
ORE BODY

Figure 5

165m, corresponding to the depth of the expected resistivity contrast between the Neogene and Mesozoic deposits.

Schlumberger soundings at the same site (Fig. 4) are so strongly affected by equivalence, that the calculated depth to target may vary within an extremely wide range of about 130m to 450m (see Fig. 4). It should be noted that such a "flexibility" of DC soundings, which, in fact, makes the interpretation almost arbitrary, has unfortunately shaken the geologists' faith in all electric methods in Israel.

TDEM Horizontal Profiling

In order to detect an ore body, two conditions must be fulfilled:

a) The anomaly above the ore body must be significant

b) It should be possible to distinguish between "true" deep anomalies and "false" anomalies which are caused by near surface conductive inhomogeneities.

Only a transient signal can distinguish between the "true" and "false" anomalies due to the fact that the signal in the time domain is sensitive to relatively less conductive and shallow objects at early times, while the effect of more conductive and deeper targets becomes dominant at later times (Fig. 5).

2. DEFINITION OF APPARENT RESISTIVITY

The concept of apparent resistivity was introduced into the time domain electromagnetic (TDEM) method to facilitate intuitive interpretation. The apparent resistivity does not contain any more

information than the original field observations, but by its shape it more clearly shows the changes in earth resistivity with depth. In other electrical methods, examining apparent resistivity curves is a quick and practical way to get a general picture of the geoelectrical structure and they also lead to a starting model for inversion. However, unlike the DC electrical sounding case, it has not been possible to find a simple time domain formula to define apparent resistivity. This is because of the different relations between the observed voltage and resistivity in a uniform earth during early and late times. Indeed, as is very well known (Kaufman and Keller, 1983), the observed voltage increases with increasing resistivity at early times ($\sim\rho$) but decreases with increasing resistivity at late times ($\sim\rho^{-3/2}$). Hence a simple proportional formula to define the apparent resistivity cannot be meaningful at both early and late times.

One useful approach to this problem is to define the apparent resistivity separately for early and late times (Kaufman and Keller, 1983). Unfortunately, neither definition is suitable in the whole time range. In fact the early stage apparent resistivity reflects the behavior of a true resistivity of the ground for only relatively shallow depths; as for the deeper parts, the early stage apparent resistivity is no better than the original voltage curves. The opposite is true for the late stage apparent resistivity which clearly shows by its shape the properties of the deep, and sometimes intermediate parts, of the geoelectrical section. However, the initial left-hand branch of the late stage

apparent resistivity curve virtually never reflects distribution of a true resistivity in the shallow part of the section.

Experience shows that the early stage definition is reasonable for the so-called long-offset (or far zone) TDEM soundings (T/R separation several times greater than the depth to the target), while the late stage definition is relevant for the short-offset (or near zone) soundings.

Since all our soundings were carried out using short-offset T/R configurations, only the late stage apparent resistivity definition was used for both the field and synthetic data. The reader should not be confused, therefore, by the "strange" behavior of the curves at early times, which results in the existence of the ascending left-hand branch in all curves. The latter does not mean that all models consist of a high resistivity upper layer, but is simply due to some mathematical transformation (namely, division of the signal by measuring time):

$$\rho_a(t) = \left(\frac{\mu^{5/2} R^2 I^2 \pi^2 r^4}{20\sqrt{\pi} t^{5/2} \varepsilon(t)}\right)^{2/3}$$

where $\varepsilon(t)$ is the electromotive force

I is a DC current in the source loop

r is the radius of the receiver coil (a small circle loop)

R is the radius of the source loop

t is a time after current switches off

μ is the magnetic permeability

3. EM37 GROUND TRANSIENT ELECTROMAGNETIC SYSTEM

The EM37, manufactured by Geonics Ltd. of Toronto, Canada, is a time domain EM prospecting system, suitable for depth penetration of up to 300 meters, depending on ground resistivity.

The system is compact and portable, requiring a four-man field crew. One of the crew should be a geophysicist or engineer familiar with geoelectrical methods. One other crew member should be a technician/operator. The other two members can be standard field hands.

The operative configuration is quite straightforward (see Fig. 1). The transmitter consists of a gasoline powered generator, a transmitter console and a turam type square loop. The receiver consists of a 1 meter diameter, air-cored coil and a console controlling pulse transmission by the transmitter and data accumulation. Synchronization between the transmitter and receiver is effected by a high stability quartz crystal oscillator. Timing is controlled by the receiver console.

The transmitter loop is laid out upon the area to be studied. The receiver coil is placed flat (horizontal) in the exact center of the transmitter loop. A steady current is generated in the loop for a period of time long enough to allow transients in the ground to decay. Upon a synchronized signal from the receiver console, the current in the loop is abruptly terminated (see Fig. 2). The turn-off time (Δt) does not exceed 400 microseconds. The abrupt termination of the primary field induces an electromotive force in the conductors. This induced emf gives rise to decaying eddy currents in the conductor. Measurement of these eddy currents

LOCATIONS AND WIDTHS OF GATES FOR BOTH 25Hz and 2.5Hz

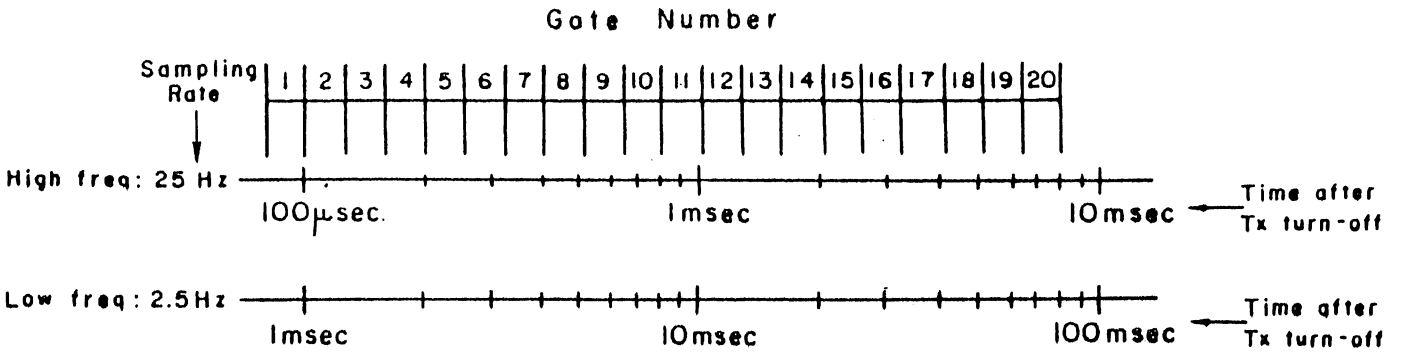


Figure 6

contains our desired information as to depth, thickness and conductivity of subsurface geoelectrical layers.

To resolve the shape of the decaying current, measurements must be made at a large number of logarithmically spaced times (Fig. 6). To improve signal-to-noise ratio, sampling is effected and averaged over many cycles.

Sampling rates is switch selectable at the transmitter to provide a high frequency sampling rate of 25Hz to permit a reading of transient decay out to 8 milliseconds (see Fig. 6) . The base frequency of operation of 25Hz was selected in order to assure that power line frequency of 50Hz will not bias the data. The transmitter also provides a low frequency sampling rate of 2.5Hz to allow transient decay readings out to 80 milliseconds.

4. EM42 - HIGH POWER DIGITAL GROUND TRANSIENT ELECTROMAGNETIC SYSTEM

The Geonics EM42 basically consists of a high power transmitter and a real-time digital acquisition system. The system has been shown theoretically to have the capacity to determine the complete geoelectric sequence to depths reaching 3000m with excellent lateral and vertical resolution.

The EM-37 system is still used in conjunction with the EM-42 in order to explore relatively shallow depth intervals. The EM-37 operates at a base frequency of 25Hz and the received signal is automatically integrated for a preset number of periods, after which it is read out as a complete transient decay from 80 μsec to

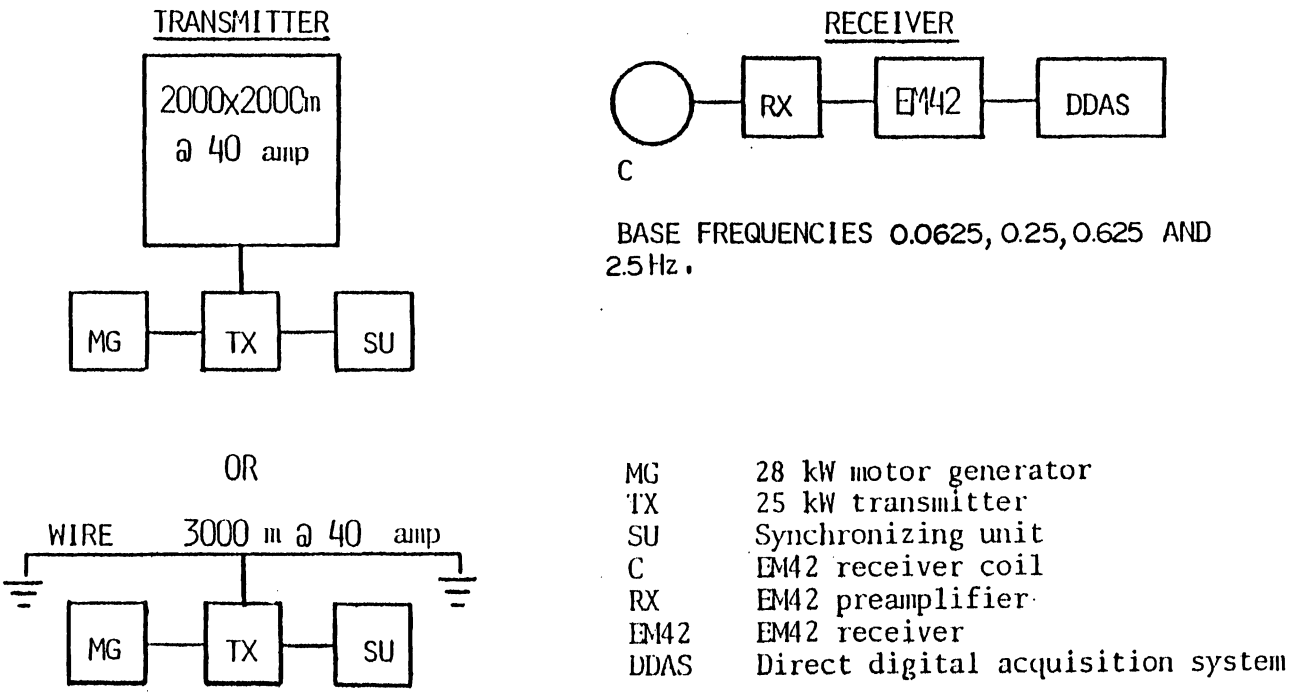
8 msec. This procedure is possible since the signal-to-noise ratio is high at a base frequency of 25Hz.

In order to extend the measurement of the transient response to further times, the base frequency must be reduced. For example, at 2.5Hz it is possible to make measurements out to 80 msec. Although the "spheric" noise power spectral density is decreasing with decreasing frequency, at 2.5Hz the measurements are made only one-tenth as often at this base frequency and the signal-to-noise ratio starts to deteriorate at late times.

Since there are many components to the noise at these low frequencies (both natural and man-made), rendering it non-random and non-Gaussian, it becomes advantageous to employ a more sophisticated detection scheme than simply integrating the response over many transmitter cycles. For base frequencies of 2.5Hz or less, it was therefore decided to record the entire transient directly in real time using a 450 μsec sampling rate. This high sampling rate is required both to adequately resolve the transient decay and to reject noise. When combined with the large dynamic range required to resolve the transient, this results in a large volume of data that must be stored at a high rate. Such consideration have been incorporated in the real time direct digital acquisition system, which essentially consists of a Compaq Plus computer operating with a 3.5" Winchester Disc as well as a 9 track tape recorder. This combination also permits rapid field analysis of the large volume of acquired data.

Finally, for later times (out to 3 seconds) it is necessary to employ a high power transmitter. Calculations have shown that

EM42 SYSTEM CONFIGURATION



BASE FREQUENCIES 0.0625, 0.25, 0.625 AND 2.5 Hz.

MG	28 kW motor generator
TX	25 kW transmitter
SU	Synchronizing unit
C	EM42 receiver coil
RX	EM42 preamplifier
EM42	EM42 receiver
DDAS	Direct digital acquisition system

Figure 7

the 25kw transmitter incorporated into the EM42 is optimum for this application.

The two configurations in which the EM42 system can be used are shown in Fig. 7.

5. PROCESSING AND INTERPRETATION OF FIELD DATA

As previously described, the measured signal is "preprocessed" by the equipment in order to improve the signal-to-noise ratio. Processing proper begins when the "preprocessed" data are transferred from the recorder to the IBM compatible PC. This process comprises the following main stages:

1. Data editing.
2. Correction for the current waveform.
3. Calculation and plotting of apparent resistivities.

The most critical stage in the data processing is the second one. In fact both the EM-37 and EM-42 transmitters provide linear-ramp current waveform; at the same time most of the currently available inversion algorithms, as well as manual interpretation techniques are based on an ideal step function excitation. Thus, there are two alternative methods of adjusting field data and interpretation techniques:

- deconvolution of the measured data for the finite ramp turn-off time; or
- convolution of the theoretical forward solution which is used in properly modified inversion algorithms.

The main advantage of the first approach is that all powerful theoretical and interpretational apparatus, developed over several

years with regard to the perfect step function excitation, can be applied to the deconvolved field data. However, our experience has shown that, in many cases, deconvolution has so seriously distorted field data that the only way to interpret the data was to employ inversion algorithms modified for the linear-ramp excitation (the second approach). The most sophisticated deconvolution algorithm was developed by Geonics Ltd. and is described by Levy (1984). In the majority of cases we corrected the measured signal using the Geonics deconvolution algorithm.

The corrected data were then converted to the late stage apparent resisitivities (see Section II, Pagagraph 2) and interpreted in terms of the required geoelectrical parameters using standard interpretational techniques, such as automatic inversion algorithms ARRTI (Stoyer, 1986) and TDINV (Goldman, 1988) or the conventional manual interpretation technique (Kaufman and Keller, 1983).

During the last stage, a synthetic apparent resistivity curve was calculated for the geoelectical parameters obtained in the interpretation. This synthetic curve was calculated using the forward solution for the linear-ramp current waveform (Fitterman and Anderson, 1987). In cases where the synthetic response did not fit well with the original (uncorrected) field data, the modified inversion algorithm was employed to the measured data (the second approach). Thus, for both approaches, all curves were compared to the original (uncorrected) field data. This allowed us on the one hand to avoid almost entirely any possible errors due to the

correction procedure and, on the other, to use all previous theoretical and interpretational developments.

Figs. 16 thru 31, 34 thru 43, 48 thru 50 and 53 thru 56 show late stage apparent resistivities calculated from the original responses, as well as appropriate synthetic curves obtained through the convolution of the standard forward solution (Fitterman and Anderson, 1987).

III. SURVEY

1. ZENIFIM RAMON AREA: DETECTION OF TOP PRECAMBRIAN BASEMENT AND HYPOTHETIC MAGMATIC INTRUSIONS

The deep TDEM sounding technique using the Geonics EM-42 equipment was tested in the area between the Zenifim structure, including the Sinaf-1 site, and the Ramon structure (Fig. 8). The target to be detected was the contact between the Precambrian Zenifim Fm. and the overlying Paleozoic and Mesozoic sediments. According to the electric log data from the only deep borehole in the area (Sinaf-1), the resistivity contrast between the Zenifim Fm. and the Paleozoic sediments seems to be sufficiently large (Fig. 9). A similar contrast was expected at the location of several magnetic anomalies interpreted by Domzalski (1967) as magmatic intrusions (Fig. 8).

Five soundings were carried out along a profile located between Zenifim and Ramon, crossing the above mentioned anomalies (Fig. 8). None of the measurements clearly showed a rise in resistivity at the expected depths (Figs. 10 thru 14). This can be explained by either insufficient penetration or too small a resistivity contrast. In these cases the main conclusion which can be drawn from the experiment is that the TDEM method is not feasible for the above particular problems, at least with the Geonics EM-42 and EM-37 equipment and the transmitter/receiver (T/R) configuration which were used for the measurements. If the problem is the penetration depth alone, then the most straightforward solution would be to use another T/R

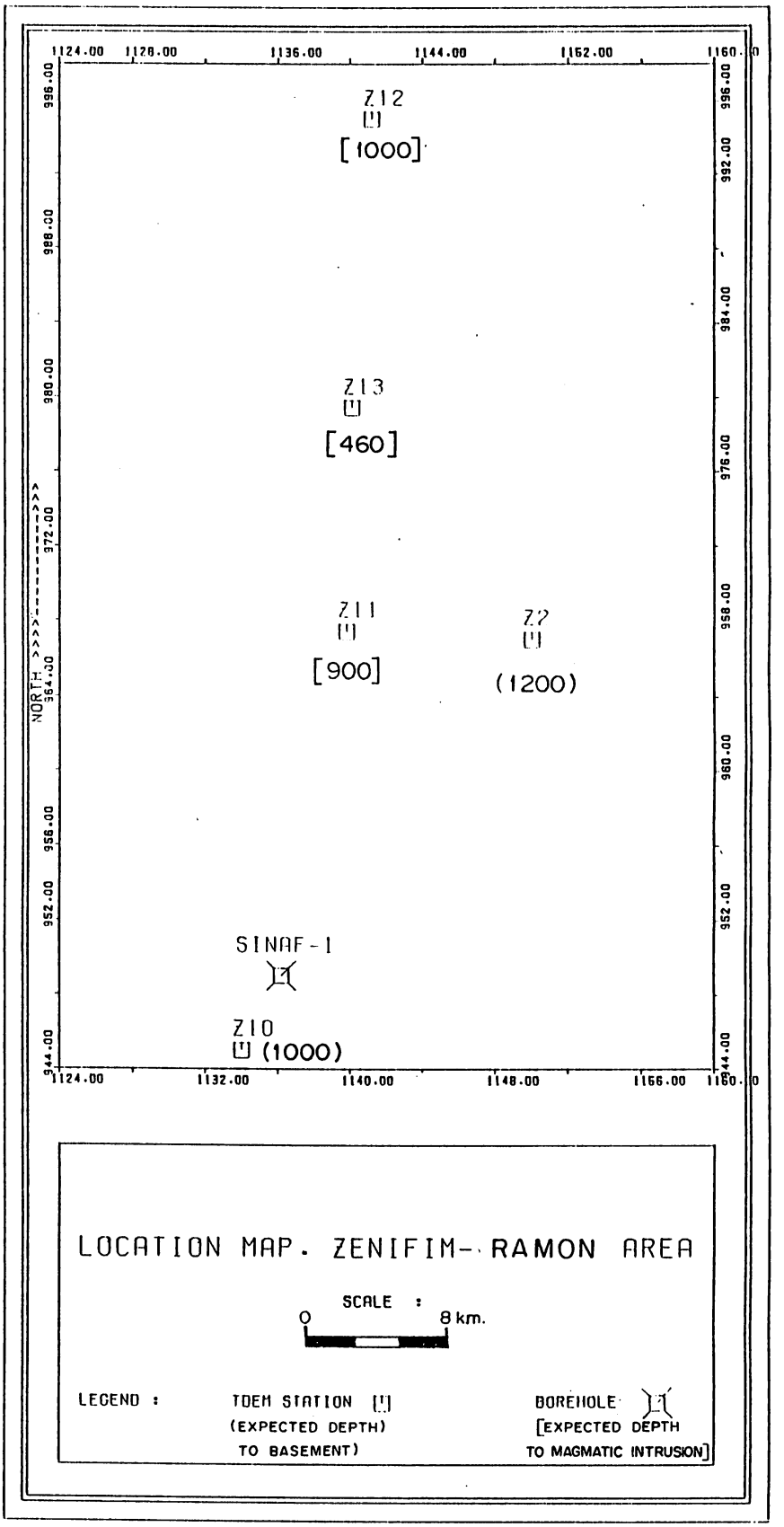

1124.00
1128.00
1136.00
1144.00
1152.00
1160.00
996.00
992.00
988.00
984.00
980.00
976.00
972.00
964.00
960.00
956.00
952.00
944.00
NORTH >>>>>>>>>>>>>>>>>>>>>>>>
Z12
[1000]
Z13
[460]
Z11
[900]
Z?
(1200)
SINAF-1
Z10
(1000)
1124.00
1132.00
1140.00
1148.00
1156.00
1160.00
LOCATION MAP. ZENIFIM-RAMON AREA
SCALE :
0
8 km.
LEGEND :
TDEM STATION
(EXPECTED DEPTH)
TO BASEMENT)
BOREHOLE
[EXPECTED DEPTH
TO MAGMATIC INTRUSION]

Figure 8

CONDENSED LOG OF SINAF NO. 1

Coord: E: 136000 N: 094883

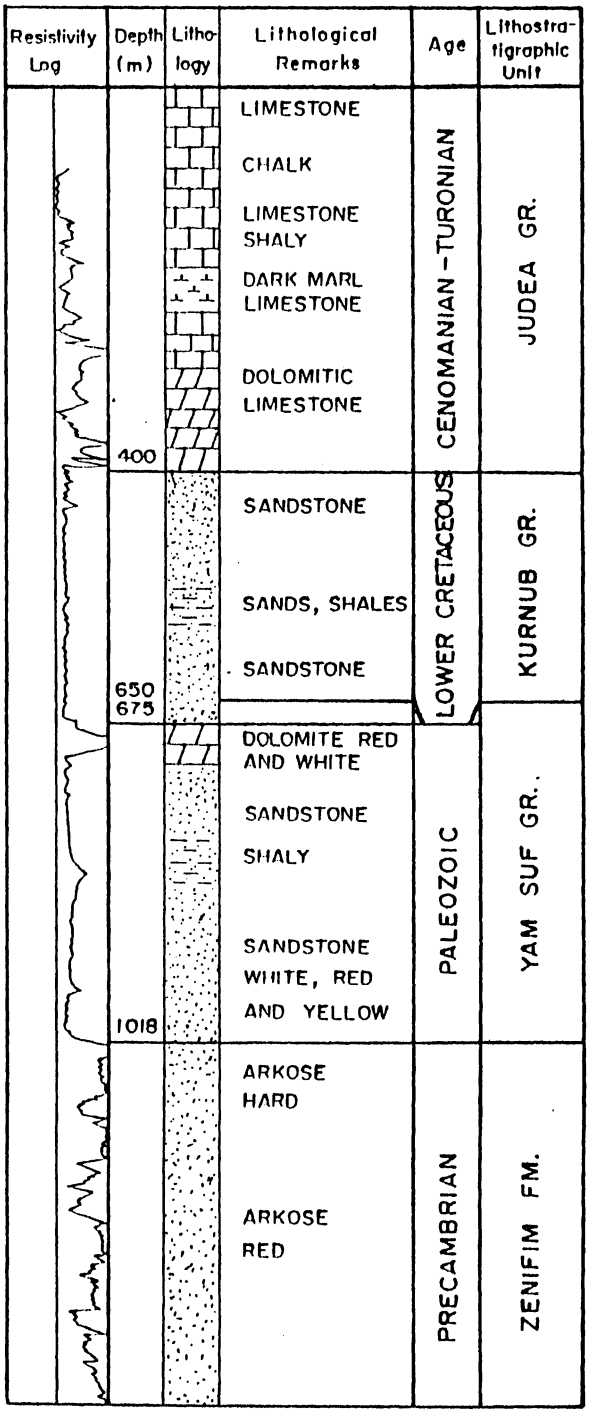


Figure 9

LINE: 001N STA.: 001E

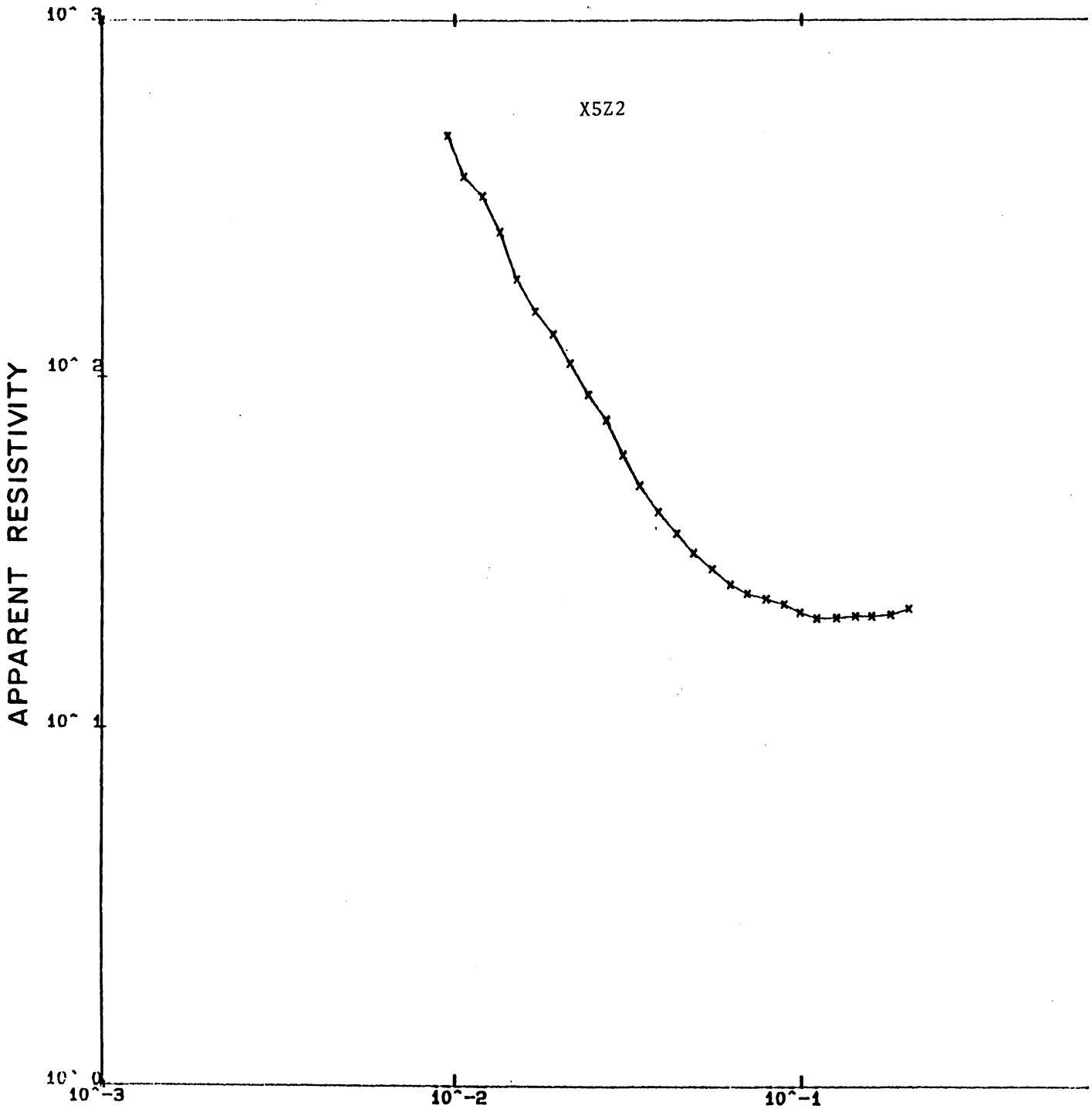


Figure 10

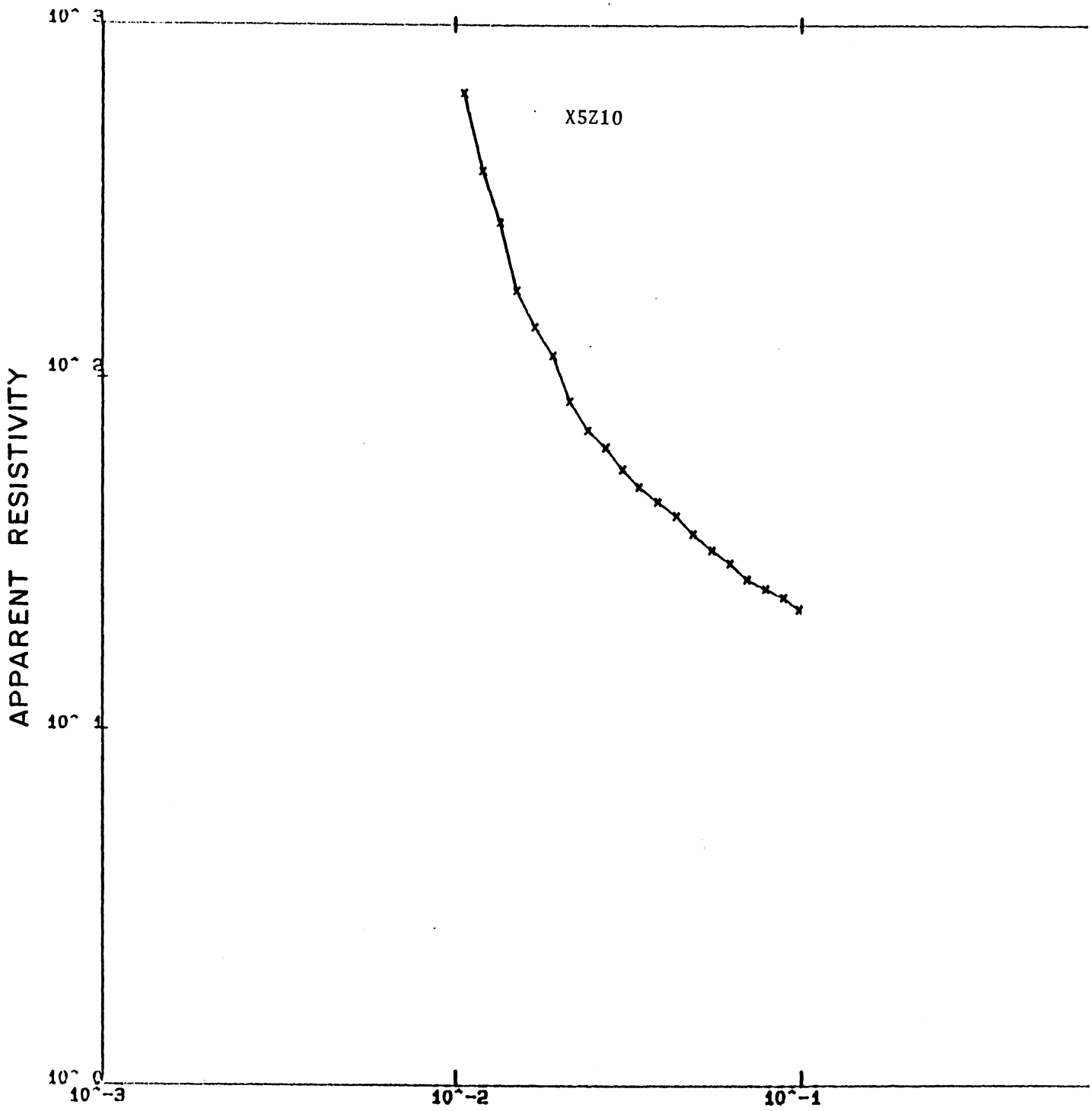
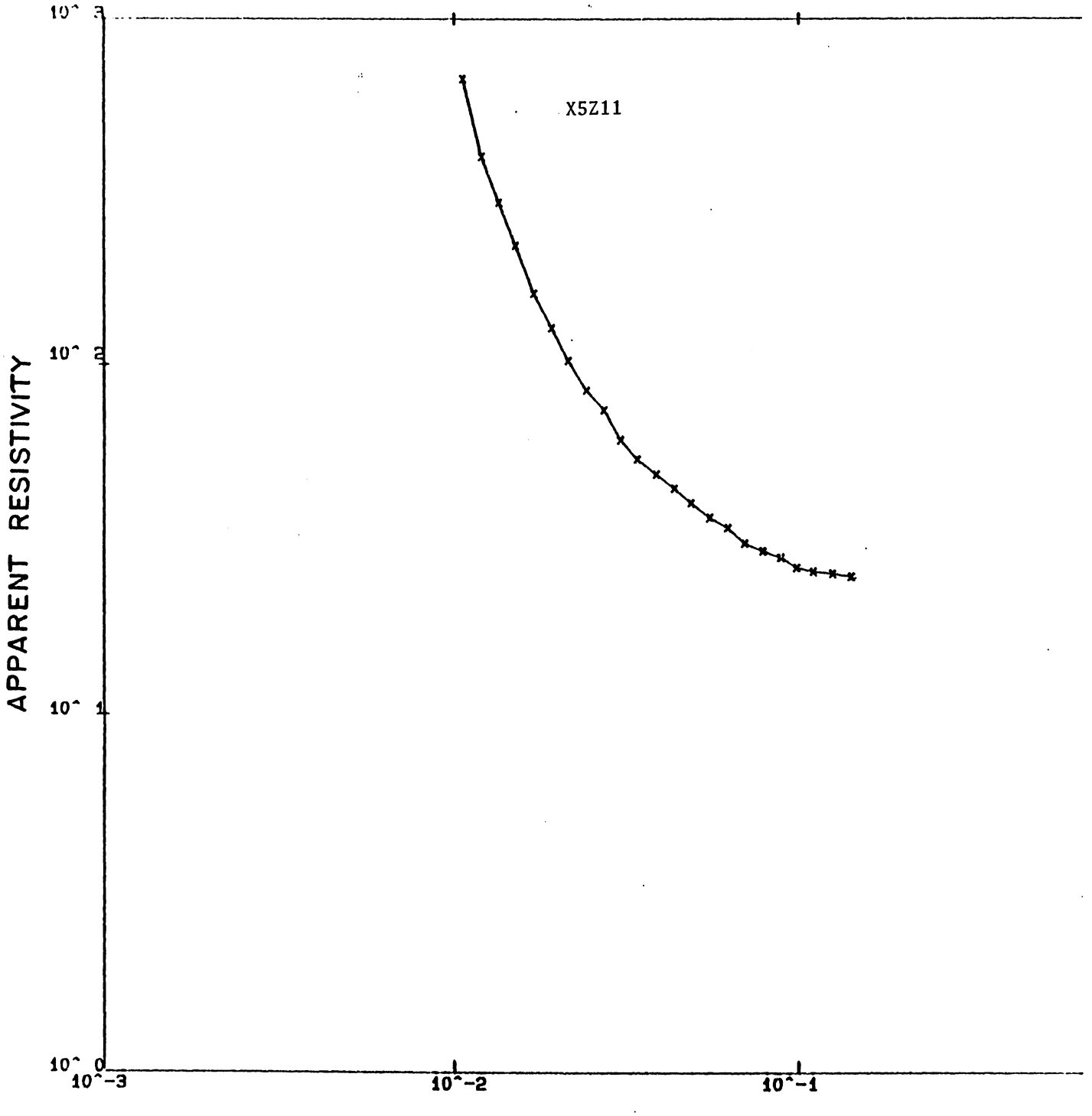

LINE: 100N
STA.: 100W
x SND# avg.
X5Z10
APPARENT RESISTIVITY
10^ 3
10^ 2
10^ 1
10^ 0
10^-3
10^-2
10^-1
TIME sec^1/2
THE EARTH TECHNOLOGY CORP.

Figure 11

LINE: 110N STA.: 110E

x SND# avg.



THE EARTH TECHNOLOGY CORP.

Figure 12

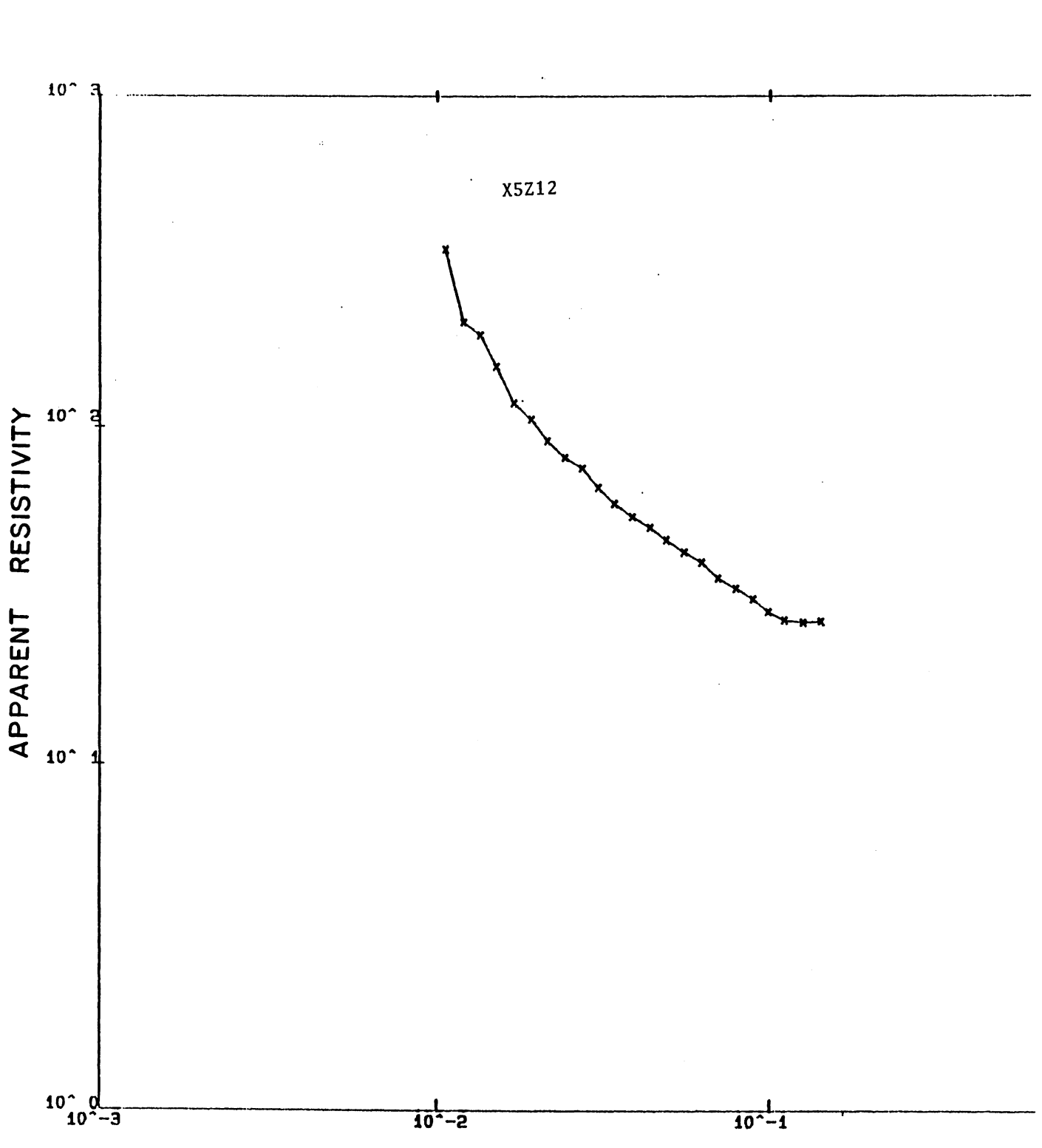

LINE: 120N
STA.: 120E
x SND# avg.
X5Z12
APPARENT RESISTIVITY
10^ 3
10^ 2
10^ 1
10^ 0
10^-3
10^-2
10^-1
TIME sec^1/2
THE EARTH TECHNOLOGY CORP.

Figure 13

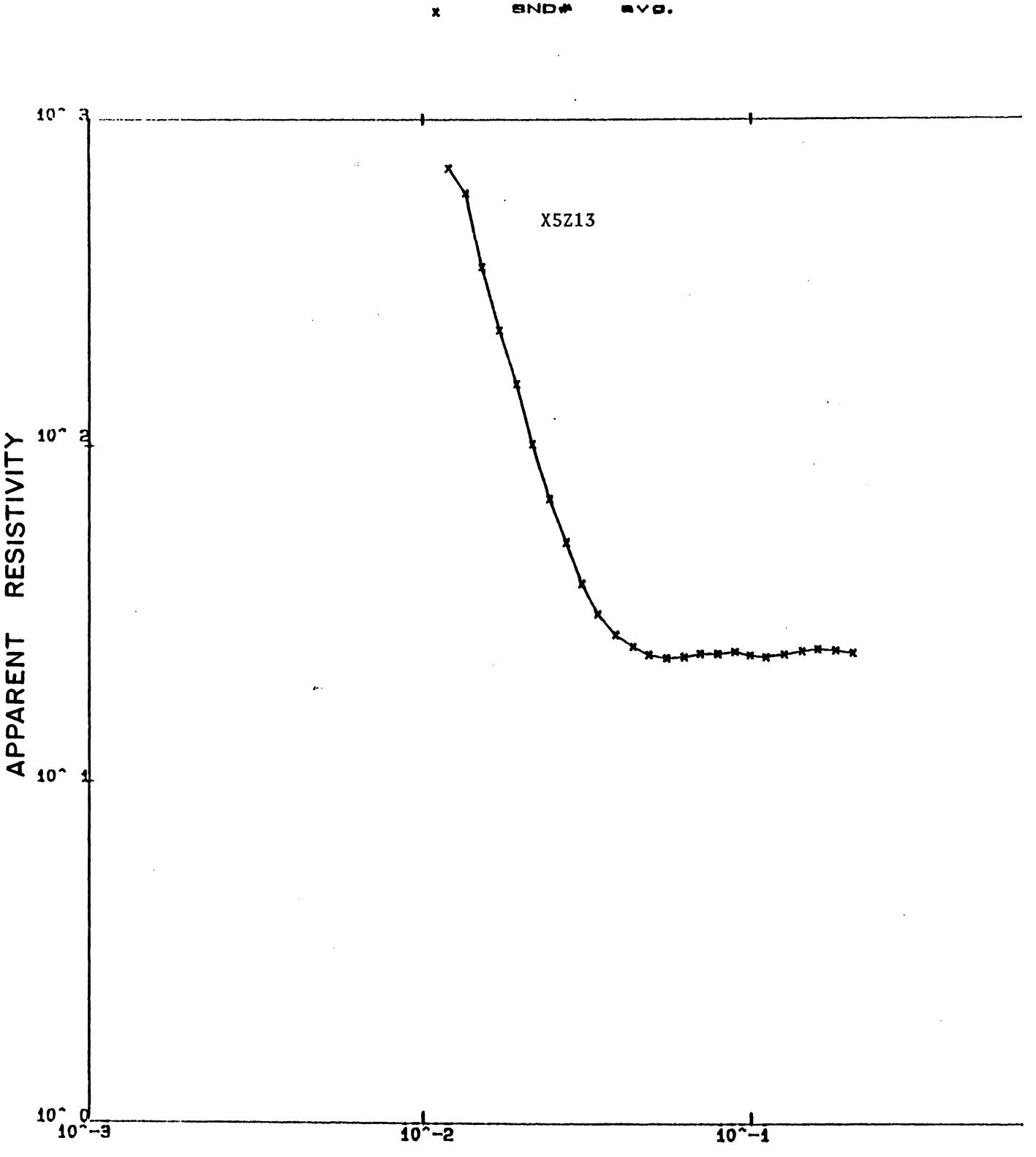

LINE: 130N
STA.: 130E
x SND# AVG.
X5Z13
APPARENT RESISTIVITY
10^ 3
10^ 2
10^ 1
10^ 0
10^-3
10^-2
10^-1
TIME sec^1/2
THE EARTH TECHNOLOGY CORP.

Figure 14

configuration, namely an AB-LOOP configuration (Fig. 7), with the T/R separation exceeding the required depth by a factor of about 3-4.

A third explanation might be that the magmatic bodies, interpreted by Domzalski, are not in their true location or are misinterpreted. In this case the real location and nature of the anomalies should be explained by an integrated geophysical survey.

2. SOUTHERN MAKHTESH-RAMON AREA: GEOMETRY OF THE QUARTZ-SYENITE INTRUSIONS

2.1 Background

The Lower Cretaceous quartz-syenite bodies along the southern flank of the Ramon (Zak, 1968) have attracted attention in recent years owing to the polymetallic hydrothermal vein type mineralization (Itamar, 1986, 1988, Itamar et al , 1983) found in them. This mineralization has been shown to be connected with the intrusion of the host rocks; thus the delineation of the extension of the quartz-syenite intrusions is important for assessing the overall potential of the region.

The magmatic rocks are exposed along the southern Ramon as a series of isolated outcrops. The quartz-syenite intrudes Triassic and Jurassic sediments. The Gavnunim outcrops ($\sim 1 km^2$, Fig. 15) have been postulated to be sill and stock-like bodies. The Shen Ramon intrusion protrudes some 150 meters above its surroundings ($\sim 1.5 km^2$). Small bodies (dikes) and alterations of the country rock occurring between Shen Ramon and Gavnunim, have been postulated to indicate a continuous subsurface magmatic body (Itamar and Baer, 1986). To the south the intrusions are cut by

the Ramon fault and, to the north, they probably vanish with a high dip.

The aim of the TDEM survey was to trace the possible subsurface extension of a large quartz-syenite intrusion between Shen Ramon and Gavnunim. Sixteen central loop TDEM soundings were carried out in the area. Based on the local geological set-up, relatively small loops, up to 200 by 200 meters, and Geonics EM37 equipment were used in the central and eastern part of the area. Larger loops, up to 400 by 400 meters, and Geonics EM42 equipment were used in the northern part.

2.2 Results

The 16 measured apparent resistivity curves can be divided into three groups:

a) Apparent resistivities measured at stations M1, M3 thru M7, M11, M13 and M15 all (except for station M7) show a clear, right-hand ascending branch. Such curve behavior undoubtedly proves the presence of a high resistivity layer in the base of the section. According to general geological knowledge of the sedimentary sequence, this layer could most likely be identified with the target. Quantitative interpretation of these curves leads to the same conclusions and, furthermore, permits the calculation of the depth to the top of the target (Figs. 16, 18, 19 thru 21, 26, 28 and 30).

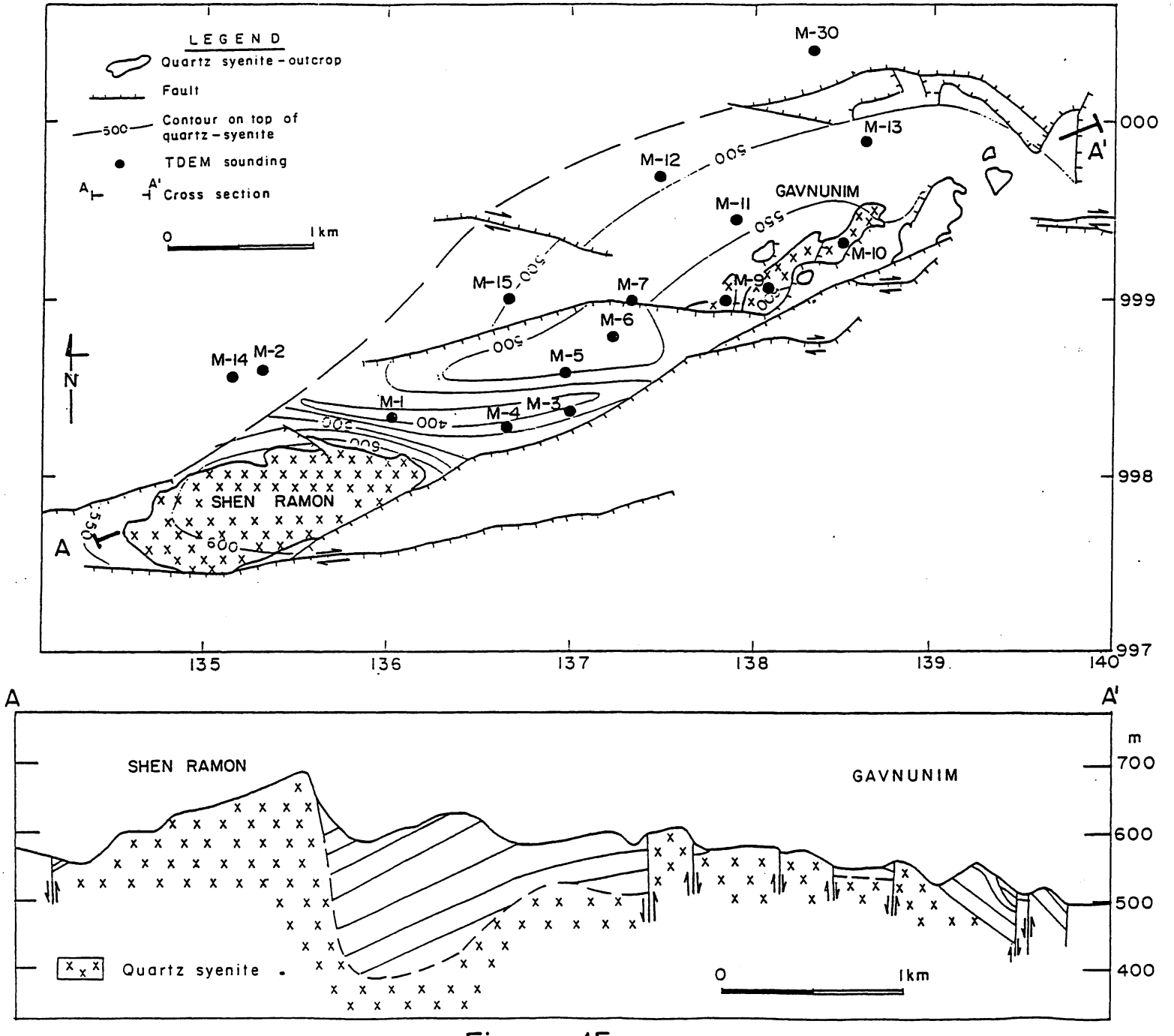

LEGEND
Quartz syenite - outcrop
Fault
Contour on top of quartz - syenite
TDEM sounding
Cross section
0
1 km
N
M-30
M-13
M-12
M-11
M-10
M-9
M-7
M-6
M-5
M-4
M-3
M-2
M-1
M-14
M-15
GAVNUNIM
SHEN RAMON
A
A'
135
136
137
138
139
140
997
998
999
000
m
700
600
500
400
Quartz syenite

Figure 15

Station M7, on the other hand, exhibits an ascending branch in the intermediate part of the curve (Fig. 22). Such behavior can be explained in two ways:

(i) within the frame of a one-dimensional model, the only possible explanation is that the thickness of the target (resistive) layer is limited to approximately 300 meters and is underlain by a highly conductive sequence;

(ii) the model is essentially two or three dimensional.

The latter explanation can be substantiated by the fact that the measuring point is located next to a known large fault (Fig. 15).

b) Stations M2 and M14, north of Shen Ramon, and station M30, situated in close proximity to the Ramon-1 borehole, show field curves with a monotonous decrease in apparent resistivity with the measuring time (Figs. 17, 29 and 31). These curves are interpreted as indicating a decrease of the true resistivity with depth, implying, in turn, the absence of a high resistivity layer to a depth of at least several hundred meters. The results correlate well with the data from the Ramon-1 borehole which did not penetrate significant layers of quartz-syenite.

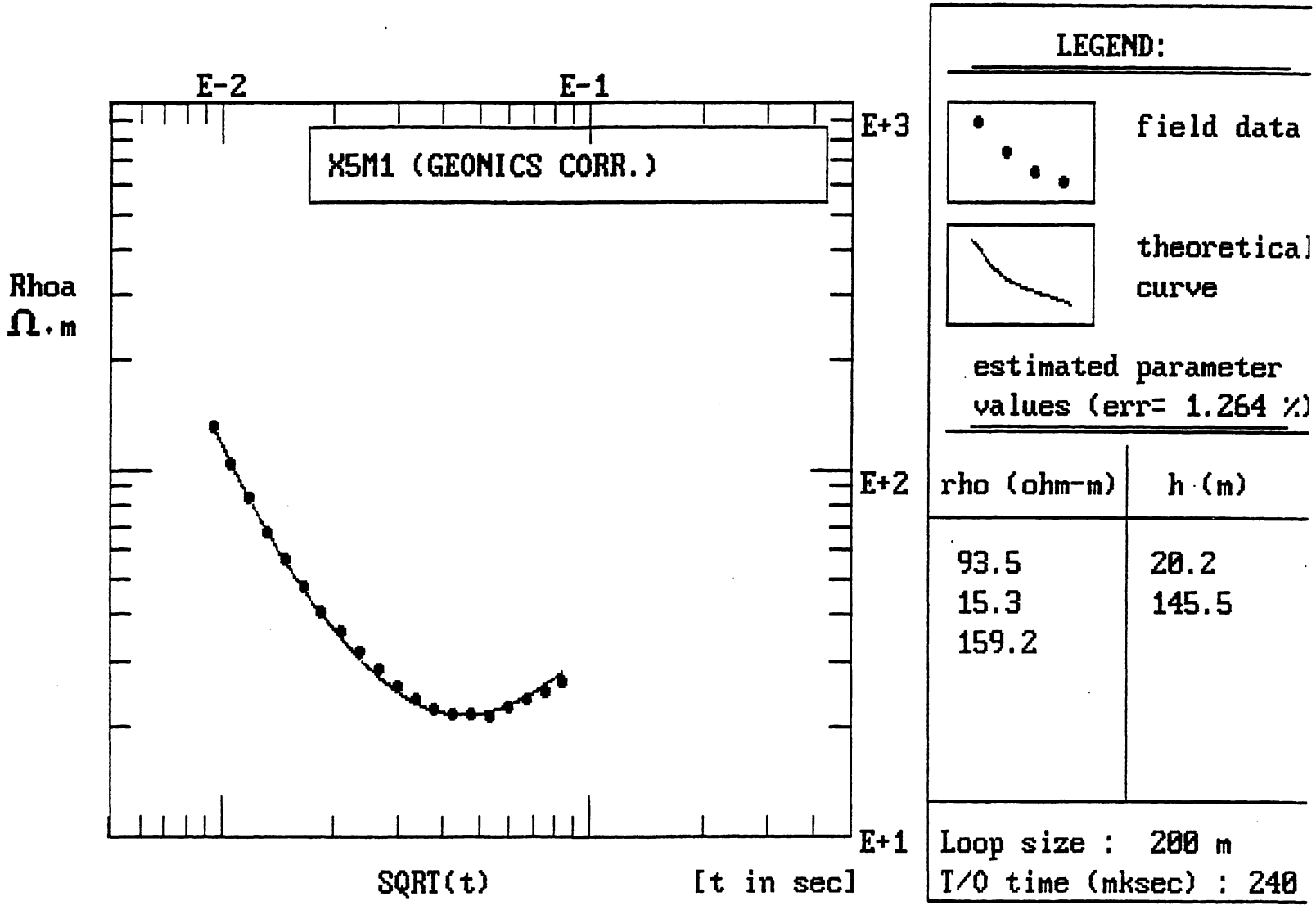

E-2
E-1
E+3
E+2
E+1
X5M1 (GEONICS CORR.)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.264 %)
rho (ohm-m)
h (m)
93.5
15.3
159.2
20.2
145.5
Loop size : 200 m
T/O time (mksec) : 240

Figure 16

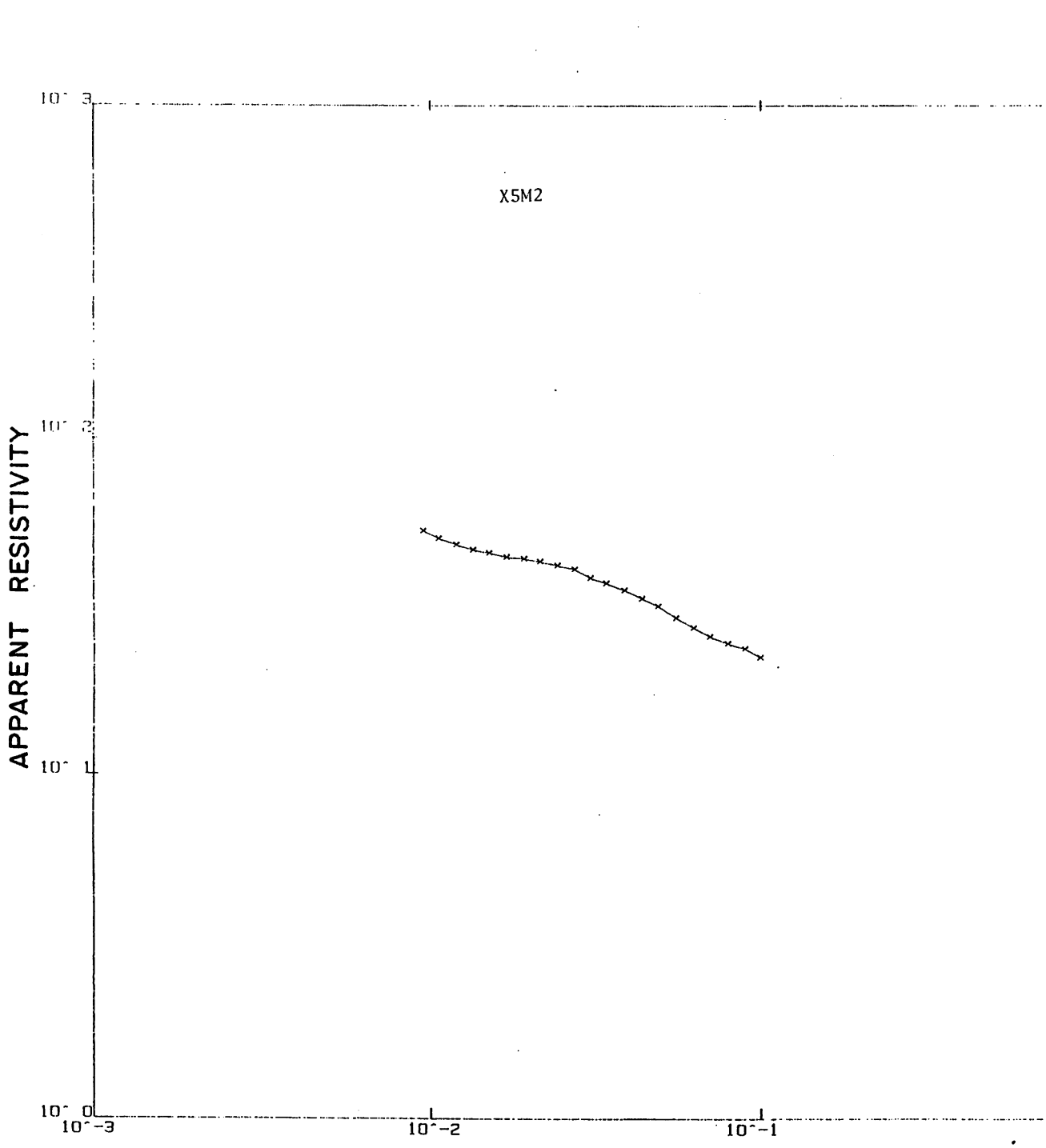


Figure 17

THE EARTH TECHNOLOGY CORP.

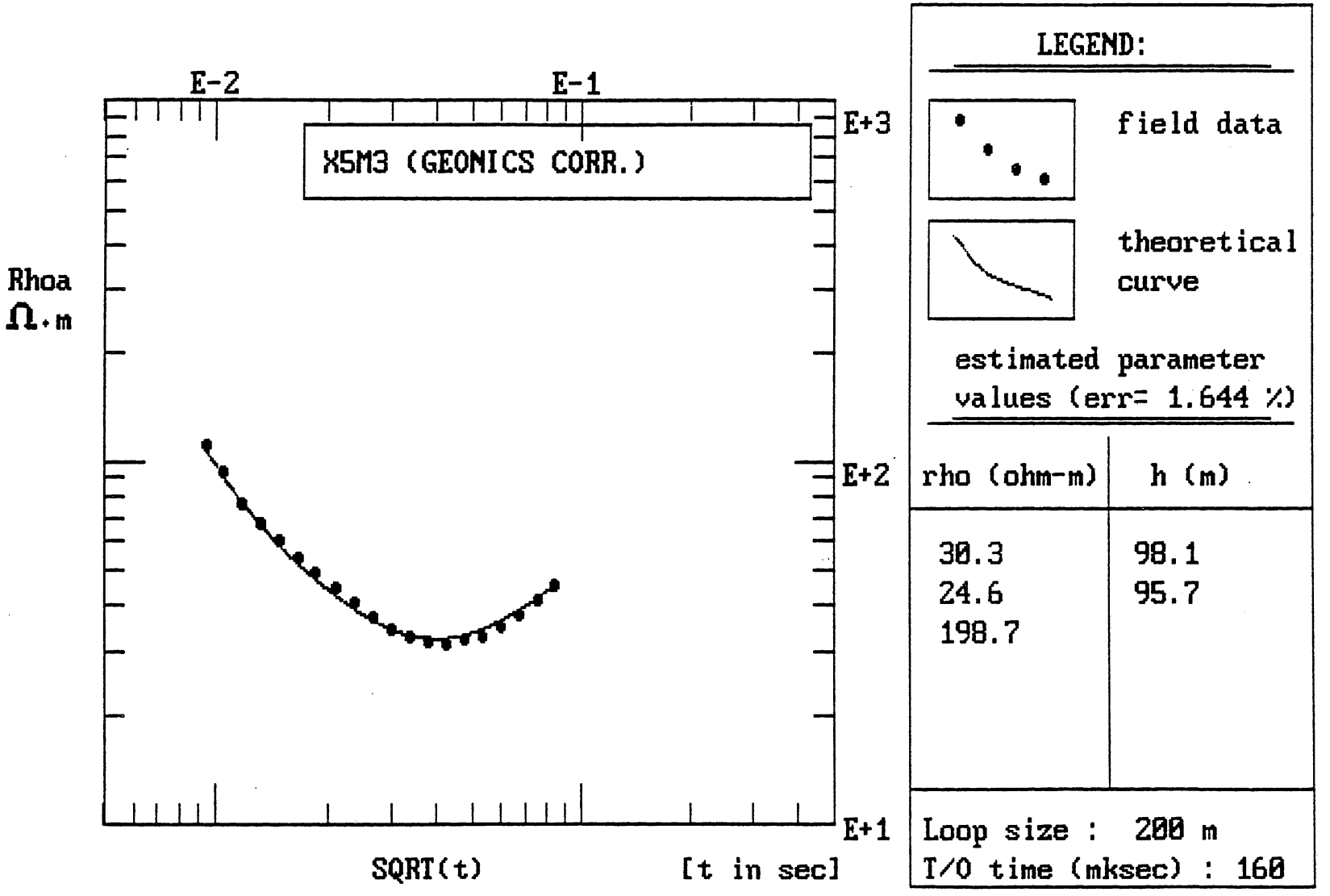

E-2
E-1
E+3
E+2
E+1
X5M3 (GEONICS CORR.)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.644 %)
rho (ohm-m)
h (m)
30.3
24.6
198.7
98.1
95.7
Loop size : 200 m
T/O time (mksec) : 160

Figure 18

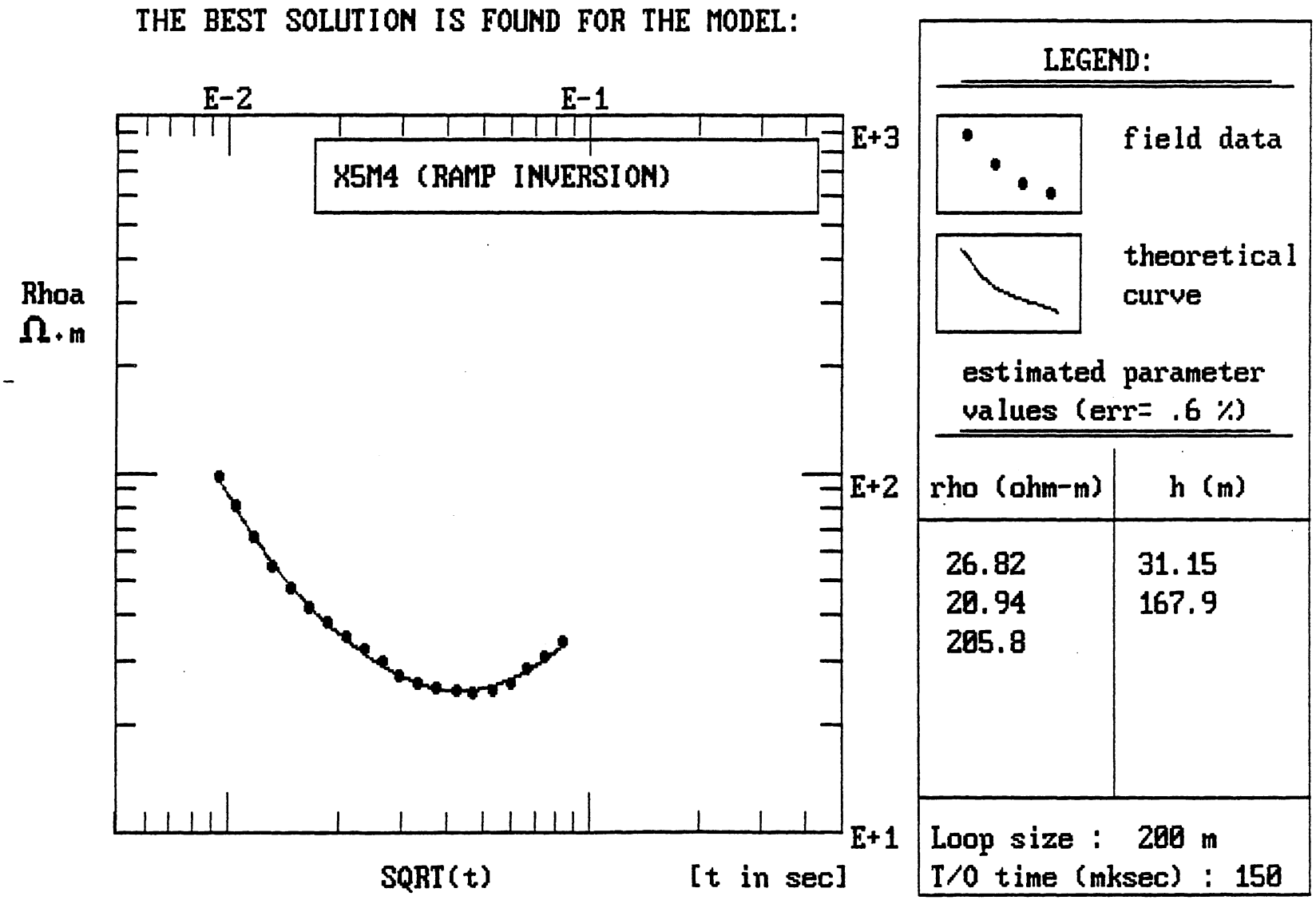

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+3
E+2
E+1
X5M4 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .6 %)
rho (ohm-m)
h (m)
26.82
20.94
205.8
31.15
167.9
Loop size : 200 m
T/O time (mksec) : 150

Figure 19

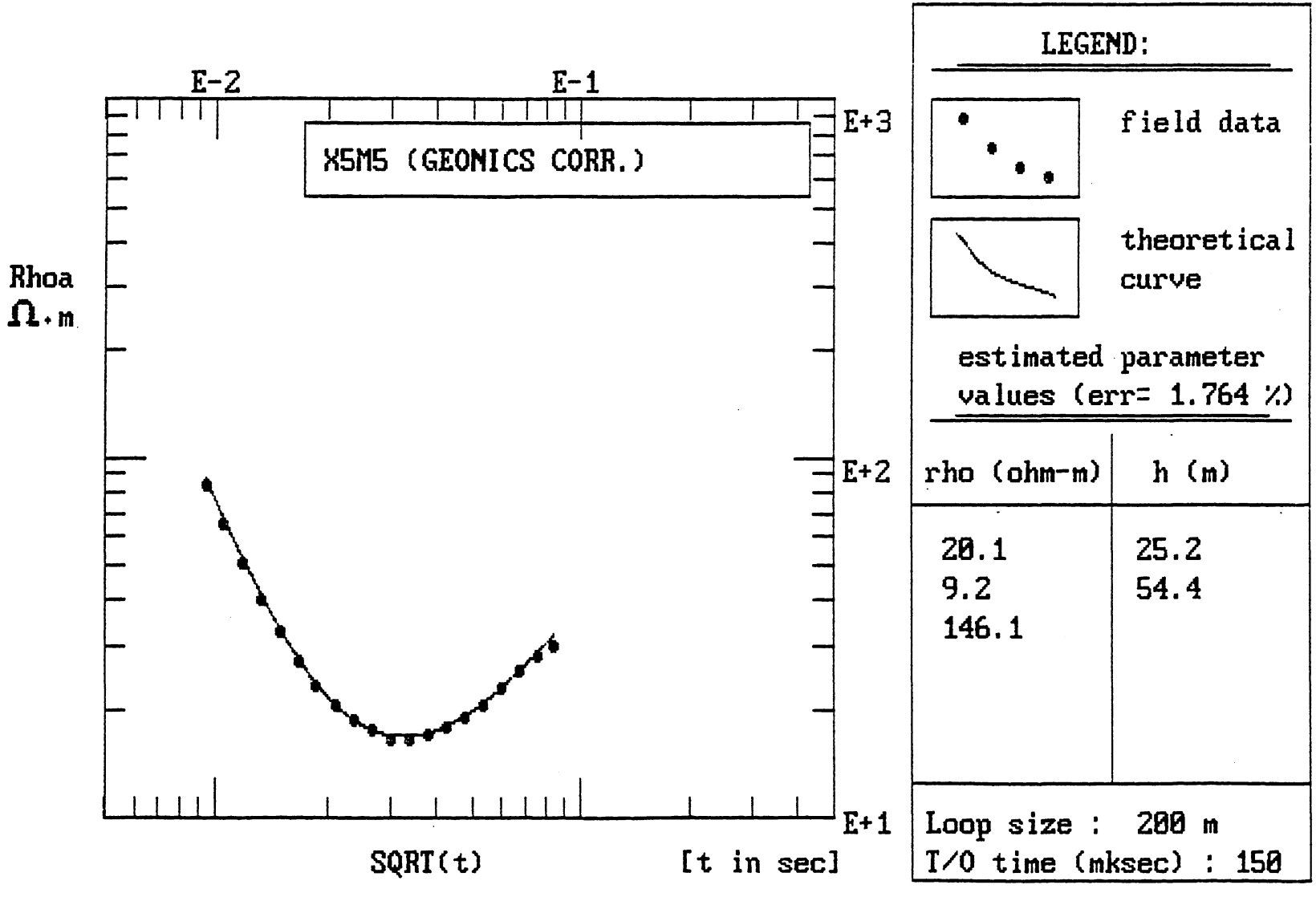

E-2
E-1
E+3
E+2
E+1
X5M5 (GEONICS CORR.)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.764 %)
rho (ohm-m)
h (m)
20.1
9.2
146.1
25.2
54.4
Loop size : 200 m
T/O time (mksec) : 150

Figure 20

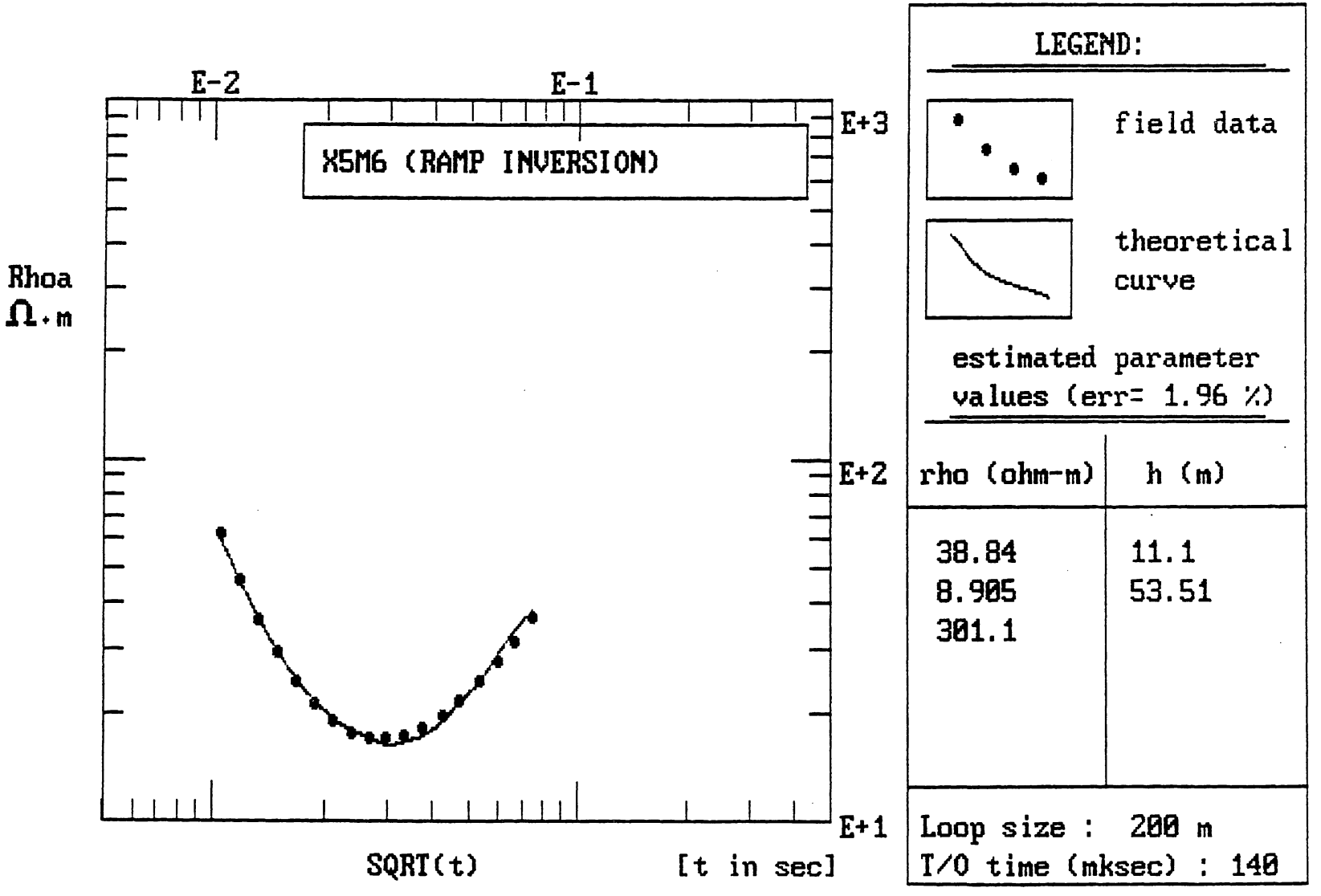

E-2
E-1
E+3
X5M6 (RAMP INVERSION)
Rhoa
Ω.m
E+2
E+1
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.96 %)
rho (ohm-m)
h (m)
38.84
8.905
301.1
11.1
53.51
Loop size : 200 m
T/O time (mksec) : 140

Figure 21

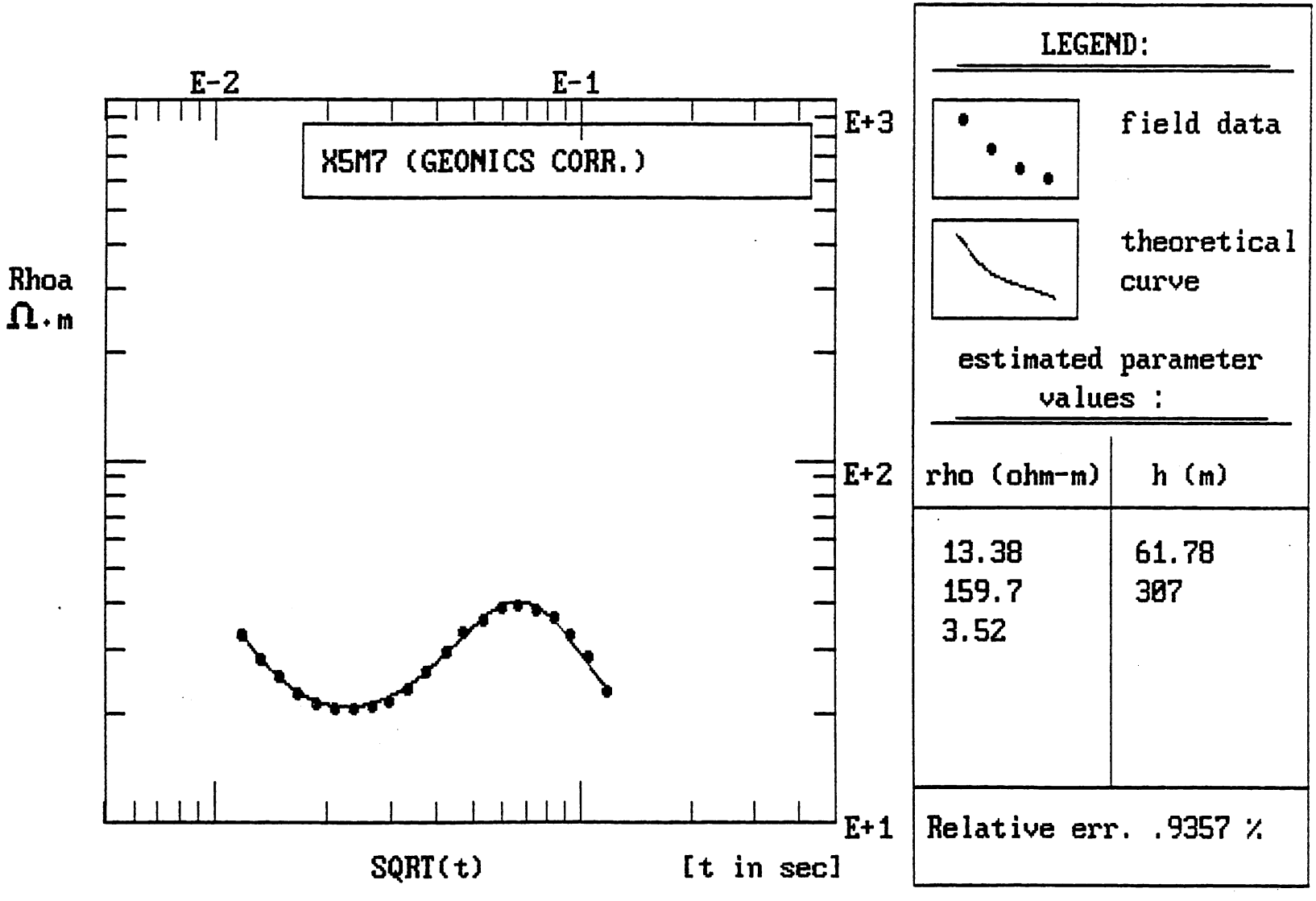

E-2
E-1
E+3
E+2
E+1
X5M7 (GEONICS CORR.)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values :
rho (ohm-m)
h (m)
13.38
159.7
3.52
61.78
307
Relative err. .9357 %

Figure 22

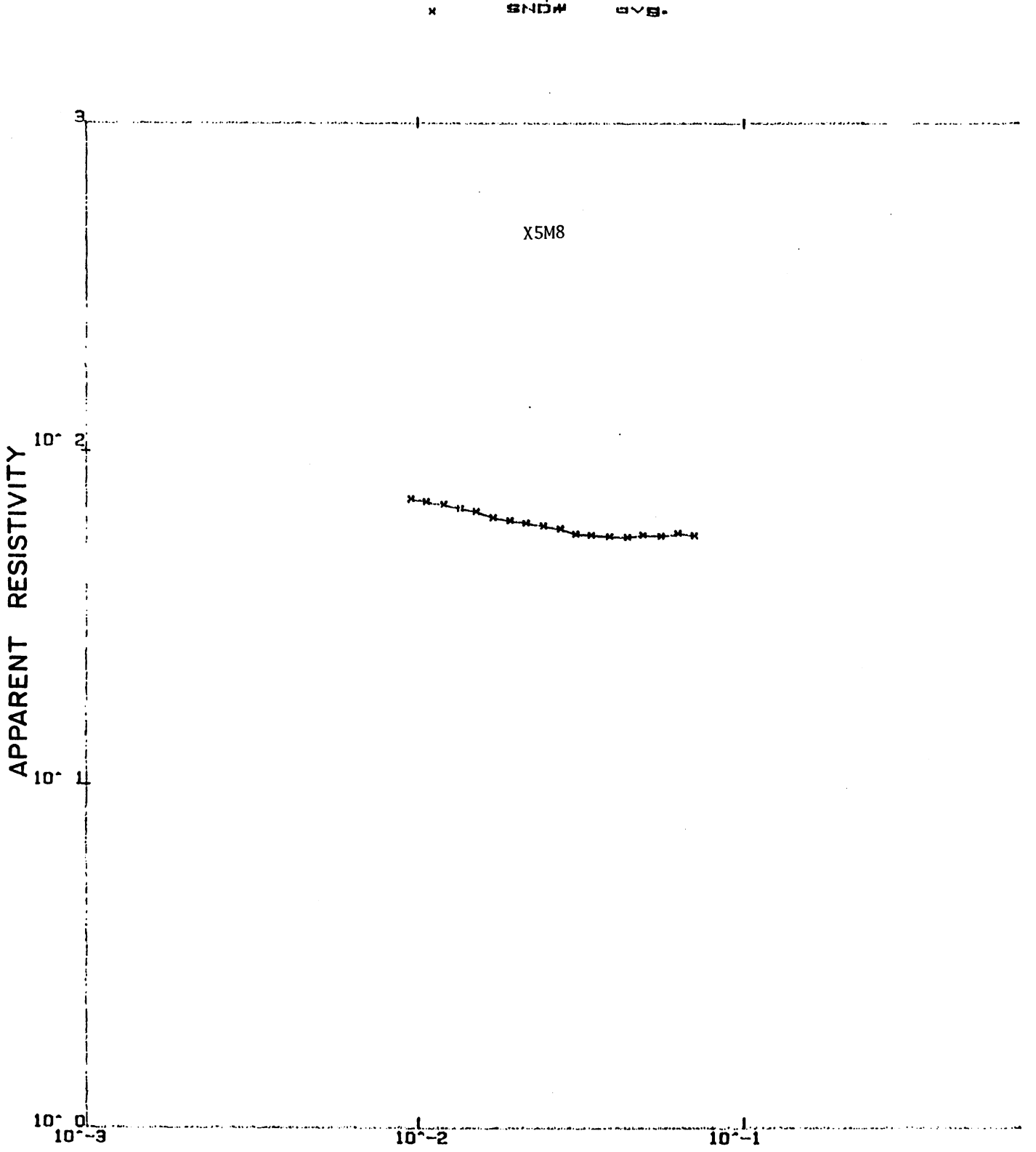

LINE: 010N
STA. : 010E
SNDG AVG.
X5M8
APPARENT RESISTIVITY
10^ 2
10^ 1
10^ 0
10^-3
10^-2
10^-1
TIME sec^1/2
THE EARTH TECHNOLOGY CORP.

Figure 23

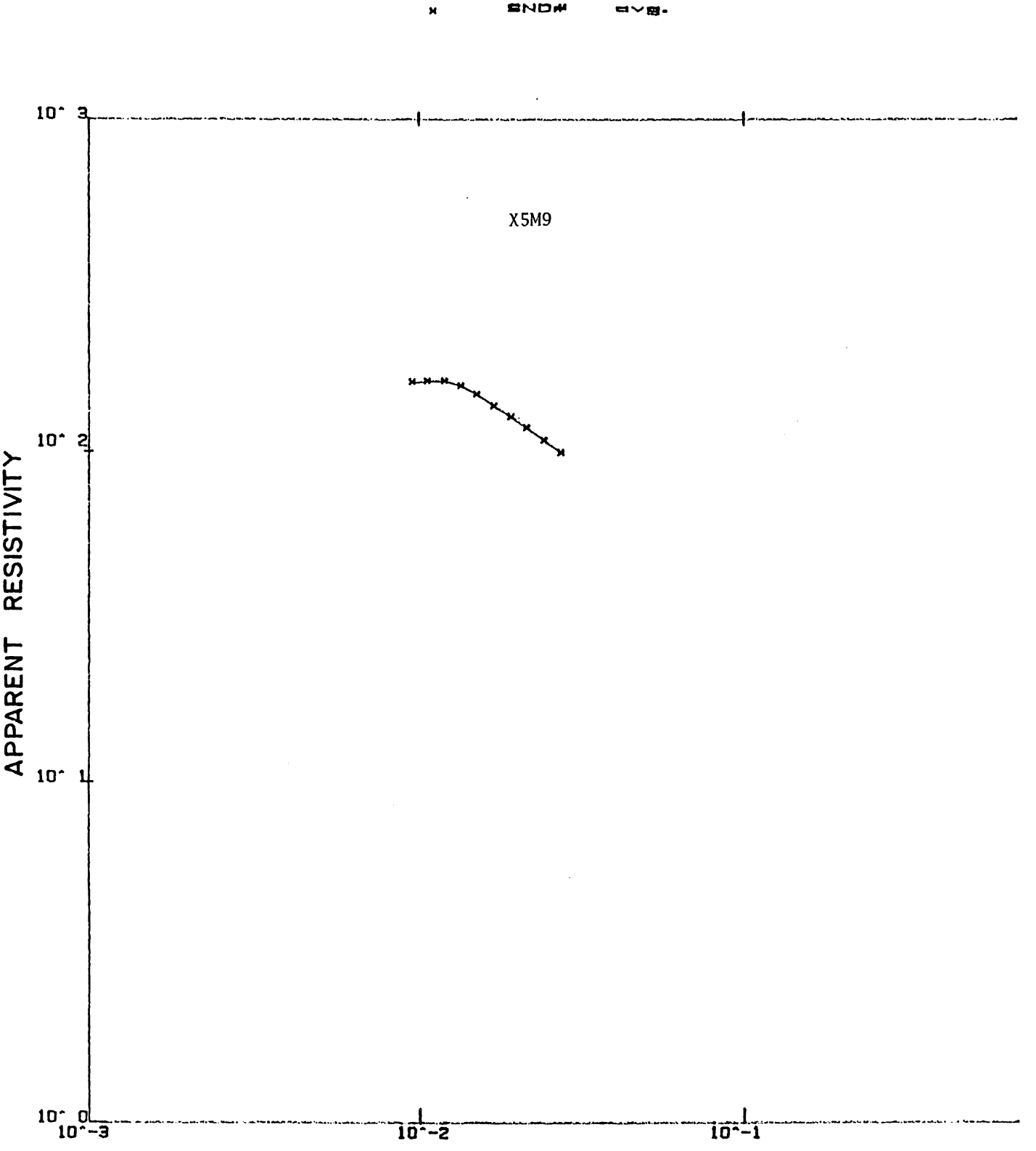
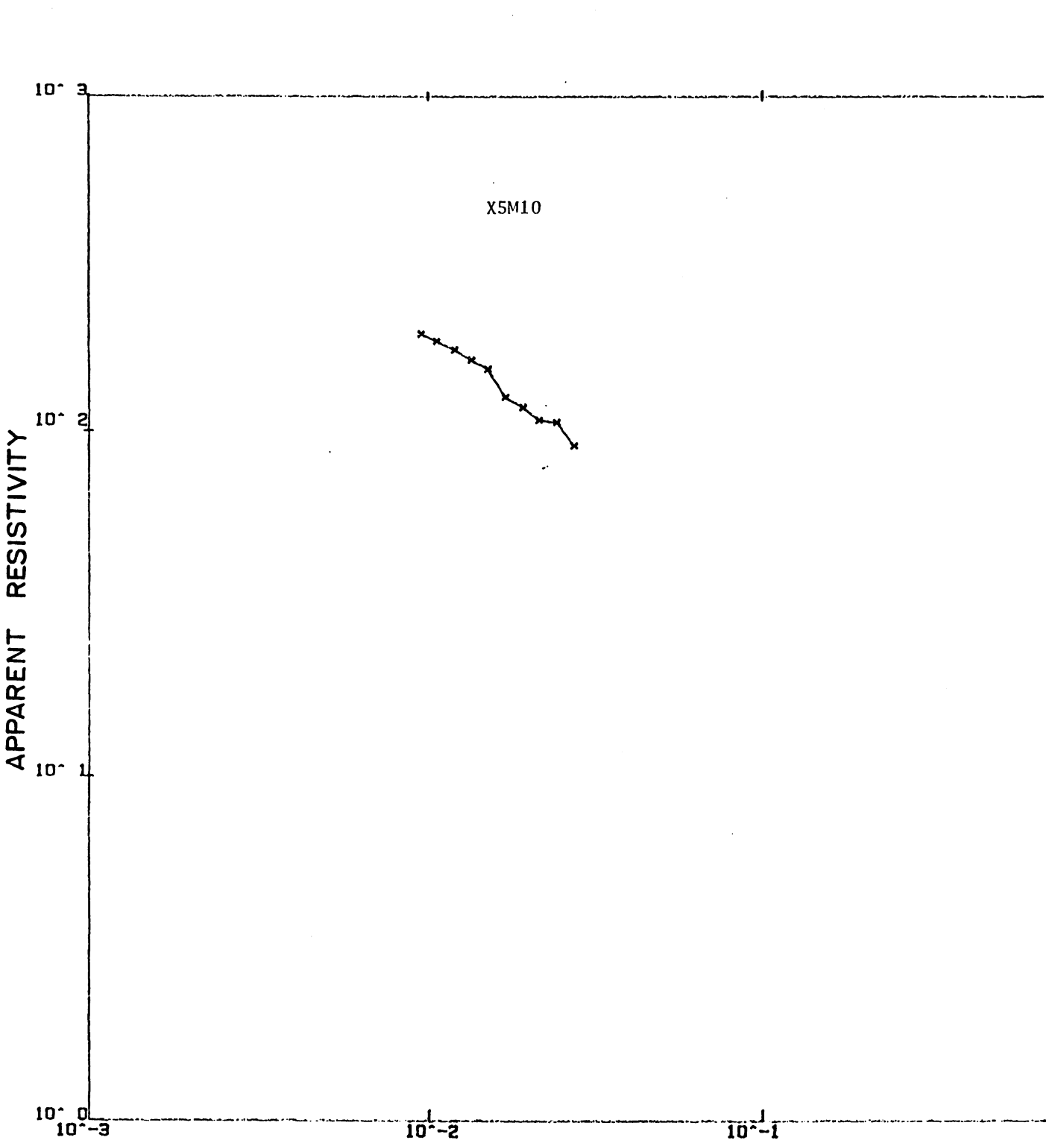

LINE: 011N
STA. : 011E
SND#
AVG.
X5M9
APPARENT RESISTIVITY
10^ 3
10^ 2
10^ 1
10^ 0
10^-3
10^-2
10^-1
TIME sec^1/2
THE EARTH TECHNOLOGY CORP.

Figure 24



THE EARTH TECHNOLOGY CORP.

Figure 25

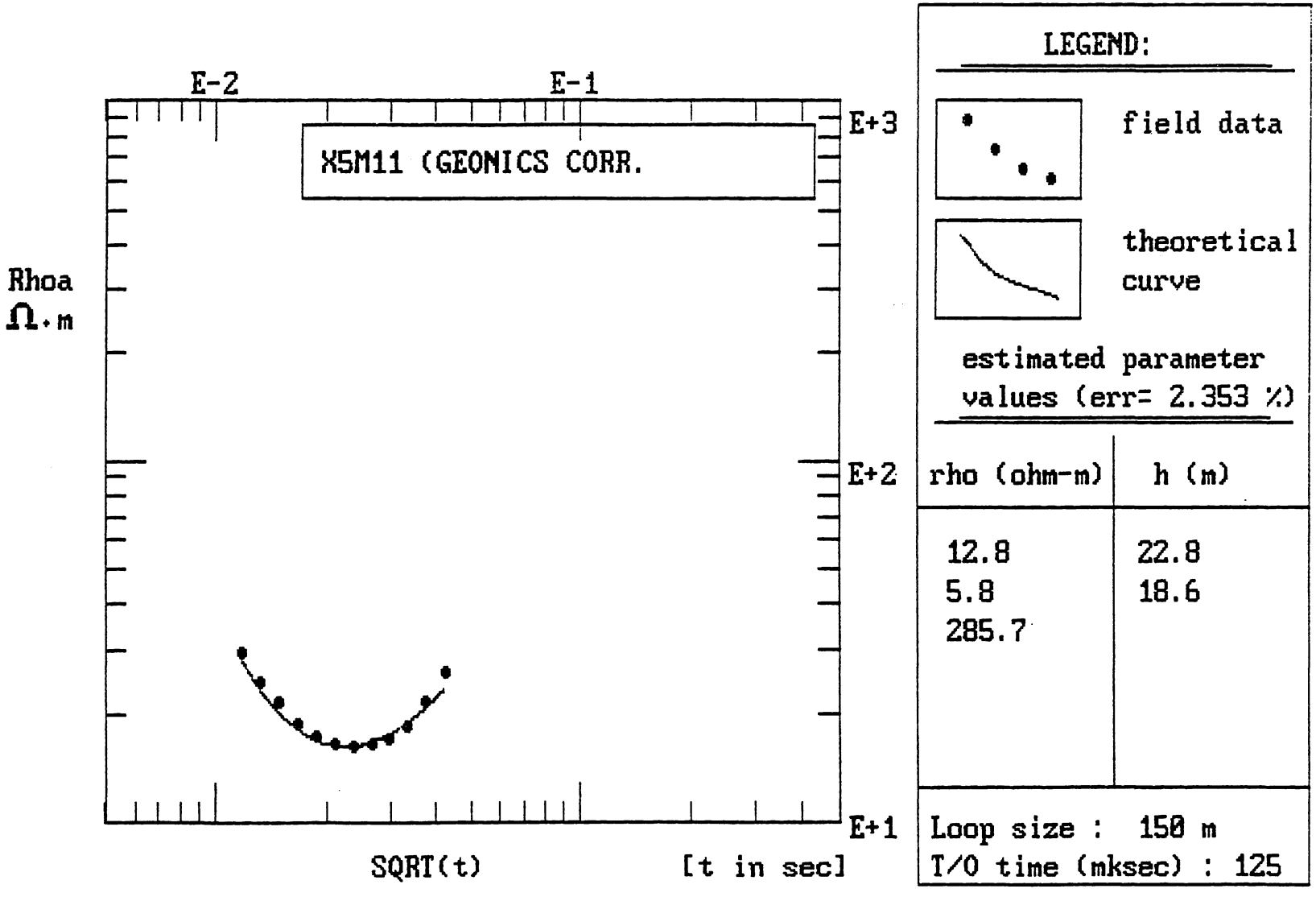

E-2
E-1
E+3
E+2
E+1
X5M11 (GEONICS CORR.
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical
curve
estimated parameter
values (err= 2.353 %)
rho (ohm-m)
h (m)
12.8
5.8
285.7
22.8
18.6
Loop size : 150 m
T/O time (mksec) : 125

Figure 26

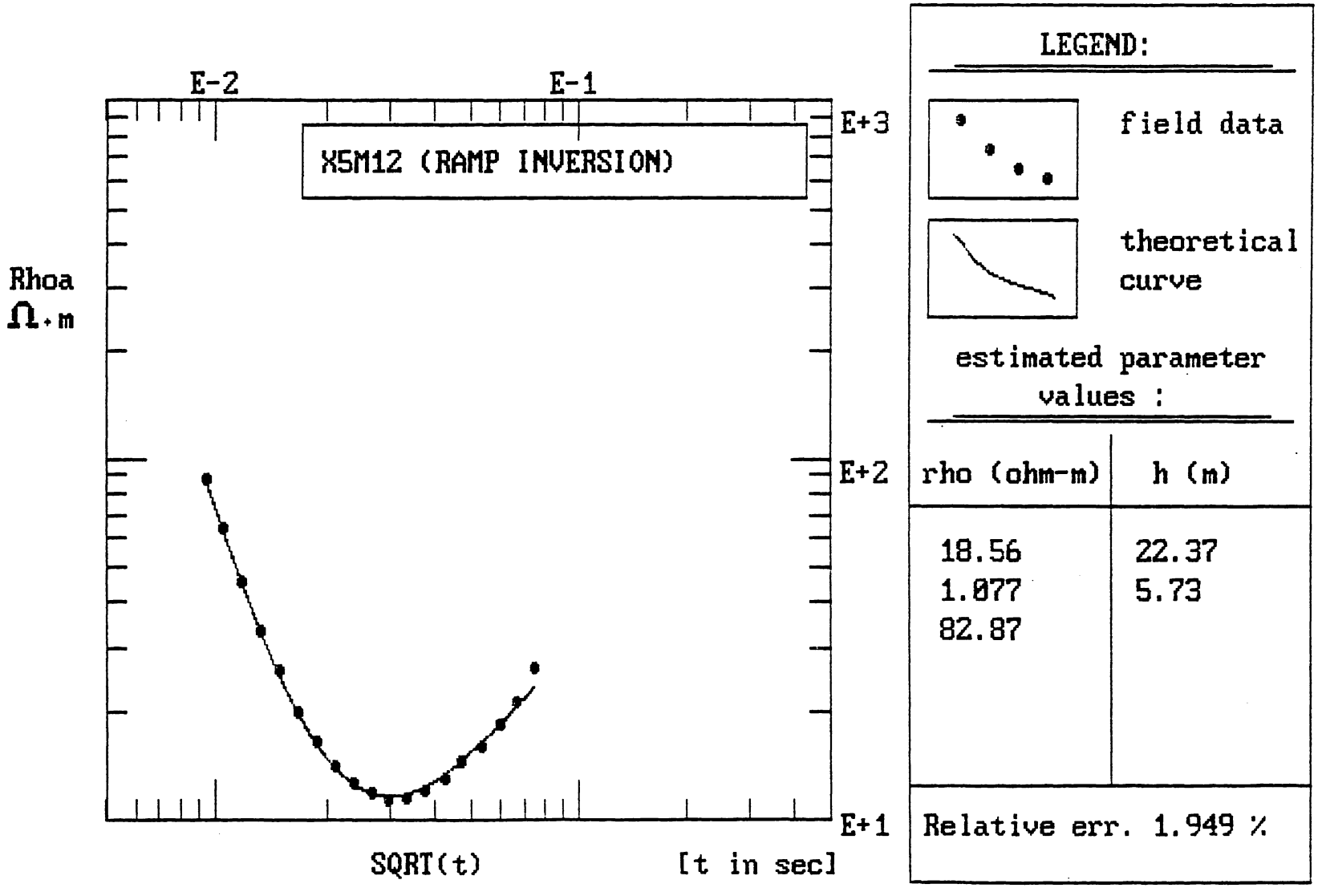

E-2
E-1
E+3
E+2
E+1
X5M12 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values :
rho (ohm-m)
h (m)
18.56
1.077
82.87
22.37
5.73
Relative err. 1.949 %

Figure 27

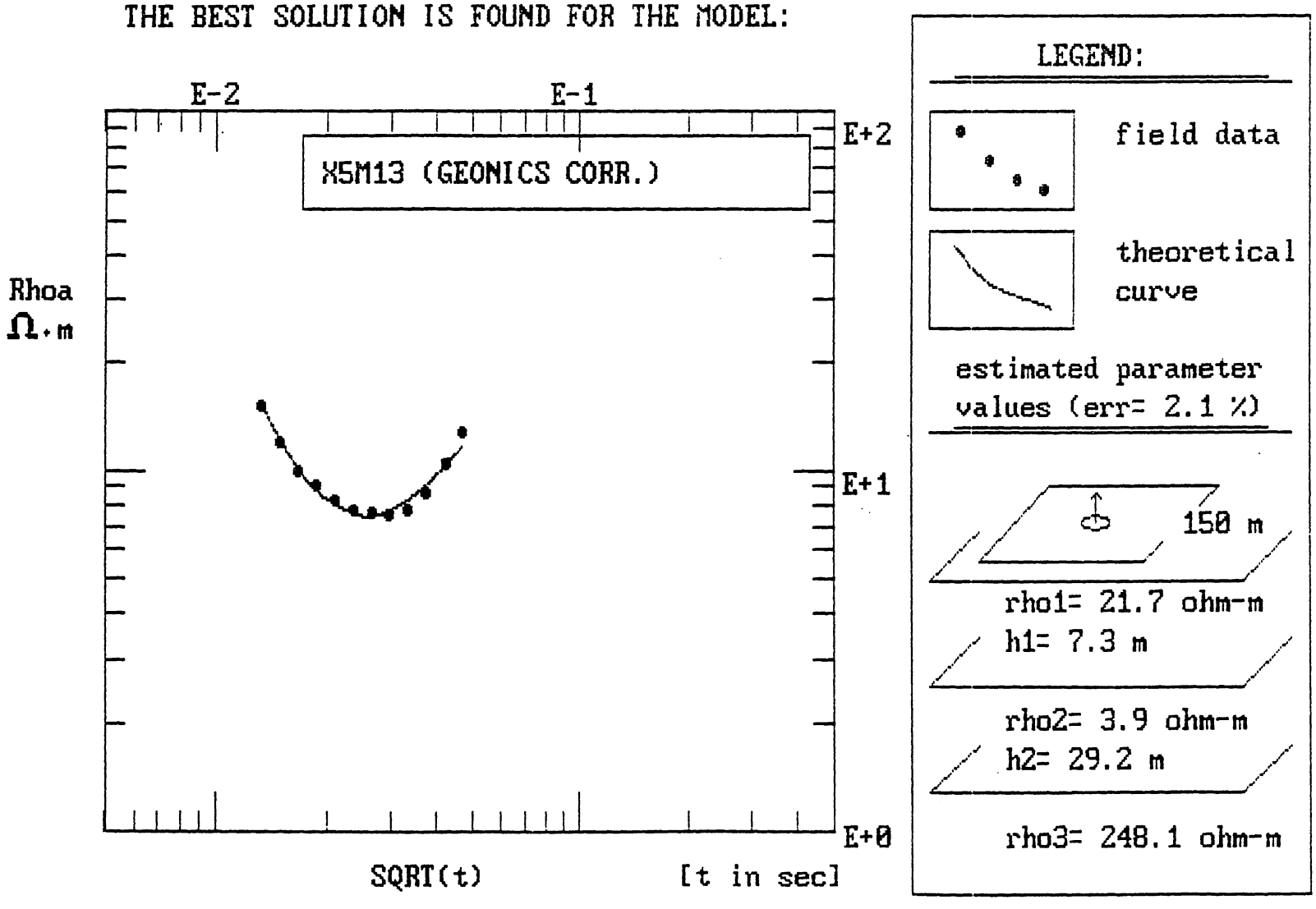
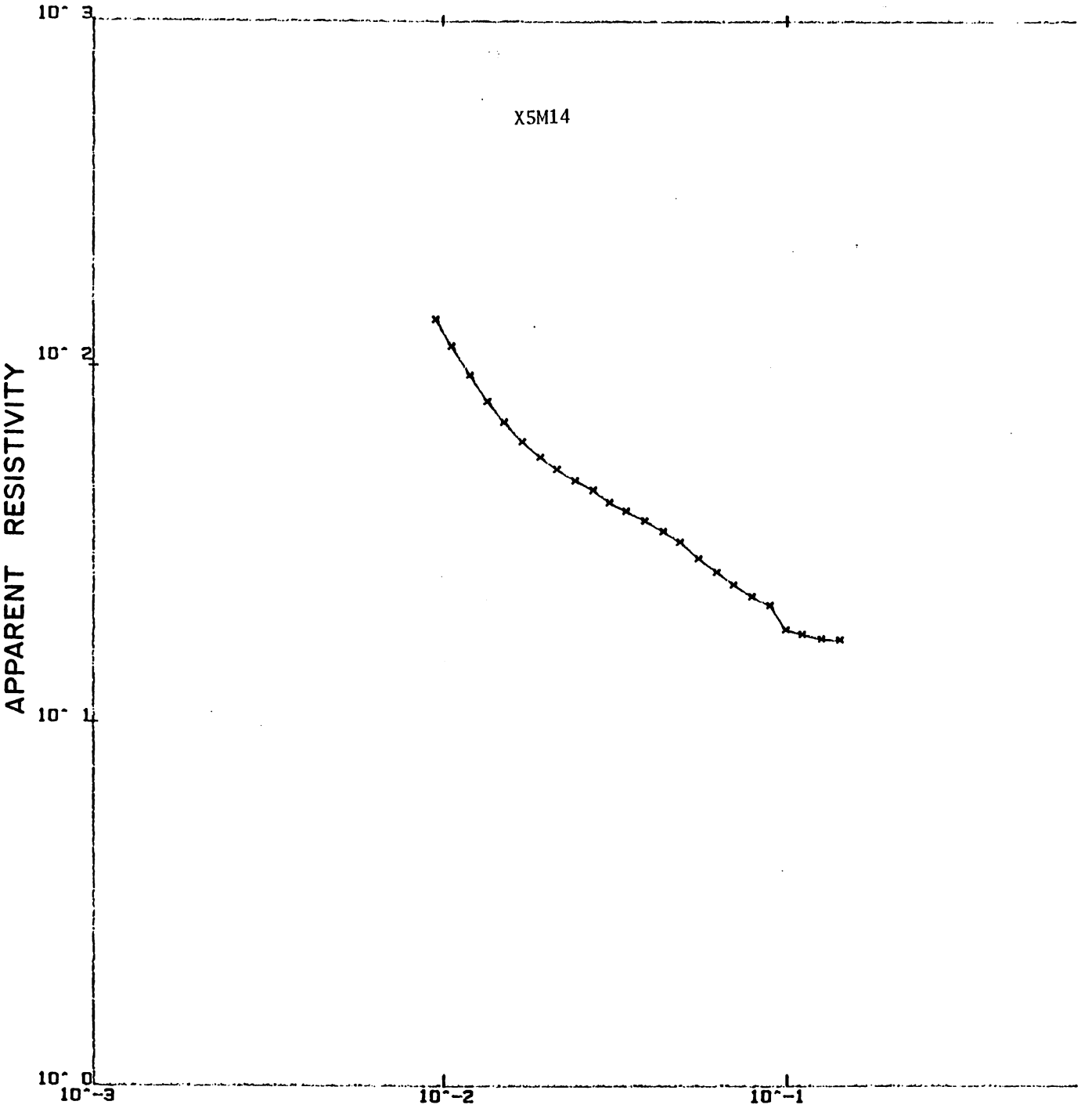

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+2
E+1
E+0
X5M13 (GEONICS CORR.)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 2.1 %)
150 m
rho1= 21.7 ohm-m
h1= 7.3 m
rho2= 3.9 ohm-m
h2= 29.2 m
rho3= 248.1 ohm-m

Figure 28

LINE: 016N STA.: 016E

x SNDM avg.



THE EARTH TECHNOLOGY CORP.

Figure 29

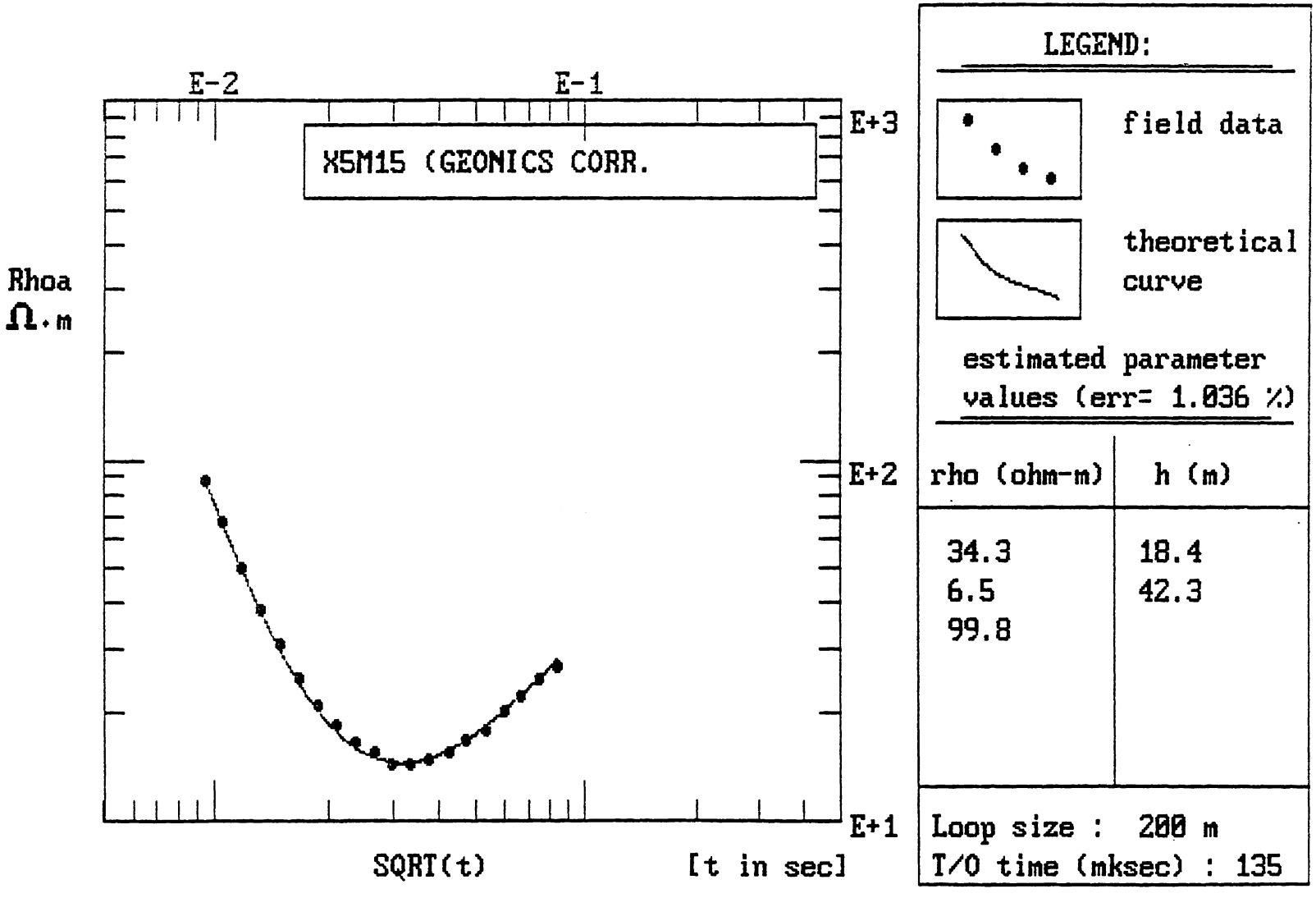

E-2
E-1
X5M15 (GEONICS CORR.
E+3
E+2
E+1
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.036 %)
rho (ohm-m)
h (m)
34.3
6.5
99.8
18.4
42.3
Loop size : 200 m
T/O time (mksec) : 135

Figure 30

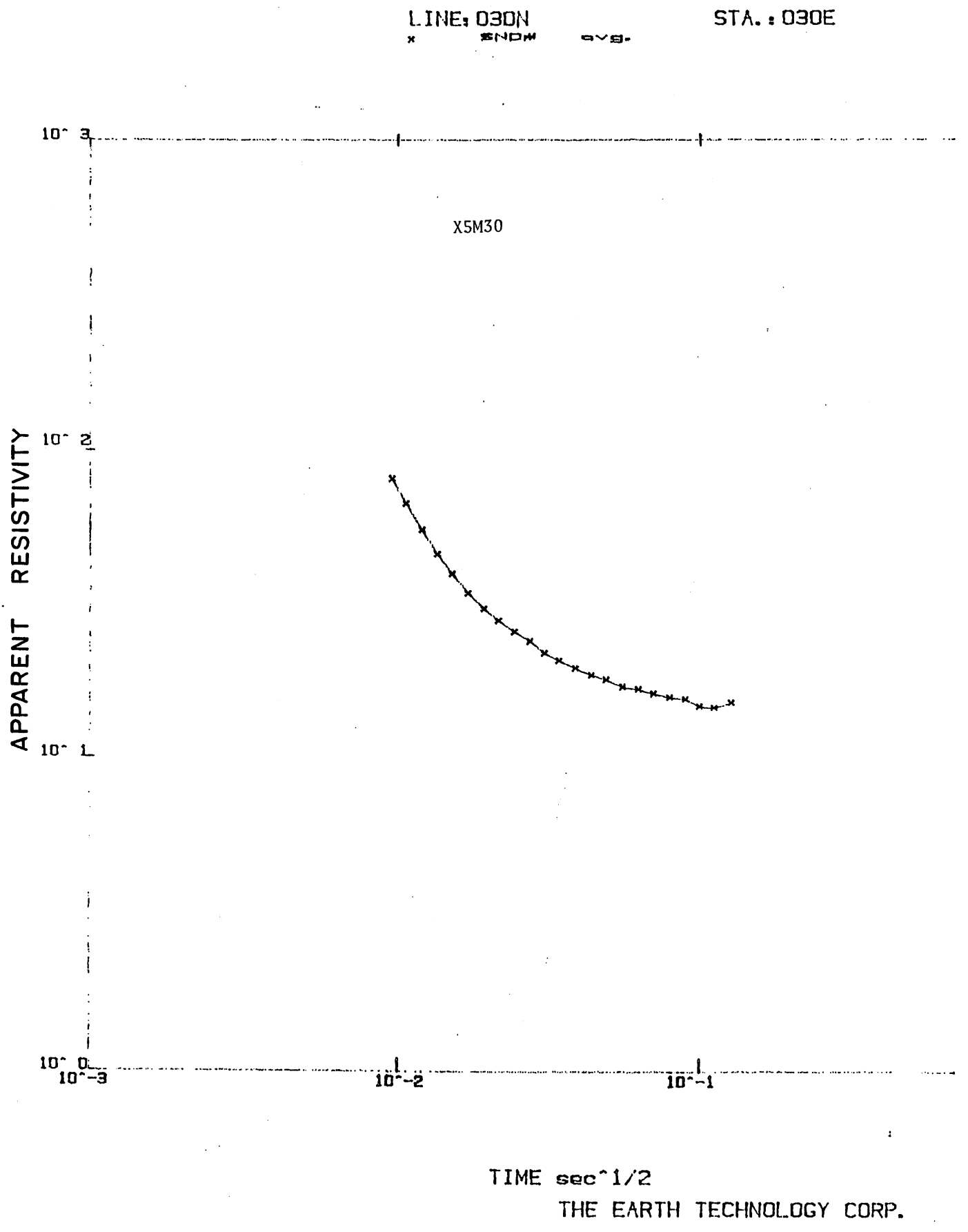

LINE: 030N
STA. : 030E
X5M30
APPARENT RESISTIVITY
10^ 3
10^ 2
10^ 1
10^ 0
10^-3
10^-2
10^-1
TIME sec^1/2
THE EARTH TECHNOLOGY CORP.

Figure 31

c) Stations M8 thru M1Ø were placed directly on exposed quartz-syenite in the Gavnunim intrusion (Fig. 15). Owing to the limitations imposed by the topography and the size of the outcrops, the transmitter loops were small, resulting in limited penetration depth. In addition, stations M9 and M1Ø were located close to the local power line which further reduced the penetration depth. However, to the limited depths obtained (<1ØØ meters) no significant changes in resisitvity were found.

2.3 Conclusions

The results of the TDEM survey prove the existence of a continuous large quartz-syenite intrusion in the southern Ramon, from Shen Ramon to Gavnunim. This intrusion dips steeply to the north, a feature which is reflected in the Ramon-1 borehole. The known areal extension of the magmatic intrusion has been enlarged by a factor of 5. A structural map on top of the magmatic body can also be presented (Fig. 15).

3. EASTERN MAKHTESH RAMON AREA: DETECTION OF A POSSIBLE INTRUSION BELOW THE RED VALLEY CAULDRON

The Red Valley is an oval N6ØW trending depression, 1km long and Ø.5km wide, bounded by 2 to 5m high undulating fault escarpments (Fig. 57). The inner, downfaulted block is composed mainly of ferruginous clay and silty clay of the Lower Member of the Hathira Formation overlying sandstone of the Inmar Formation.

The uplifted area surrounding the Red Valley consists of sandstone of the Inmar Formation. A small basaltic plug is exposed along the western slopes of Nahal Harerim, 400m southeast of the Red Valley and a large volcanic center, named Giv'at Ga'ash (Mazor, 1978), is located 8km northwest of the Red Valley.

The Red Valley was first described as a graben (Mazor, 1978). Baer (1987) suggested that the structure is a collapse caldera (cauldron), formed by the deflation of a previously active magma chamber below, which caused subsidence along irregular ring fractures in the overlying strata.

The attribution of a subvolcanic origin to this structure could have been verified if a volcanic intrusion were found only below the Red Valley and not below its close vicinity. A preliminary TDEM survey was therefore performed in the Red Valley. Only one 200 by 200 meter loop was deployed in the southeastern part of the depression (Fig. 57).

The apparent resistivities measured in this station (Fig. 58) show a clear right-hand ascending branch, proving the existence of a high resisitivity layer at a depth of about 100m below the surface. Since there is no indication of a thick highly resistant layer within the known stratigraphic sequence, it might indicate the existence of volcanic rocks of high resistivity below the Red Valley. However, it should be emphasized that this conclusion is based solely on one measurement inside the valley and must be confirmed by more measurements taken inside and outside the Red Valley.

EASTERN MAKHTESH RAMON AREA

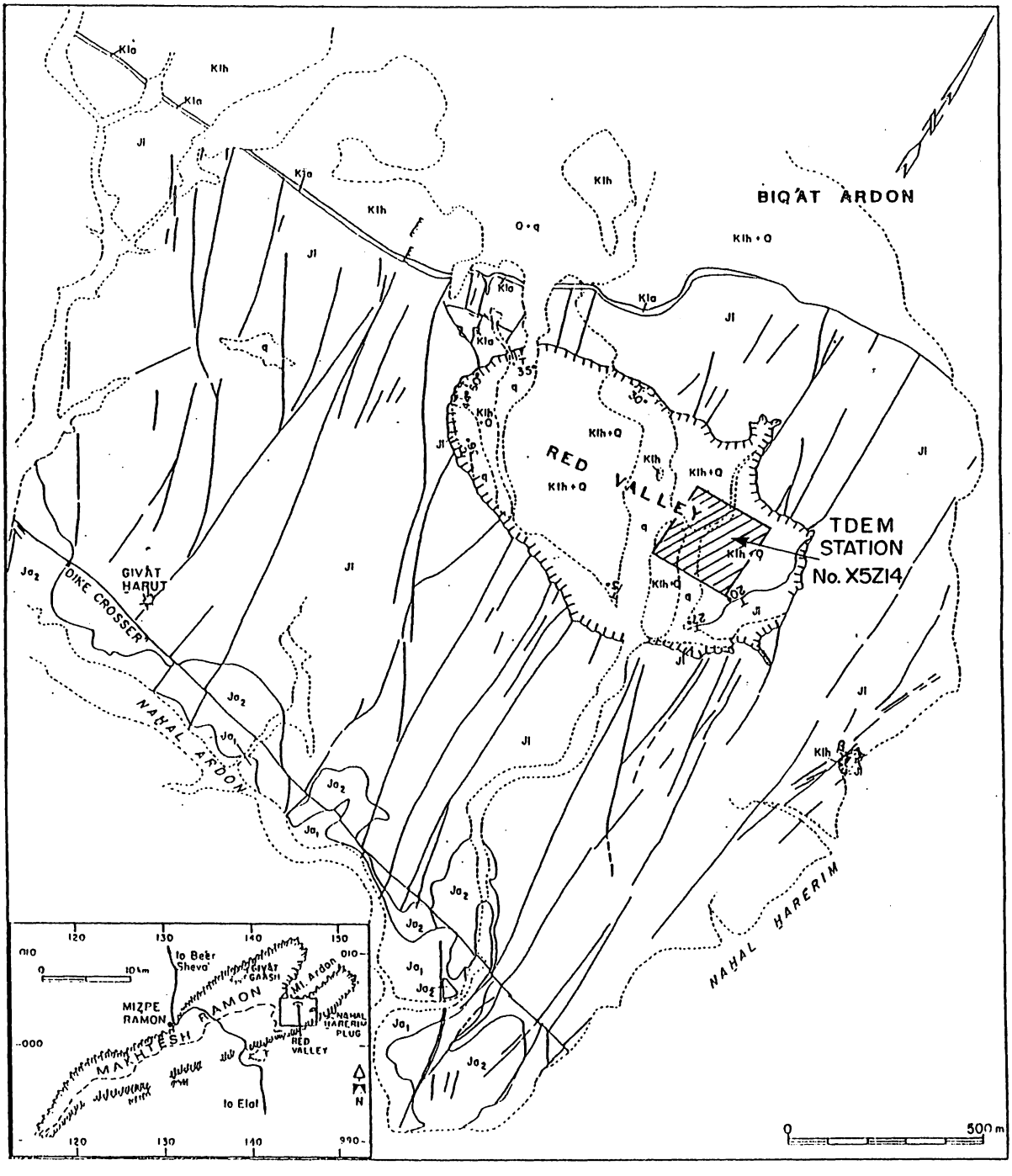


Figure 57. Photogeological map of the Red Valley and its vicinity (from Baer, 1987). Heavy lines represent dikes, thin lines represent lithological contacts. Stratigraphic symbols: Ja - Ardon Formation, Jurassic; Ji - Inmar Formation, Jurassic; Kla - Arod Formation, Lower Cretaceous; Klh - Hathira Formation, Lower Cretaceous; Q - Elevated terraces; q - alluvium. Hatched area marks the location of TDEM station No. X5 Z 14.

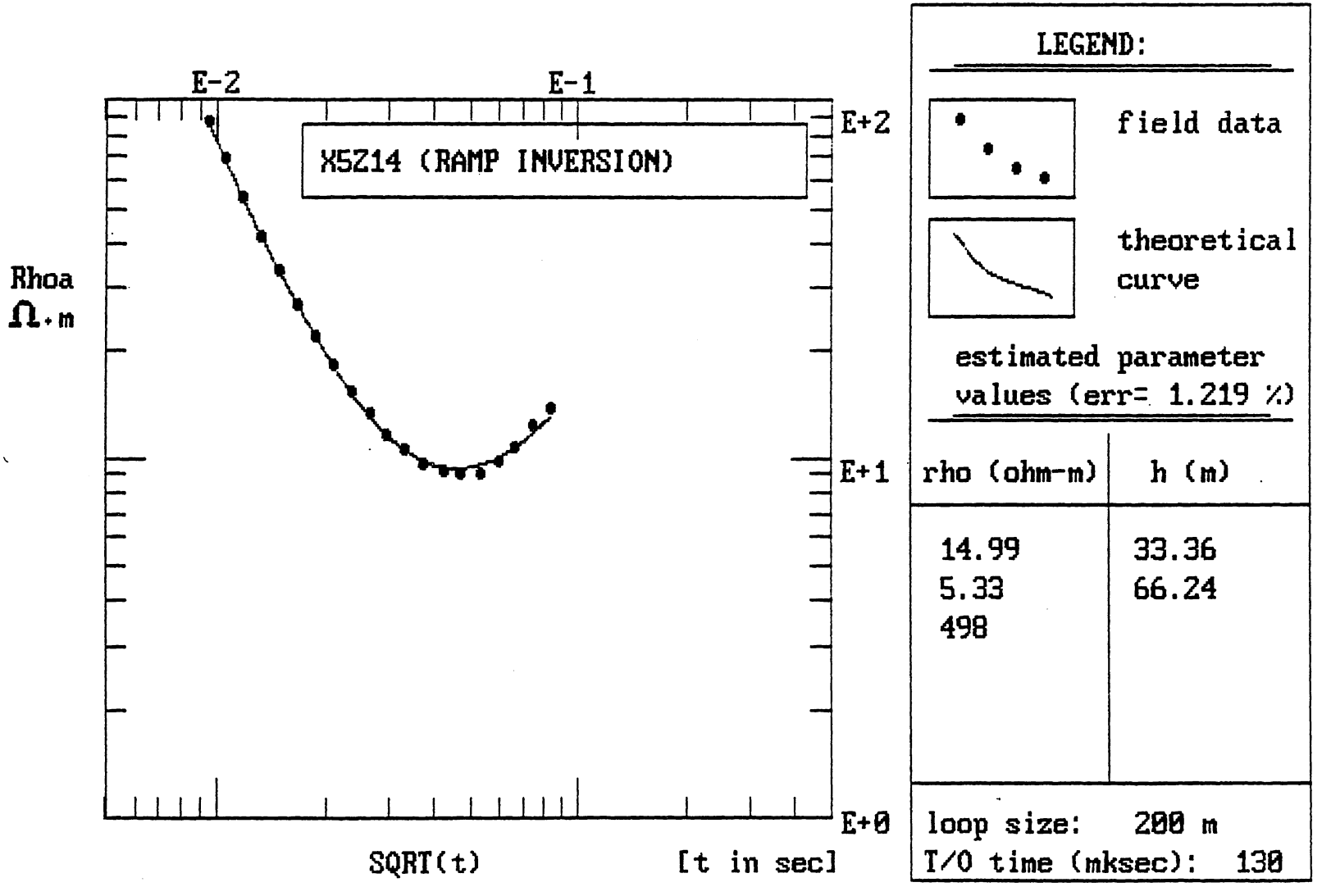

E-2
E-1
E+2
X5Z14 (RAMP INVERSION)
Rhoa
Ω.m
E+1
E+0
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.219 %)
rho (ohm-m)
h (m)
14.99
5.33
498
33.36
66.24
loop size: 200 m
T/O time (mksec): 130

Figure 58

4. RAMAT KORAZIM AND CENTRAL GOLAN AREA: DETECTION OF LITHOLOGICAL BOUNDARIES WITHIN THE NEOGENE SEDIMENTARY AND BASALTIC SEQUENCE

4.1 Background

Estimation of the thickness and nature of lithological alternations within the widespread Neogene sedimentary and volcanic sequence of northeastern Israel and the Golan Heights is of great practical importance since almost no direct observations are available in the area. The existing data for the Golan Heights are only from the outcrops overlying the carbonate Avedat Group close to the Jordan river in the Hula Valley and along the eastern shores of Lake Kinneret (Fig. 32). The Rosh Pina block is the only area where deep drillholes penetrated the carbonate sequence of the Judea Group (Fig. 32a). Three of the ten TDEM soundings were, therefore, carried out in close proximity to the boreholes (KN-1, KN-3 and RP-1, Fig. 33) to provide geological control of the interpretation of the TDEM data. A further survey of five soundings was carried out along the profile between Kfar Hanassi and Kazrin. The last two TDEM stations were established next to the boreholes of Alonei Habashan (AB-1 and AB-2, Fig. 33) in order to check the feasibility of the TDEM method in the detection of the Scoria-basalt boundaries in the volcanics of the Bashan Group. The latter problem is important for groundwater exploration in the Golan (Shaliv, personal communication).

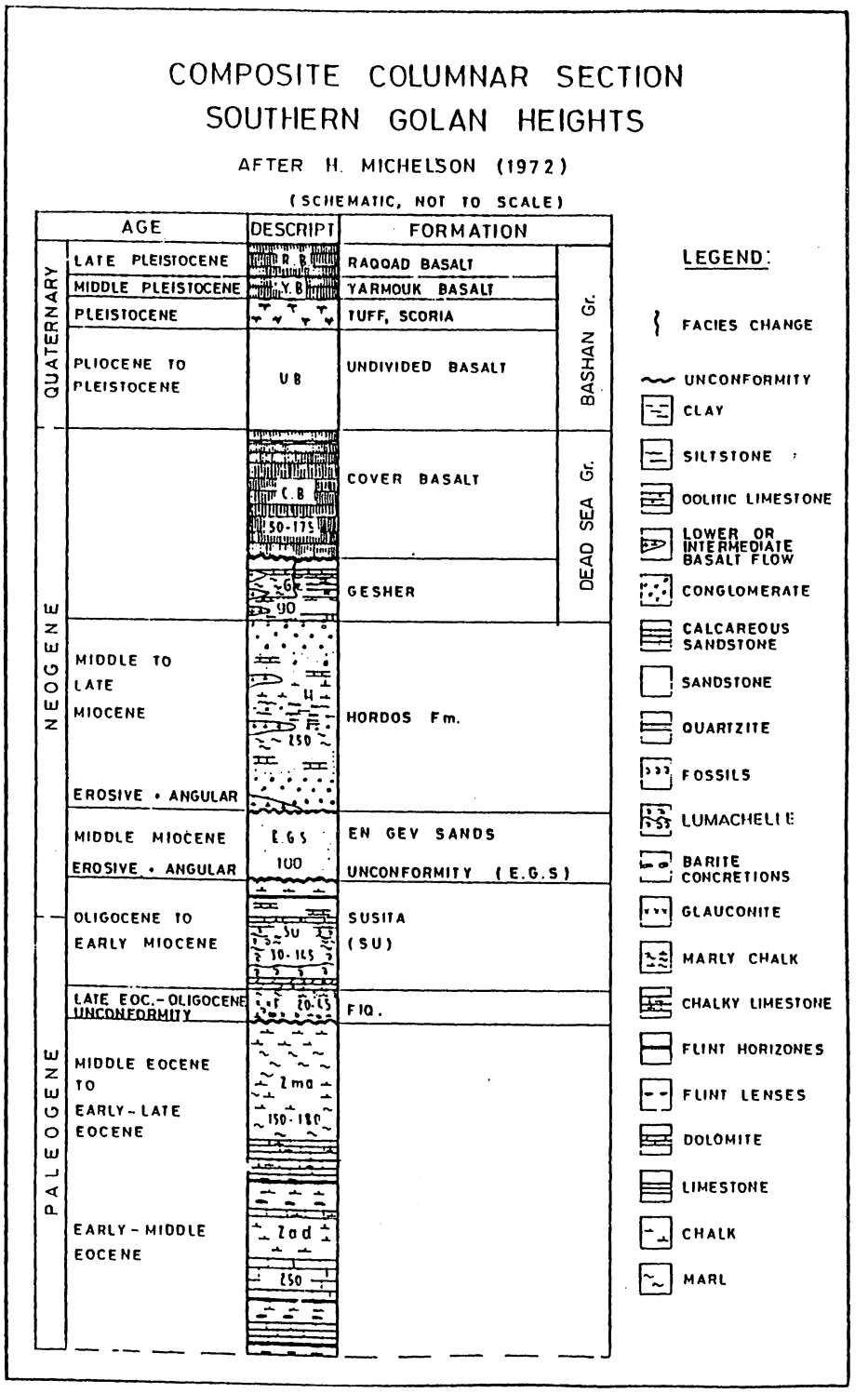

COMPOSITE COLUMNAR SECTION
SOUTHERN GOLAN HEIGHTS
AFTER H. MICHELSON (1972)
(SCHEMATIC, NOT TO SCALE)
AGE
DESCRIPT
FORMATION
QUATERNARY
NEOGENE
PALEOGENE
LATE PLEISTOCENE
MIDDLE PLEISTOCENE
PLEISTOCENE
PLIOCENE TO PLEISTOCENE
MIDDLE TO LATE MIOCENE
EROSIVE • ANGULAR
MIDDLE MIOCENE
EROSIVE • ANGULAR
OLIGOCENE TO EARLY MIOCENE
LATE EOC.-OLIGOCENE UNCONFORMITY
MIDDLE EOCENE TO EARLY-LATE EOCENE
EARLY-MIDDLE EOCENE
R.B
Y.B
UB
C.B
50-175
Gr
90
150
E.G.S
100
SU
10-145
10-15
Zma
150-180
Zad
150
RAQQAD BASALT
YARMOUK BASALT
TUFF, SCORIA
UNDIVIDED BASALT
COVER BASALT
GESHER
HORDOS Fm.
EN GEV SANDS
UNCONFORMITY (E.G.S)
SUSITA (SU)
FIQ.
BASHAN Gr.
DEAD SEA Gr.
LEGEND:
FACIES CHANGE
UNCONFORMITY
CLAY
SILTSTONE
OOLITIC LIMESTONE
LOWER OR INTERMEDIATE BASALT FLOW
CONGLOMERATE
CALCAREOUS SANDSTONE
SANDSTONE
QUARTZITE
FOSSILS
LUMACHELLE
BARITE CONCRETIONS
GLAUCONITE
MARLY CHALK
CHALKY LIMESTONE
FLINT HORIZONES
FLINT LENSES
DOLOMITE
LIMESTONE
CHALK
MARL

Figure 32

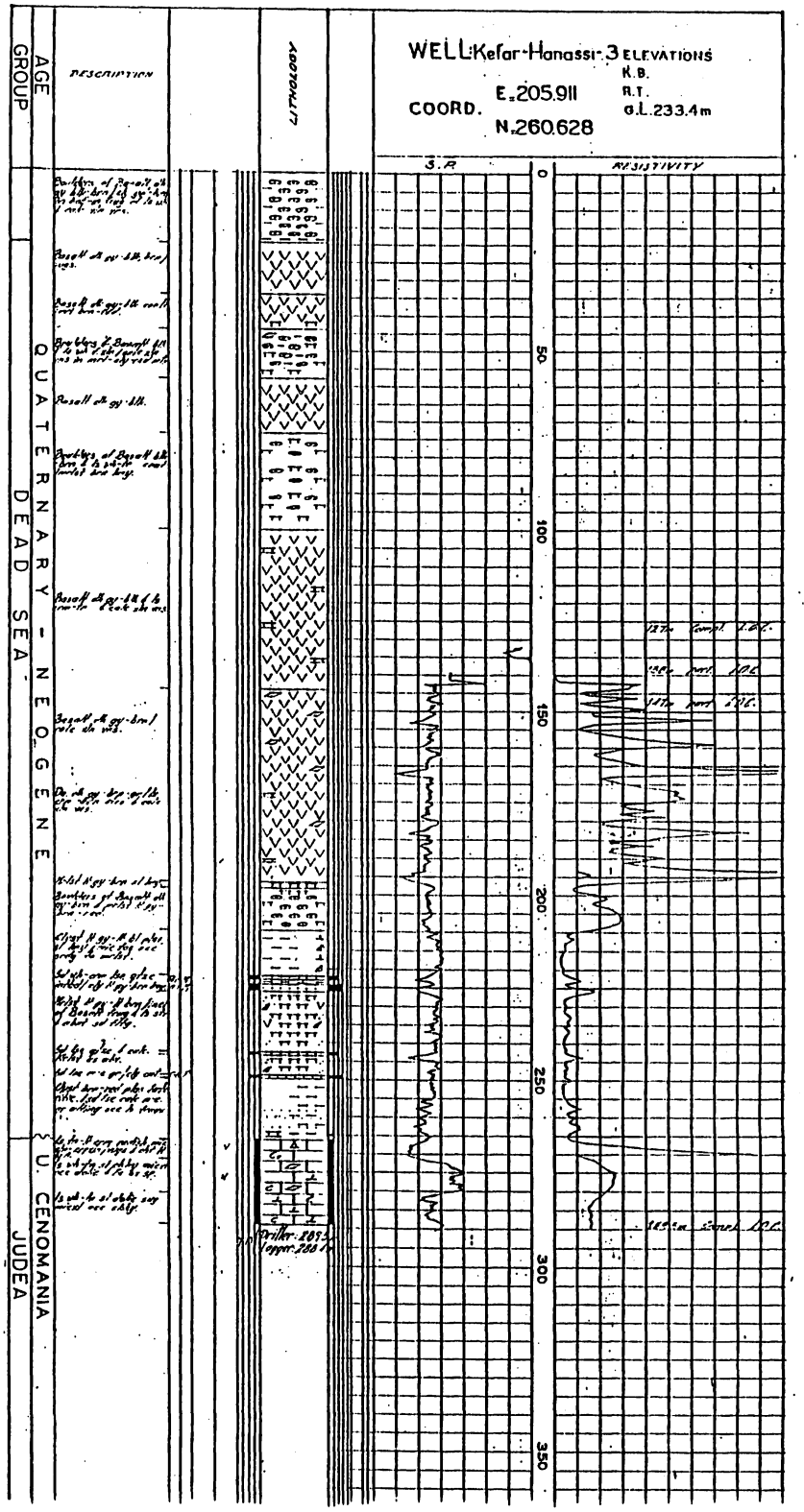
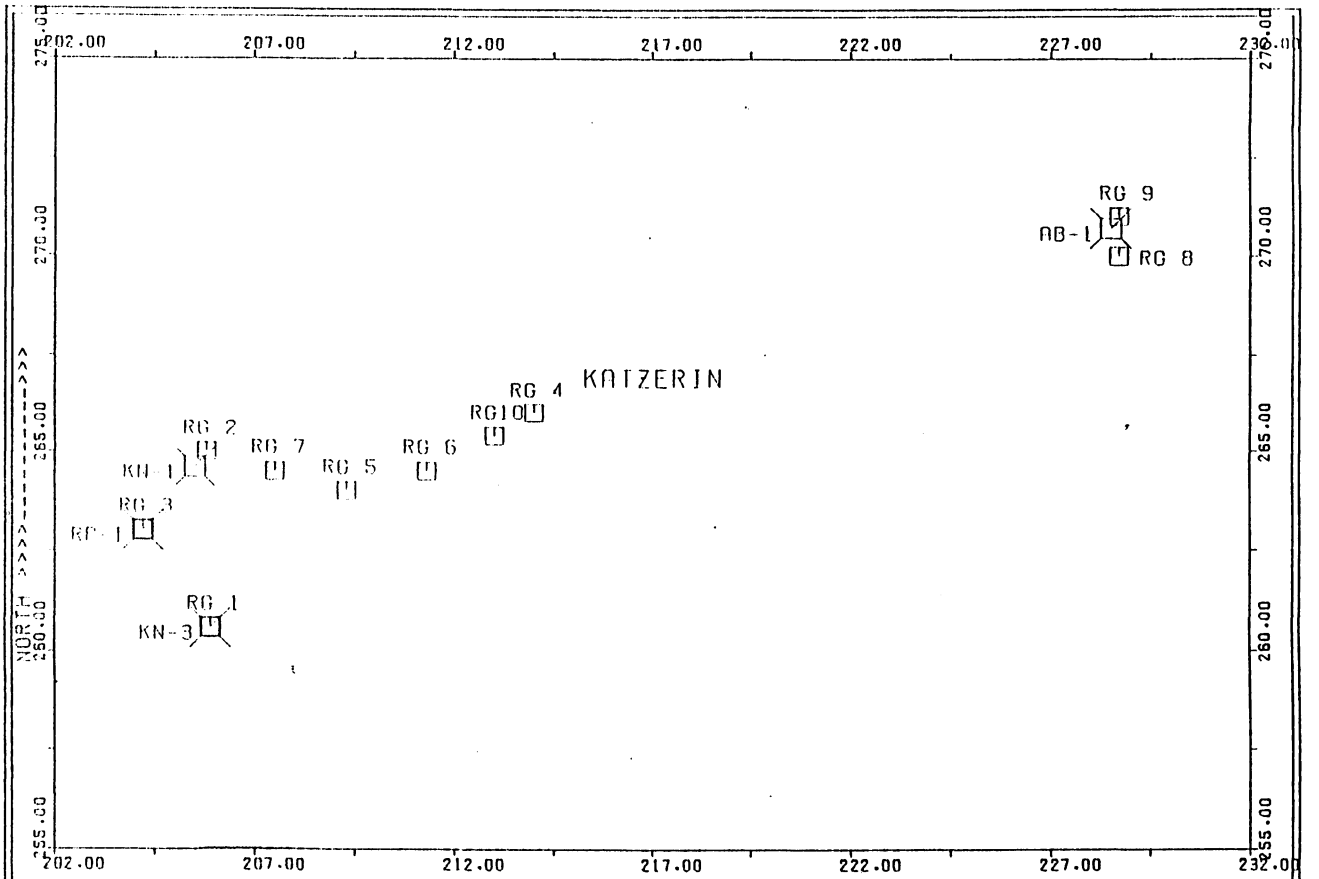

WELL: Kefar-Hanassi-3
ELEVATIONS
K.B.
R.T.
G.L. 233.4m
COORD.
E. 205.911
N. 260.628
AGE
GROUP
DESCRIPTION
LITHOLOGY
S.P.
RESISTIVITY
QUATERNARY - NEOGENE
DEAD SEA
U. CENOMANIA
JUDEA
0
50
100
150
200
250
300
350

Figure 32a


202.00
207.00
212.00
217.00
222.00
227.00
232.00
275.00
270.00
265.00
260.00
255.00
NORTH >>>>

4.2 Results

All ten apparent resistivity curves (Figs. 34 thru 43) have a descending branch in either the intermediate section (RG1 and RG2) or the right hand side (remainder of the curves). Qualitatively this means that all measurements indicate the presence of a relatively low resistivity layer; in most cases this layer is found in the base of the section, except for RG1 and RG2 stations where this layer is underlain by more resistive units. Results of a quantitative interpretation of the TDEM measurements, as well as appropriate borehole data and some other relevant parameters, are presented in Table 1.

A comparative analysis of the TDEM and borehole data for the first three stations shows that:

a) Resistivity contrasts between the Neogene basaltic sequence and the clastic rocks of the Hordos Fm. and the Dead Sea Group can be clearly detected by the TDEM measurements. A similar conclusion is true for the contact between the Upper Cretaceous limestone/dolomite unit (Judea Gr.) and the above mentioned clastic rocks.

b) There is no clear resistivity contrast between the basaltic and the Mesozoic-Cenozoic limestone/dolomite sequences.

c) In the case of a good resistivity contrast, the TDEM method permits determination of the depth to the interface within a margin of error of about 10-15%; an increasing resistivity contrast can be detected more accurately than a decreasing one.

T A B L E 1

Station	Borehole	Coordinates	TDEM 1st Break (m)	Appropriate Lithological Contact (m) (Borehole)	TDEM 2nd Break (m)	Appropriate Lithological Contact (m) (Borehole)	TDEM 3rd Break (m)	Appropriate Lithological Contact (m) (Borehole)
RG1	Kfar Hanassi-3 (KN-3)	2059/2606	19 ↑	20 (a/b)	173 ↓	197 (b/c)	267 ↑	264 (c/J)
RG2	Kfar Hanassi-1 (KN-1)	2058/2650	82 ↑	78 (c/b)	197 ↓	175 (b/c)	233 ↑	218 (c/b)
RG3	Rosh Pinna-1 (RP-1)	2042/2630	18 ↑	0-50 (a/b)	69 ↓	50-94 (b/c)	155 ↑	152-183 (c/b)
RG4		2140/2660	334 (b/c)? ↓					
RG5		2093/2640	214 (b/c)? ↓					
RG6		2113/2645	393 (b/c)? ↓					
RG7		2075/2645	107 (c/b)? ↑		326 (b/c)? ↓			
RG8	Alonei Habashan-1 (AB-1)	2287/2700	108 ↓	105 (b/s)	154 ↑	150 (s-b)	280 ↓	
RG9		2287/2710	86 (s/b)? ↑		187 (b/s)? ↓			
RG10		2130/2654	213 (c/b)? ↓		367 (b/c)? ↓			

DESIGNATIONS

J - Judea Group
a - alluvium
b - basalt of Dead Sea and Bashan Groups
c - clastic rocks of Hordos Fm. and Dead Sea Gr.
s - scoria
↑ - increasing resistivity contrast
↓ - decreasing resistivity contrast

d) The deep resistivity contrast at 378m depth which was found at the RG3 TDEM station (Fig. 36) could not be correlated with the nearby RP-1 borehole, owing to the loss of the circulation zone in the corresponding depth at the borehole. Since this interface is situated within the Judea Gr. sequence, it might be attributed to the Sakhnin/Dir Hanna Fms. boundary.

e) In the case of relatively thin alternations of basalts and clastic rocks (20-30m), it is not possible to detect the appropriate resistivity alternations using the TDEM measurements.

The correlation of the borehole data and TDEM results from nearby sites permitted geological interpretation of TDEM results for remote areas which are believed to show similar stratigraphic conditions. For example, the sharp decrease in resistivity in the depth range of 200-400m obtained at stations RG4 through RG7 and RG10 (see Table 1 and Figs. 37 thru 40 and 43) is most likely caused by the basalt-clastic rocks (b/c) contact. It should be noted that such a decrease in resistivity was found independently at the same range of depths by Audio Magnetotelluric (AMT) measurements (Ballestracci, 1984). However, the geological interpretation of this interface given by Ballestracci essentially differs from the above interpretation.

The consistency in appearance of the boundary in question permits its use as a marker for future structural mapping of the

area. The TDEM method can be used as a simple and reliable tool for this purpose.

As regards the basalt/scoria contact in the vicinity of t Alonei Habashan area (stations RG8 and RG9, Figs. 41 and 42), it seems that this boundary can be detected by the TDEM method; moreover, since it is characterized by a significant drop in resistivity, one would expect a fairly accurate determination of its depth from TDEM measurements (see Table 1). However, this latter conclusion is based on the only borehole in the area and, therefore, is not yet considered as statistically significant. The resistivity break at a depth of 280m (station RG8, Table 1 and Fig. 41) is still questionable.

4.3 Conclusions

The TDEM method is suitable for the detection of the alluvium-basalt contact, as well as for the differentiation of the sufficiently thick Neogene sedimentary sequences from the basalt layers and from the carbonate rocks of the Judea and Avedat Groups in northern Israel.

Detection of boundaries within the basaltic sequence itself (e.g. basalt/scoria) still require further investigation.

It seems that the contrast between the volcanics and carbonates of the Judea Group does not provide sufficient resistivity contrast and is, therefore, not detectable by the TDEM method.

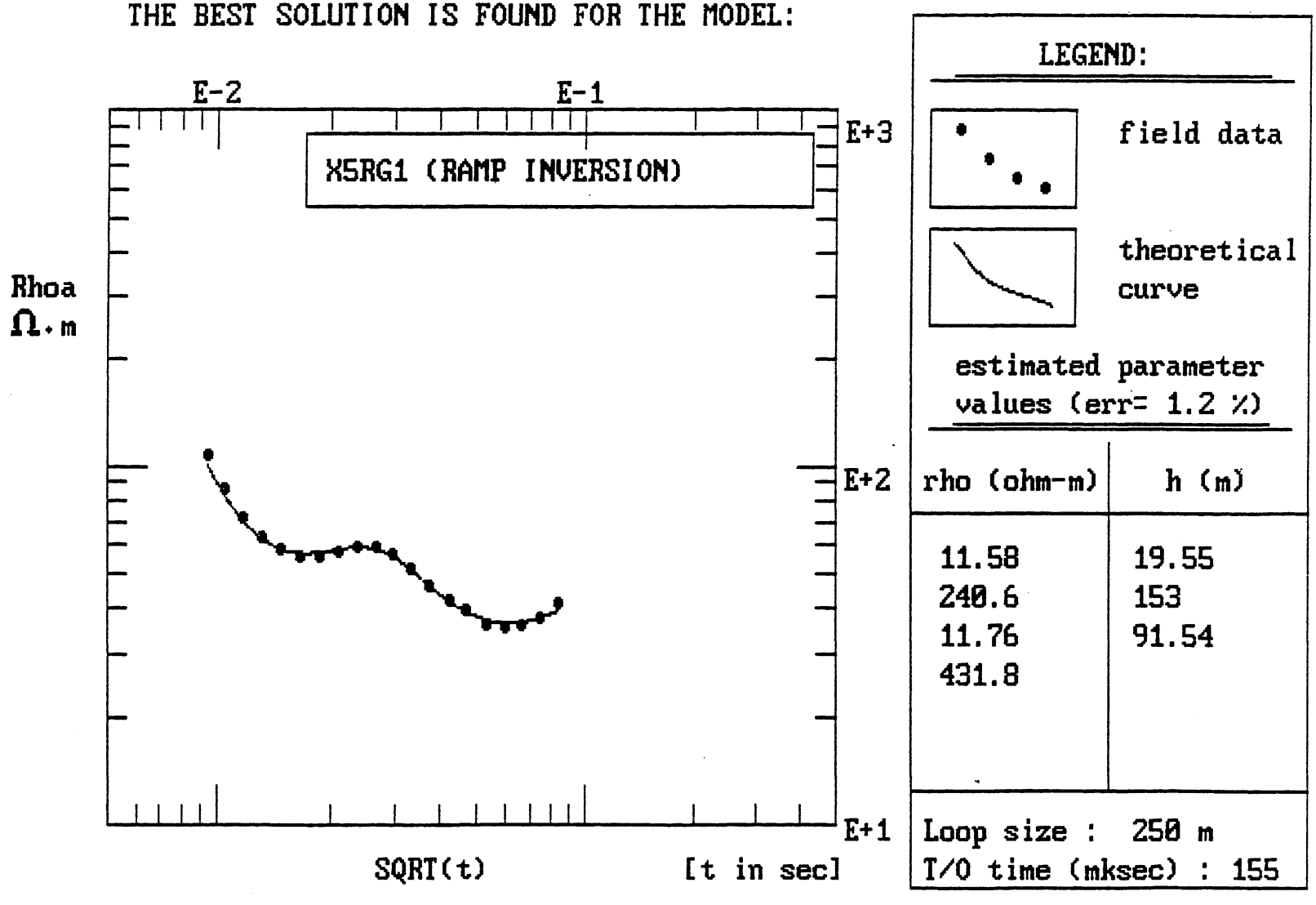

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+3
E+2
E+1
X5RG1 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.2 %)
rho (ohm-m)
h (m)
11.58
240.6
11.76
431.8
19.55
153
91.54
Loop size : 250 m
T/O time (mksec) : 155

Figure 34

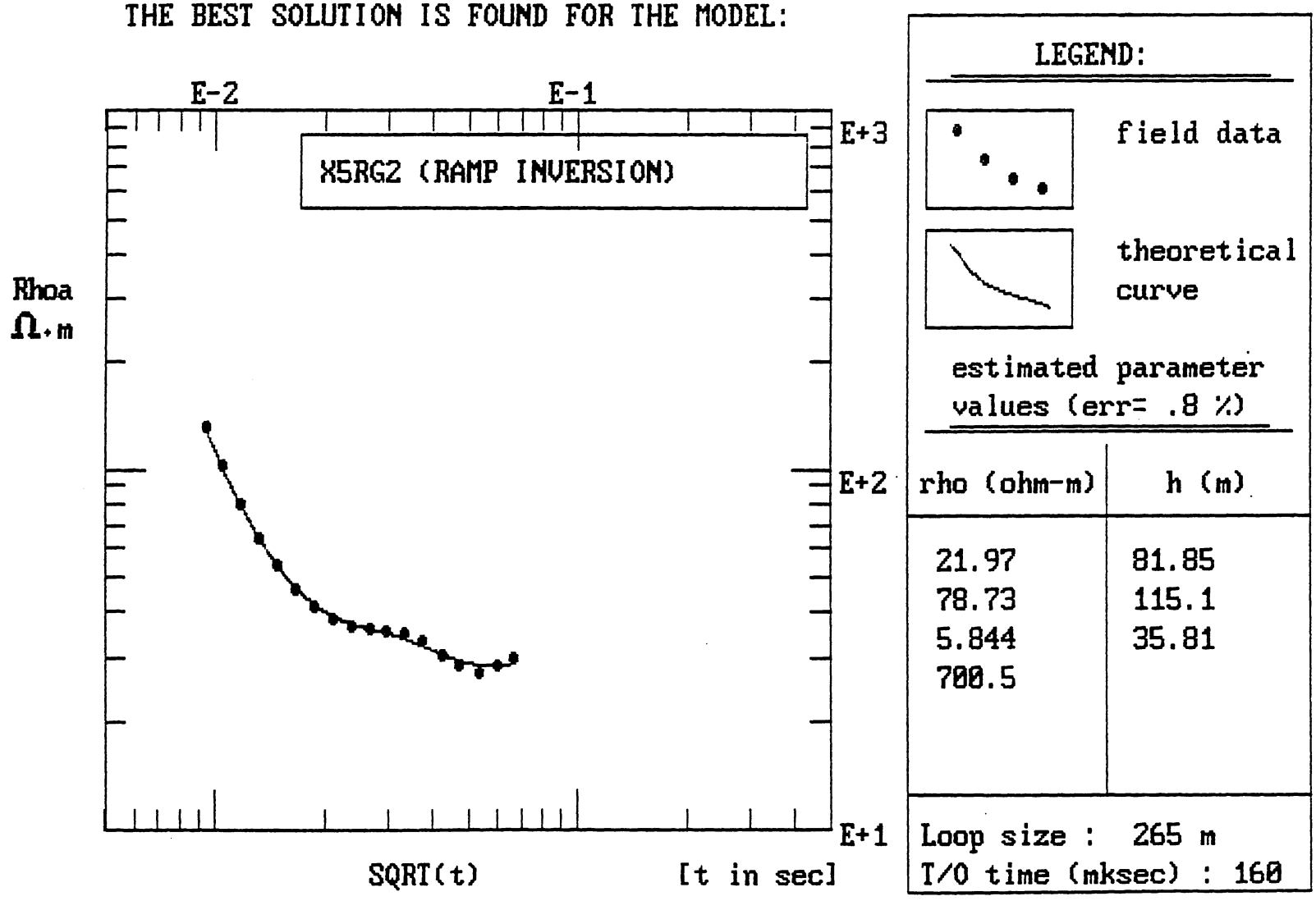

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+3
E+2
E+1
X5RG2 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .8 %)
rho (ohm-m)
h (m)
21.97
78.73
5.844
700.5
81.85
115.1
35.81
Loop size : 265 m
T/O time (mksec) : 160

Figure 35

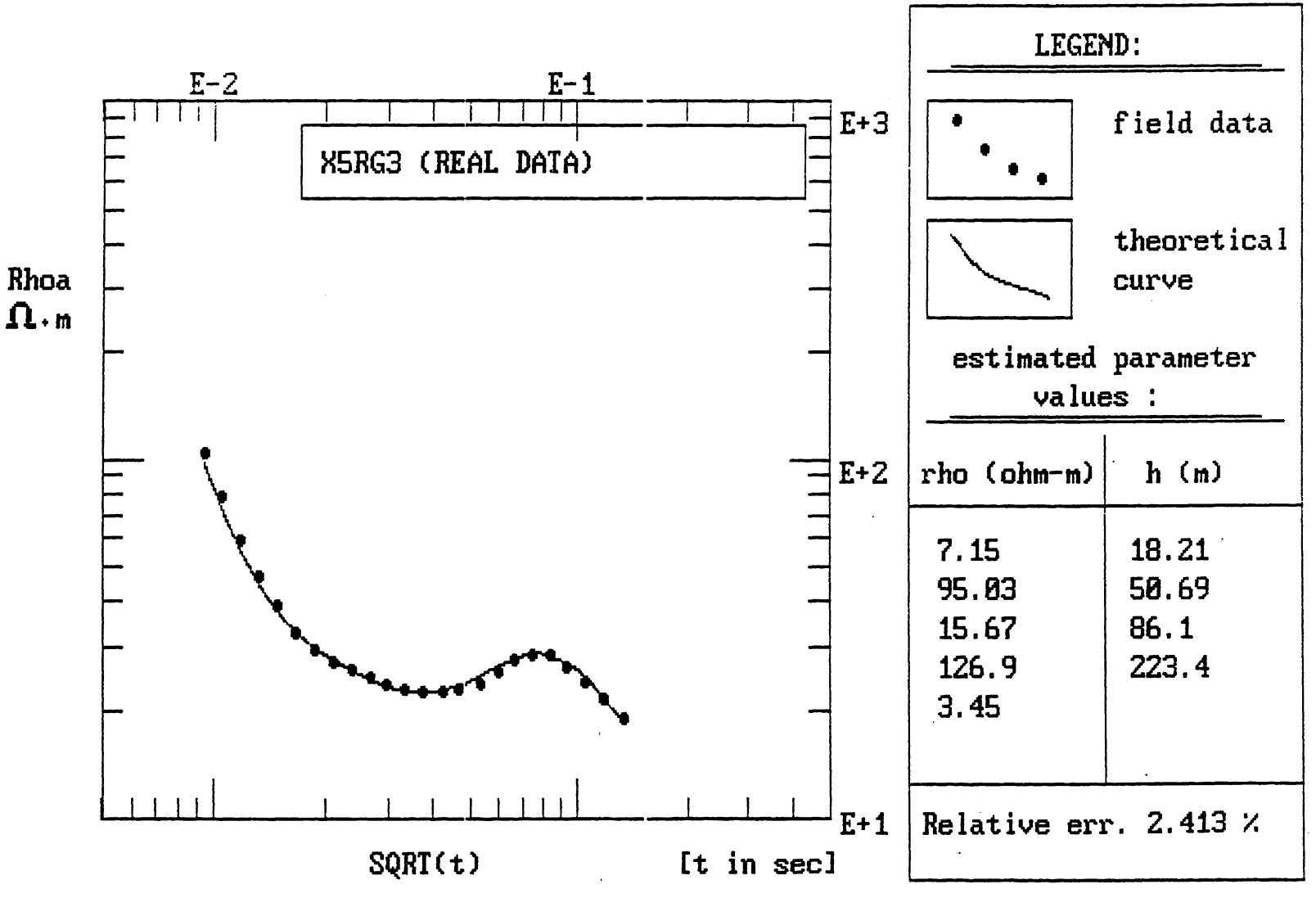

E-2
E-1
E+3
E+2
E+1
X5RG3 (REAL DATA)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values :
rho (ohm-m)
h (m)
7.15
95.03
15.67
126.9
3.45
18.21
50.69
86.1
223.4
Relative err. 2.413 %

Figure 36

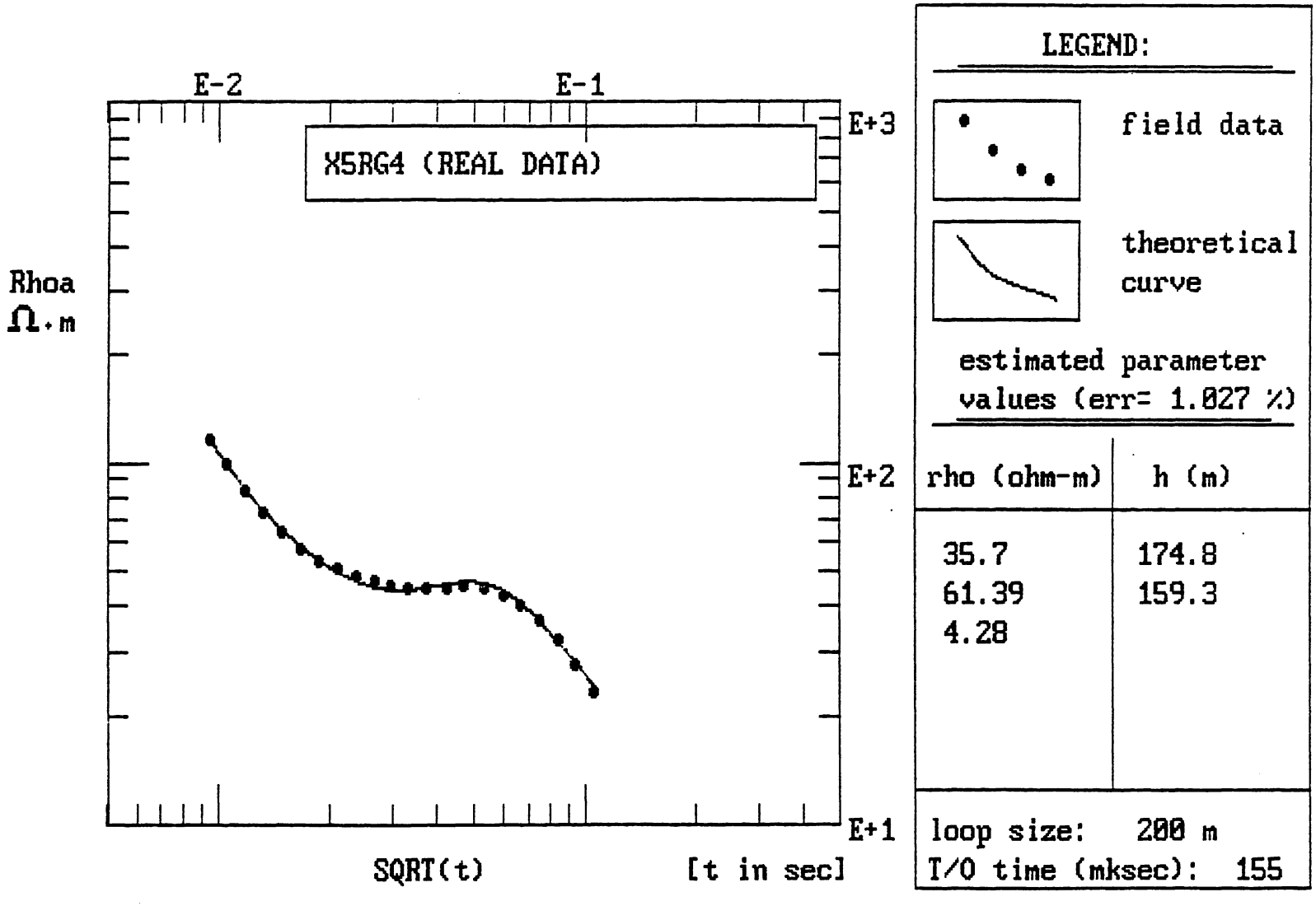

E-2
E-1
E+3
E+2
E+1
X5RG4 (REAL DATA)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.027 %)
rho (ohm-m)
h (m)
35.7
61.39
4.28
174.8
159.3
loop size: 200 m
T/O time (mksec): 155

Figure 37

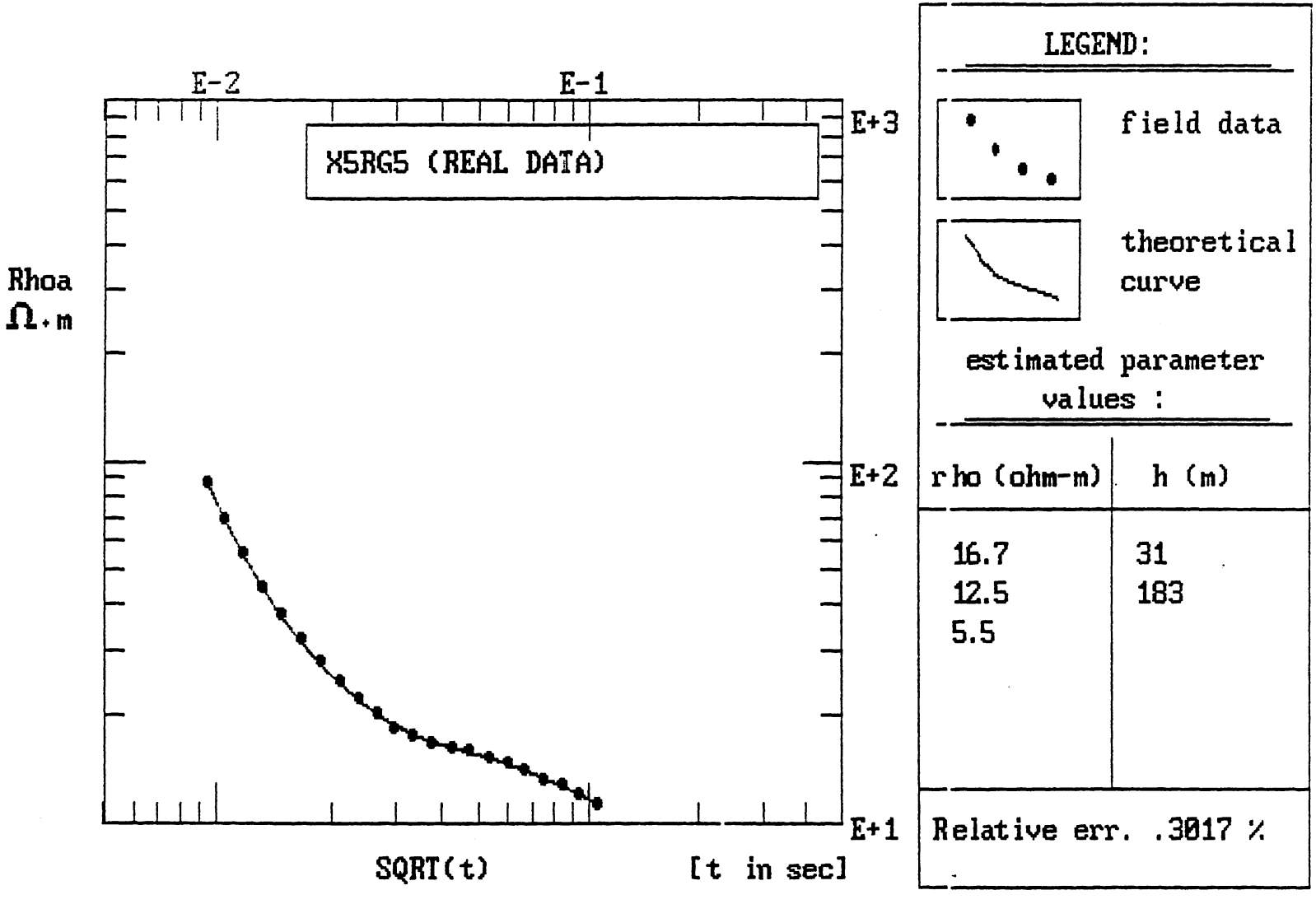

E-2
E-1
E+3
E+2
E+1
X5RG5 (REAL DATA)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values :
rho (ohm-m)
h (m)
16.7
12.5
5.5
31
183
Relative err. .3017 %

Figure 38

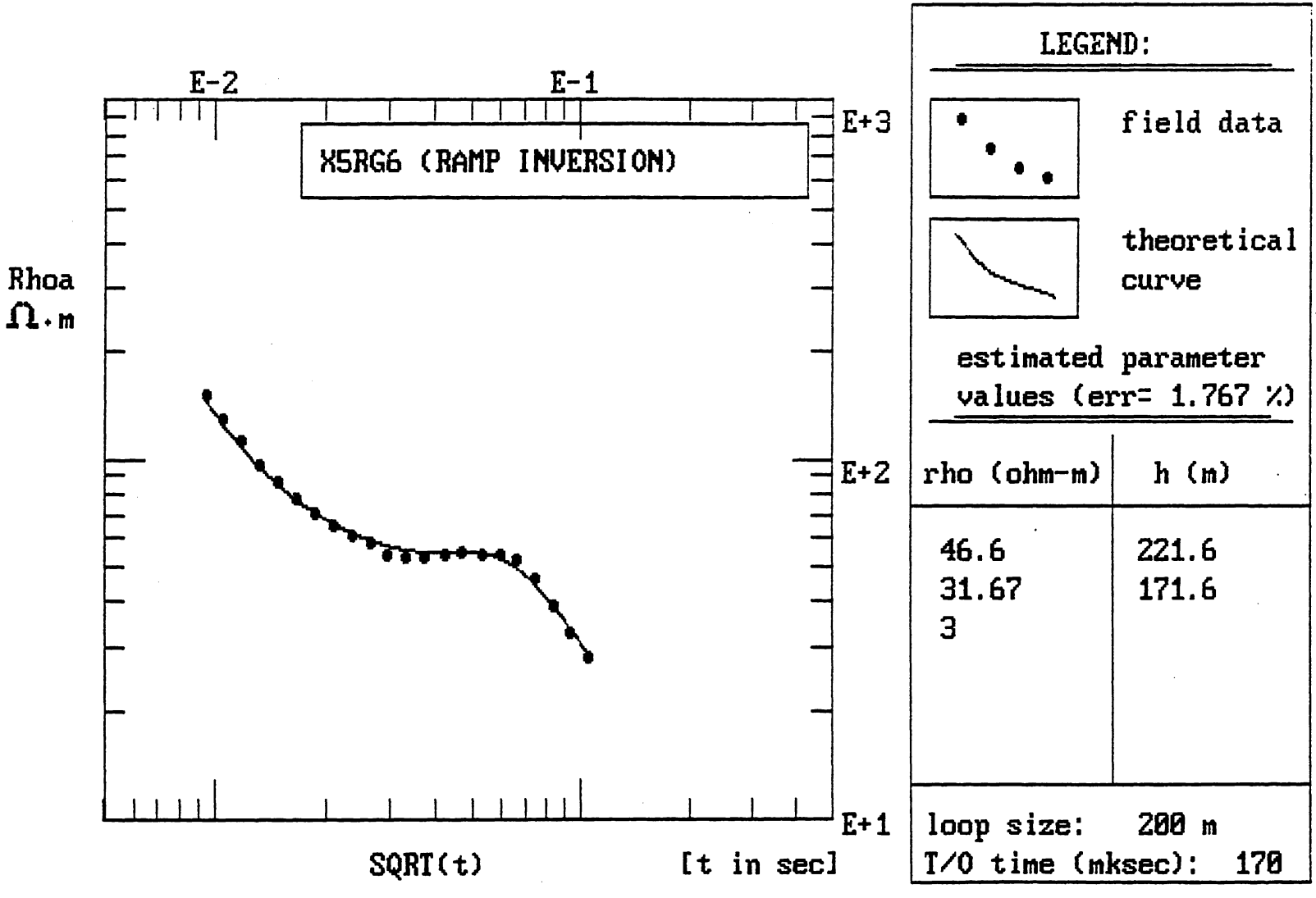

E-2
E-1
E+3
X5RG6 (RAMP INVERSION)
Rhoa
Ω.m
E+2
E+1
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.767 %)
rho (ohm-m)
h (m)
46.6
31.67
3
221.6
171.6
loop size: 200 m
T/O time (mksec): 170

Figure 39

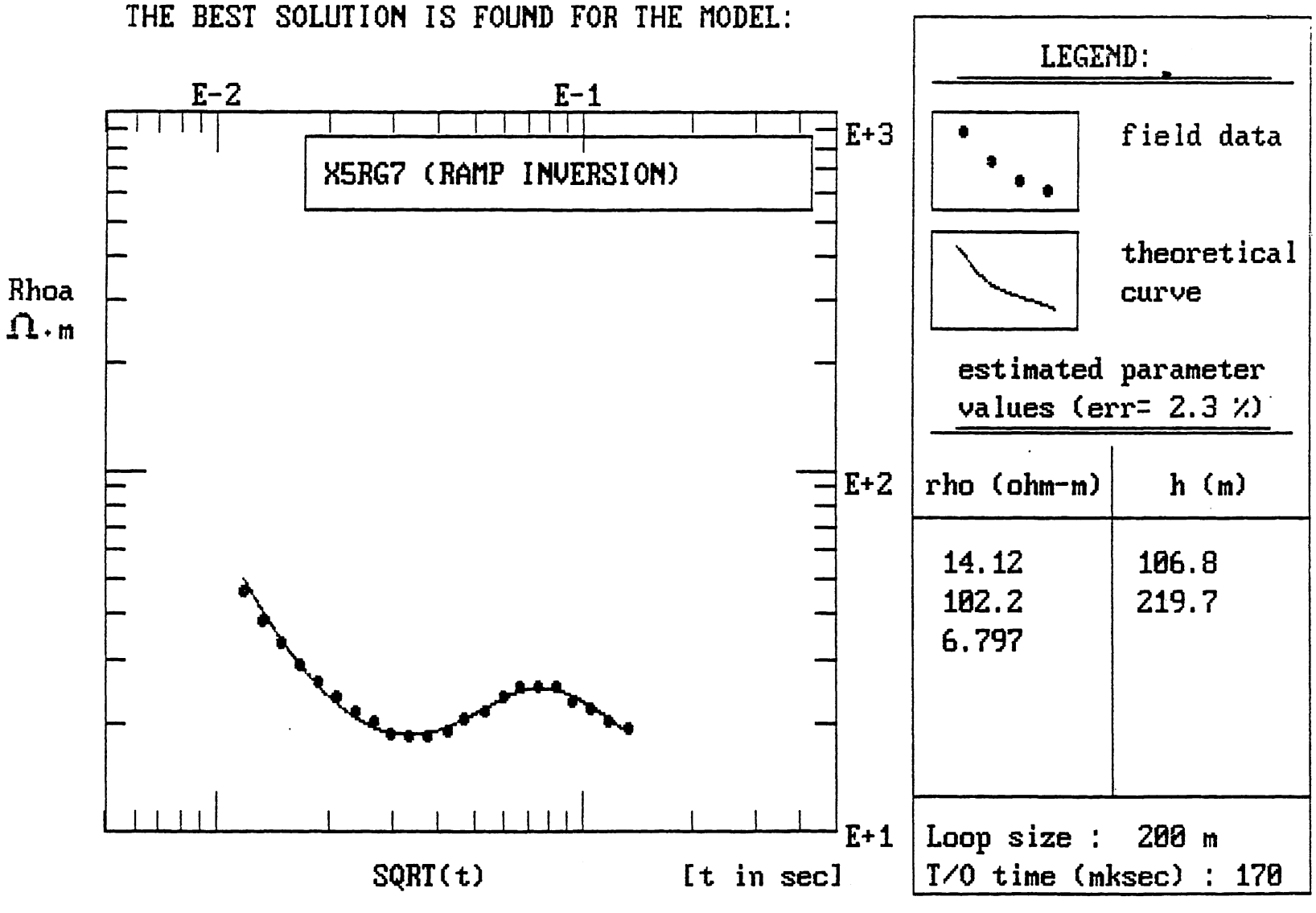

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+3
E+2
E+1
X5RG7 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 2.3 %)
rho (ohm-m)
h (m)
14.12
102.2
6.797
106.8
219.7
Loop size : 200 m
T/O time (mksec) : 170

Figure 40

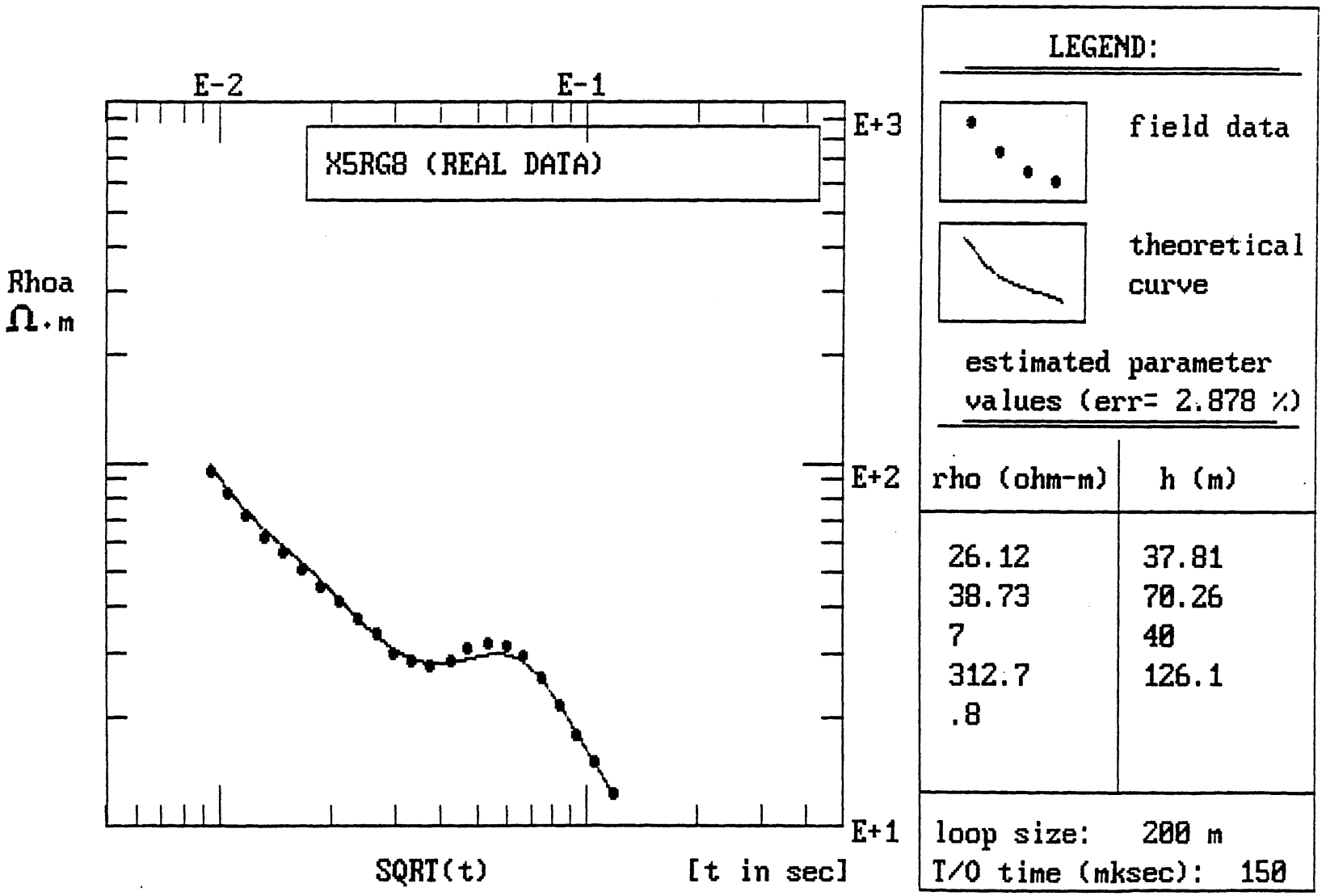

E-2
E-1
E+3
E+2
E+1
X5RG8 (REAL DATA)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 2.878 %)
rho (ohm-m)
h (m)
26.12
38.73
7
312.7
.8
37.81
70.26
40
126.1
loop size: 200 m
T/O time (mksec): 150

Figure 41

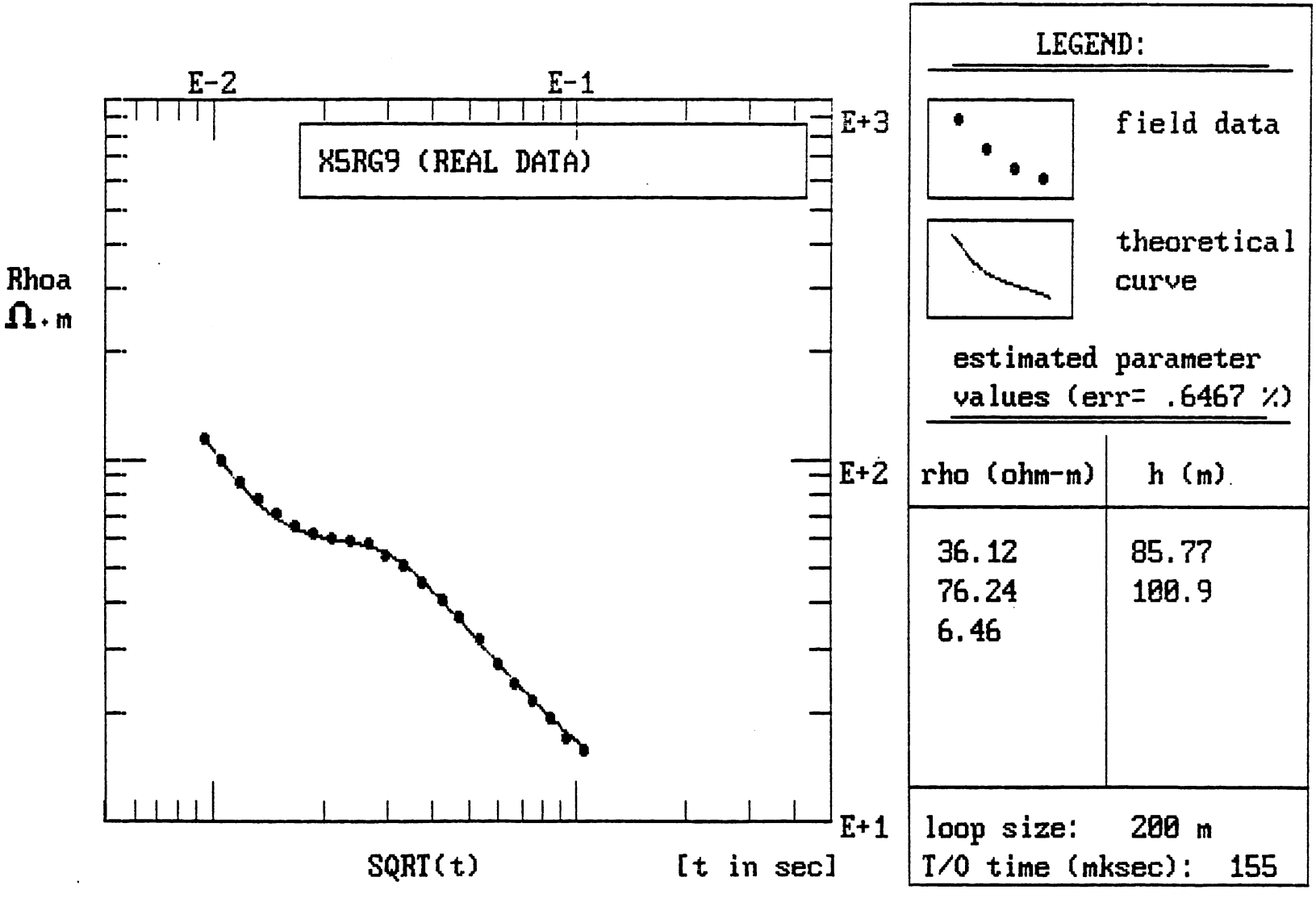

E-2
E-1
E+3
E+2
E+1
X5RG9 (REAL DATA)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .6467 %)
rho (ohm-m)
h (m)
36.12
76.24
6.46
85.77
100.9
loop size: 200 m
T/O time (mksec): 155

Figure 42

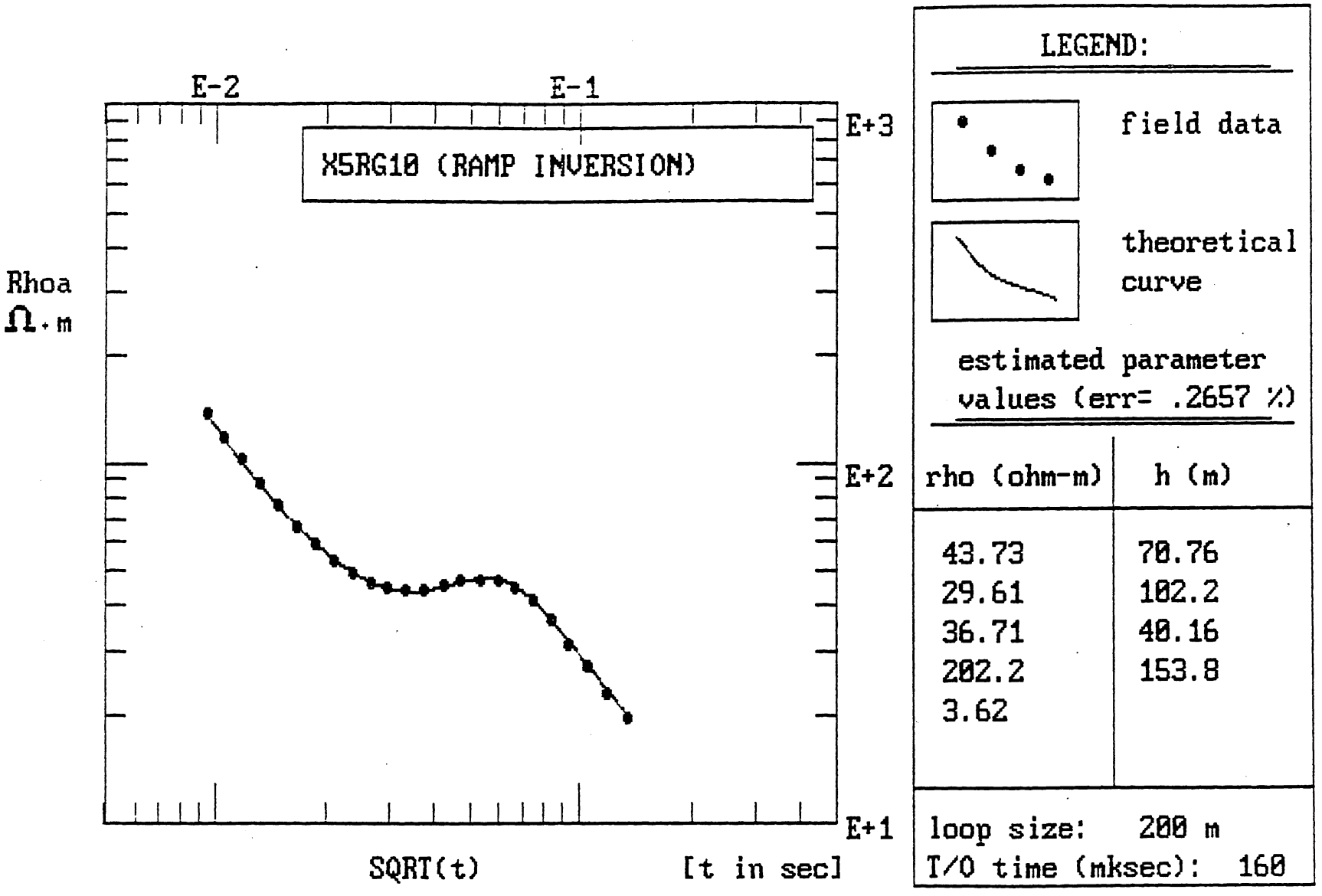

E-2
E-1
E+3
E+2
E+1
X5RG10 (RAMP INVERSION)
Rhoa
Ω·m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .2657 %)
rho (ohm-m)
h (m)
43.73
29.61
36.71
202.2
3.62
70.76
102.2
40.16
153.8
loop size: 200 m
T/O time (mksec): 160

Figure 43

5. MOUNT CARMEL AREA: FRESHWATER/SEAWATER INTERFACE DETECTION

5.1 Background

The resistivity of water-bearing rocks is a function of the water content, water salinity and the manner in which the water is distributed throughout the rock. The water content is determined by porosity and degree of saturation. The resistivity of a saturated aquifer is related to both the salinity of the water-bearing formation and to the waters included. The resistivities for a given aquifer decrease as the salinity increases. This physical relationship between pore water salinity and formation resistivity allows us to use surface electrical and/or EM methods to map changes in groundwater salinity and, in particular, to detect a fresh/saline water interface. These methods are applicable mostly in shallow aquifers where considerable salinity contrasts are expected.

The limitations of the methods usually stem from the configuration spreads which integrate lateral lithological transitions (facies, different blocks) and/or salinity variations. From this point of view the TDEM method is preferable because of its excellent lateral resolution. Another serious problem, which is common to all electrical and EM methods, is that interpretation of the field data is often ambiguous with regard to whether the resistivity changes result from lithological changes or from salinity differences.

To avoid such ambiguity, a preliminary attempt was made herein to detect a freshwater/seawater interface of high contrast within a roughly homogeneous, known lithology at a reasonable

depth by employing the TDEM method. The Mount Carmel region was selected as the study area for the following reasons:

a) The regional aquifer is that of the well known carbonate, partly karstic Judea Group, particularly the dolomitic Yagur Formation, which is characterized by a relatively high electric resistivity. This aquifer is 200-300m thick and contains no considerably thick aquicludal intercalations of low resistivity, which might mislead by yielding a resistivity drop similar to that obtained by a fresh/saline water interface.

b) The overlying Upper Cenomanian thick chalk aquiclude (Isfiya and Hureibe chalks) has a pronounced low resistivity, clearly differentiating it from the Yagur dolomite.

c) Suggested current subsurface seawater intrusion in northern Mount Carmel along the Yagur fault and in the Zevulun Plain (Arad et al , 1975) and below most of Mount Carmel (Kafri and Arad, 1979), should make detection of the freshwater/seawater interface in this area plausible.

d) Since the stratigraphy as well as the depth of water table are known from boreholes, the soundings could be planned adjacent to them for correlation purposes. Even though the actual steady-state interface was not encountered in the boreholes, the water level data, through the Ghyben-Herzberg equation, facilitate estimation of the theoretical freshwater/seawater interface depth.

5.2 Procedure

Four sites were selected for TDEM soundings as follows:

K-1 (coordinates 1483/2355) (Fig. 44)

The TDEM station was located close to the Nahal Oren-2 borehole, which was shutdown prior to the survey to allow recovery to the static water level.

The surface elevation of the borehole is around 55m above MSL. The accurate water level in this borehole is not known. Water level measurements in the nearby Nahal Oren-3 in the dry season of 1984 yielded a figure of 3.66m above MSL. Since the 1987 season was very wet, in fact above the average, one can assume a water level of some 4m above MSL in the Nahal Oren-2 borehole. Assuming that the interface depth below sea level for normal seawater is some 35-40m for each meter head of freshwater, the depth of the interface was calculated as being:

$$55m + 4m\ (35\text{-}40) = 195\text{-}215m \text{ depth}$$

K-2 (coordinates 1467/2306) (Fig. 44)

The TDEM station was located close to the Hof Carmel-4 borehole. The surface elevation of the borehole is 36m above MSL; the static water level is 3m above MSL. The calculated theoretical interface depth is:

$$36m + 3m\ (35\text{-}40) = 141\text{-}156m \text{ depth}$$

MOUNT CARMEL AREA

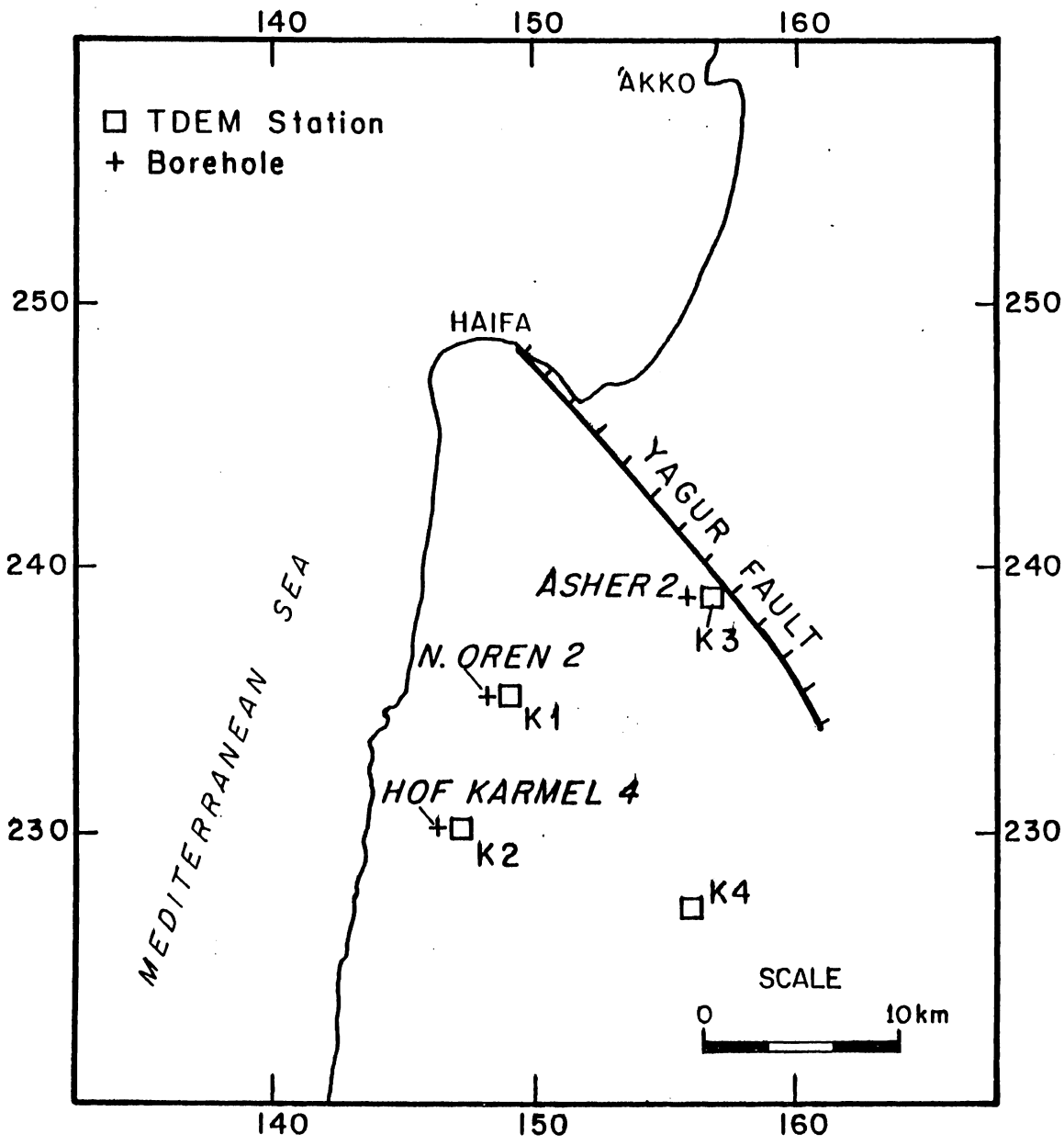


Figure 44

K-3 (coordinates 1563/2396) (Fig. 44)

The TDEM station was located close to the Asher-2 wildcat. The surface elevation of the borehole is some 20m above MSL; the static water level is 4.5m above MSL. The calculated theoretical interface depth is:

20m + 4.5m (35-40) = 177-200m depth

K-4 (coordinates 1557/2268) (Fig. 44)

The TDEM station was located close to the top of the Judea Group (top Bina Fm.) in the region of the groundwater divide, but with no adjacent correlation borehole. The surface elevation is around 200m above MSL; the static water level is 10-12m above MSL. The freshwater/seawater interface is expected at a depth of 550-680m.

This particular sounding required a deeper penetration. Unfortunately, because of a high voltage power line in the neighborhood which generated strong background noise, no results could be obtained.

5.3 Results

The results and correlation of the different soundings by relevant borehole stratigraphy and interface location are shown in Figs. 45-47.

a) K-1 data yield a four-layer geoelectrical model (Figs. 45 and 48). The two upper layers correspond to lithological units and the boundaries roughly coincide

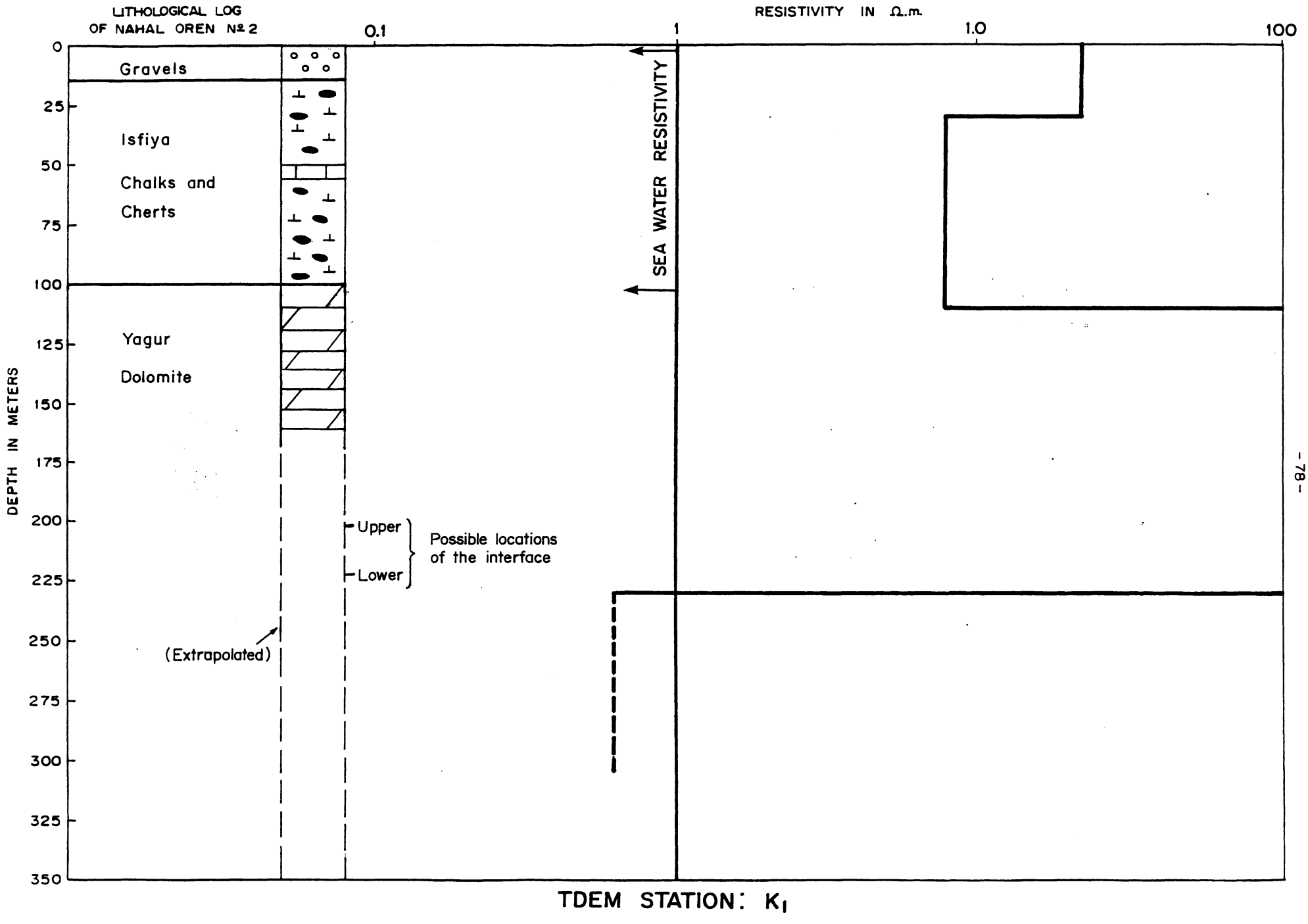

LITHOLOGICAL LOG
OF NAHAL OREN Nº 2
RESISTIVITY IN Ω.m.
0.1
1
1.0
100
0
25
50
75
100
125
150
175
200
225
250
275
300
325
350
Gravels
Isfiya
Chalks and
Cherts
Yagur
Dolomite
SEA WATER RESISTIVITY
DEPTH IN METERS
Upper
Lower
Possible locations
of the interface
(Extrapolated)
TDEM STATION: K1

Figure 45

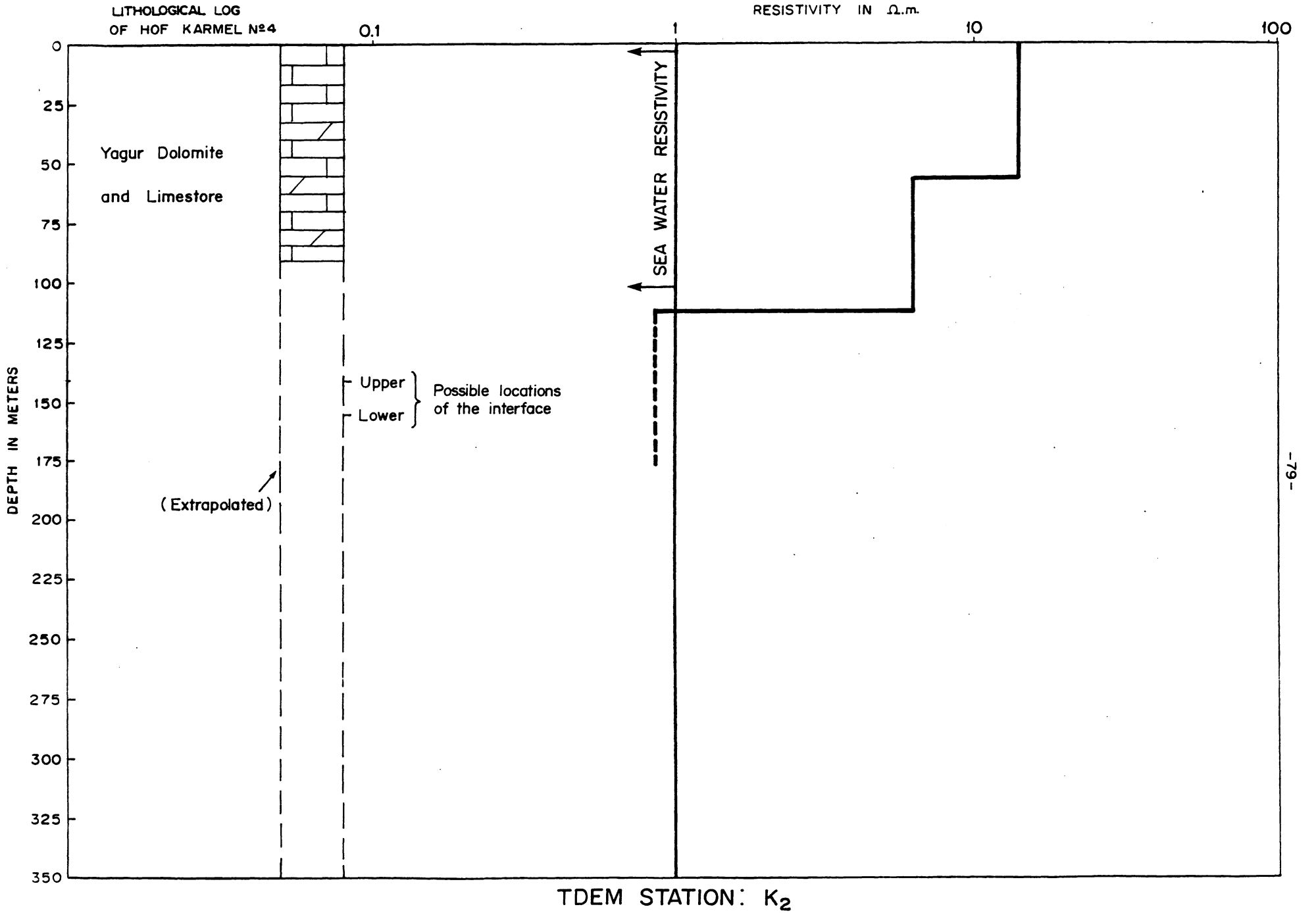

LITHOLOGICAL LOG
OF HOF KARMEL Nº4
RESISTIVITY IN Ω.m.
0.1
1
10
100
0
25
50
75
100
125
150
175
200
225
250
275
300
325
350
DEPTH IN METERS
Yagur Dolomite
and Limestore
SEA WATER RESISTIVITY
Upper
Lower
Possible locations
of the interface
(Extrapolated)
TDEM STATION: K2

Figure 46

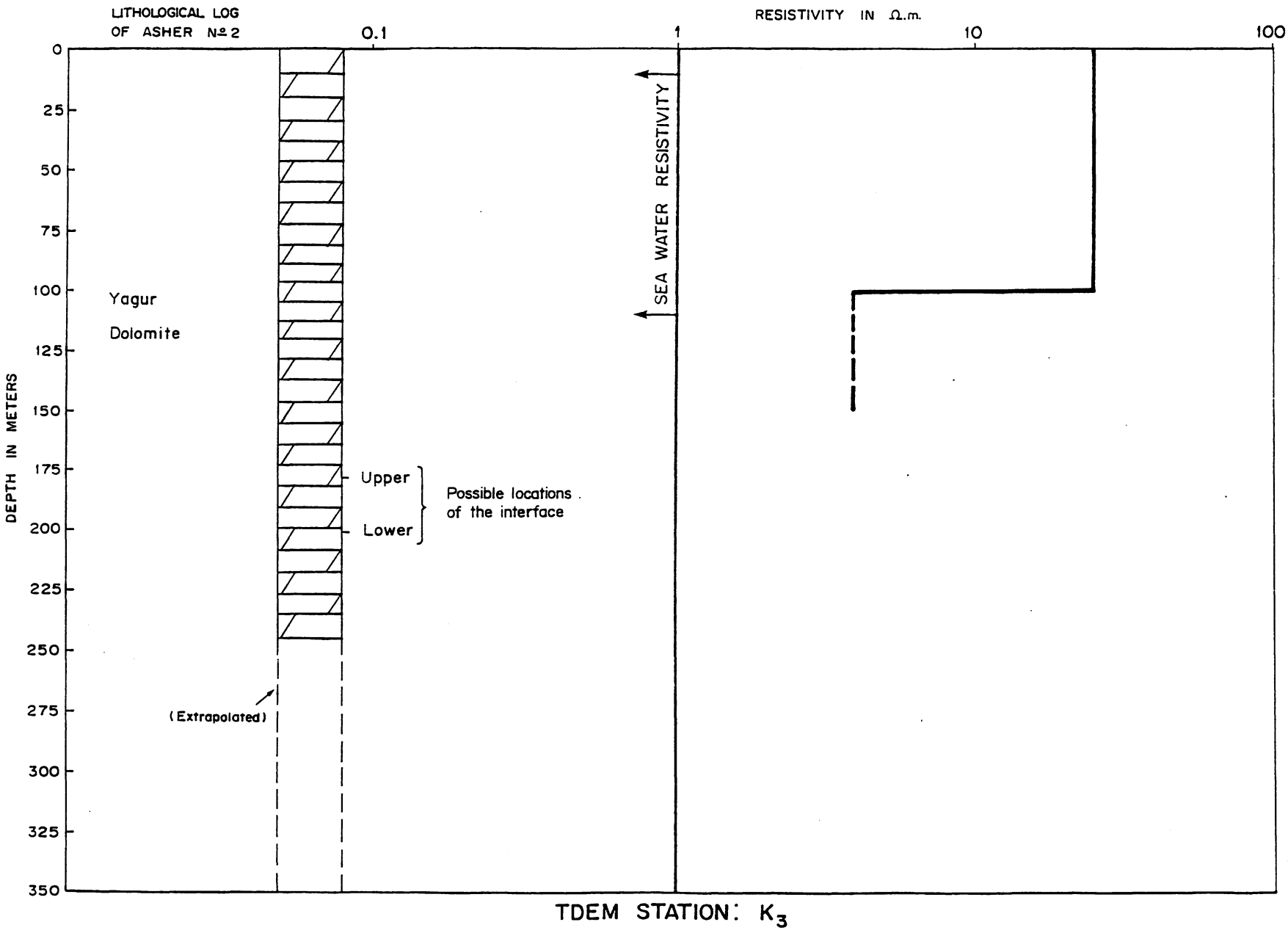

LITHOLOGICAL LOG
OF ASHER Nº2
RESISTIVITY IN Ω.m.
0.1
1
10
100
0
25
50
75
100
125
150
175
200
225
250
275
300
325
350
DEPTH IN METERS
Yagur
Dolomite
SEA WATER RESISTIVITY
Upper
Lower
Possible locations
of the interface
(Extrapolated)
TDEM STATION: K3

Figure 47

with the lithological contacts. The boundary between the lower layers has no lithological significance, as noted in the adjacent correlation boreholes, since it is found within a continuous dolomite sequence. It appears that the resistivity drop correlates with the depth of the calculated freshwater/seawater interface. Furthermore, resistivity values in the range of Ø.5 to 1.5 ohm.m are typical for normal seawater intrusions elsewhere (Israel coastal aquifer, etc.).

b) K-2 data yield a three-layer geoelectrical model (Figs. 46 and 49) entirely within an approximately homogeneous dolomite sequence.

The lower electrical boundary approximately coincides with the depth of the calculated interface and the resistivity value of the lower layer, less than 1 ohm/m, is attributed to seawater intrusion.

Since there is no lithological explanation for the upper boundary, it is assumed that it represents the upper limit of the presumed mixing zone between the upper fresh, and lower sea, waters. The resistivity value in this presumed mixing zone is indeed intermediate between the fresh and saline ones.

c) The K-3 site is located in difficult topography, thus permitting only a limited size of transmitter loop. Owing to the limited penetration depth, a two-layer model only was obtained for the K-3 data within a continuous dolomite sequence (Figs. 47 and 50). The sounding did not reach the depth of the calculated interface at 200m, which is recognized elsewhere in the electric resistivity log of the Asher-2 wildcat. The boundary at a 100m depth has no lithological significance and is therefore interpreted, as for K-2, as the upper boundary of the mixing zone.

5.4 Conclusions

1. The present study should be regarded as preliminary, since the number of soundings is too small to be statistically significant.
2. Nevertheless, it seems that reasonably accurate detection of a freshwater/seawater interface is possible within a homogeneous lithological sequence.
3. Resistivity values of less than 1 ohm·m (average between 0.5-1.5 ohm·m) seem to be solely indicative of seawater-intruded aquifers, whereas low resistivity lithologies usually have values significantly exceeding 1 ohm·m.

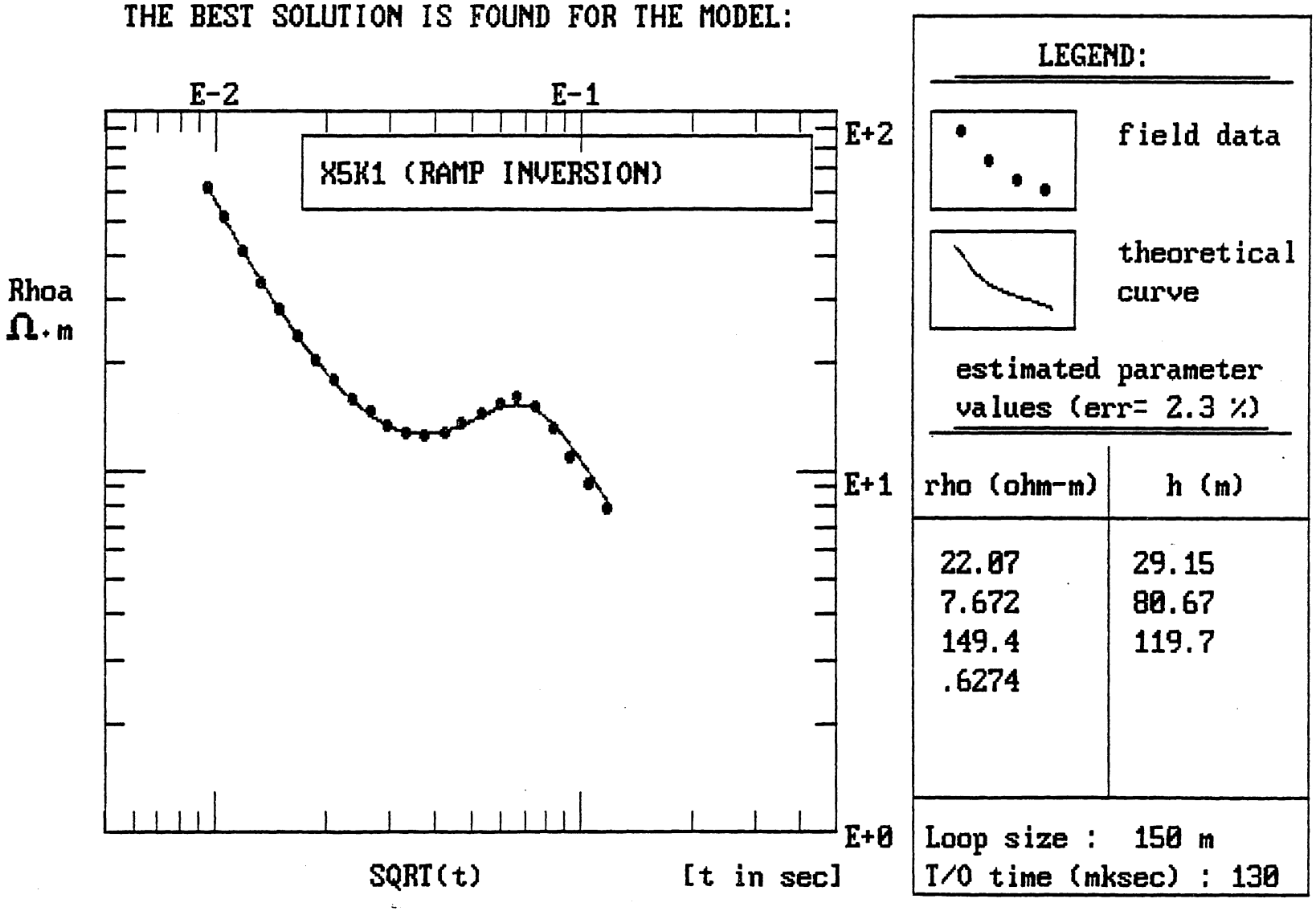

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+2
E+1
E+0
X5K1 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 2.3 %)
rho (ohm-m)
h (m)
22.07
7.672
149.4
.6274
29.15
80.67
119.7
Loop size : 150 m
T/O time (mksec) : 130

Figure 48

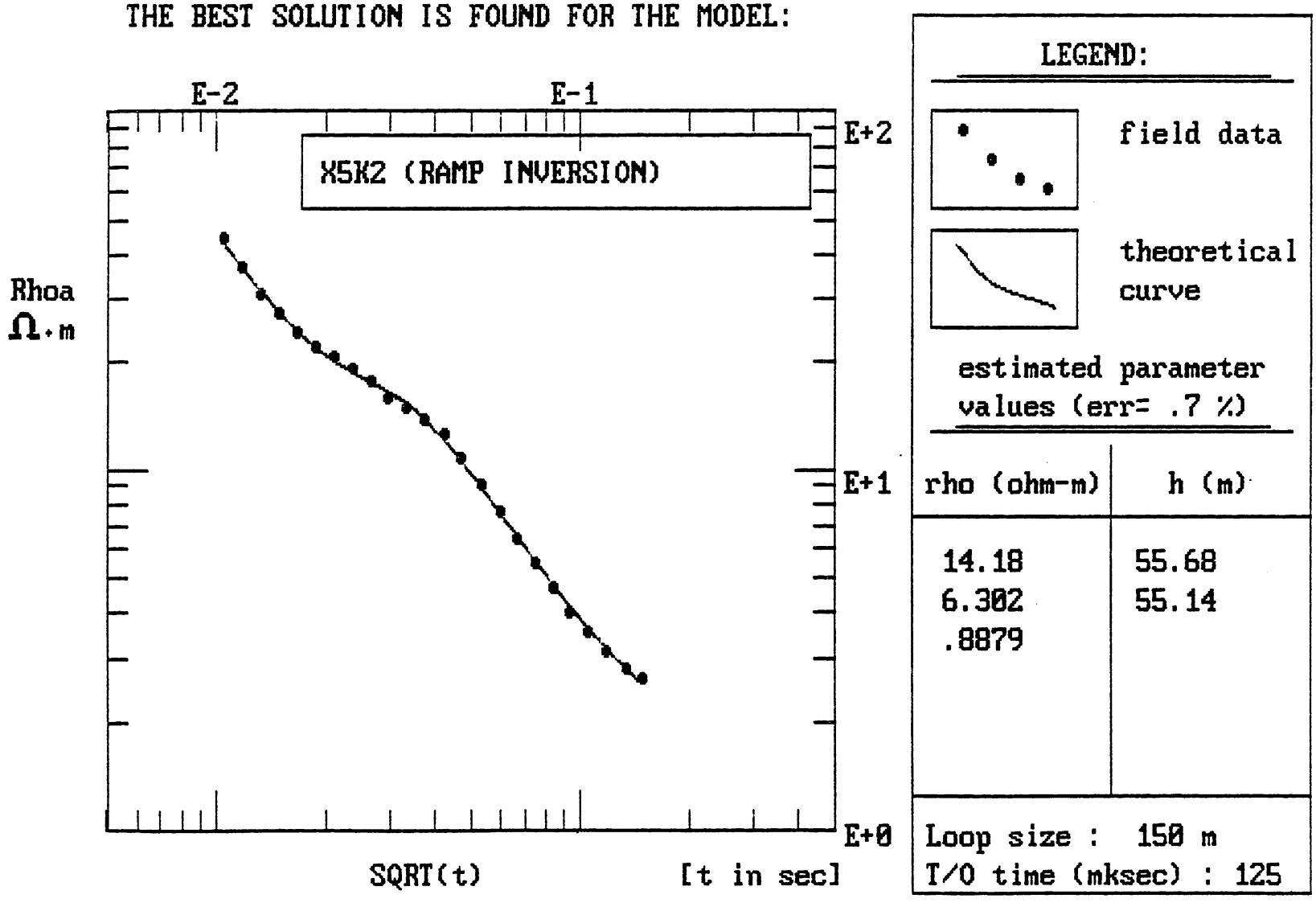

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+2
E+1
E+0
X5K2 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .7 %)
rho (ohm-m)
h (m)
14.18
6.302
.8879
55.68
55.14
Loop size : 150 m
T/O time (mksec) : 125

Figure 49

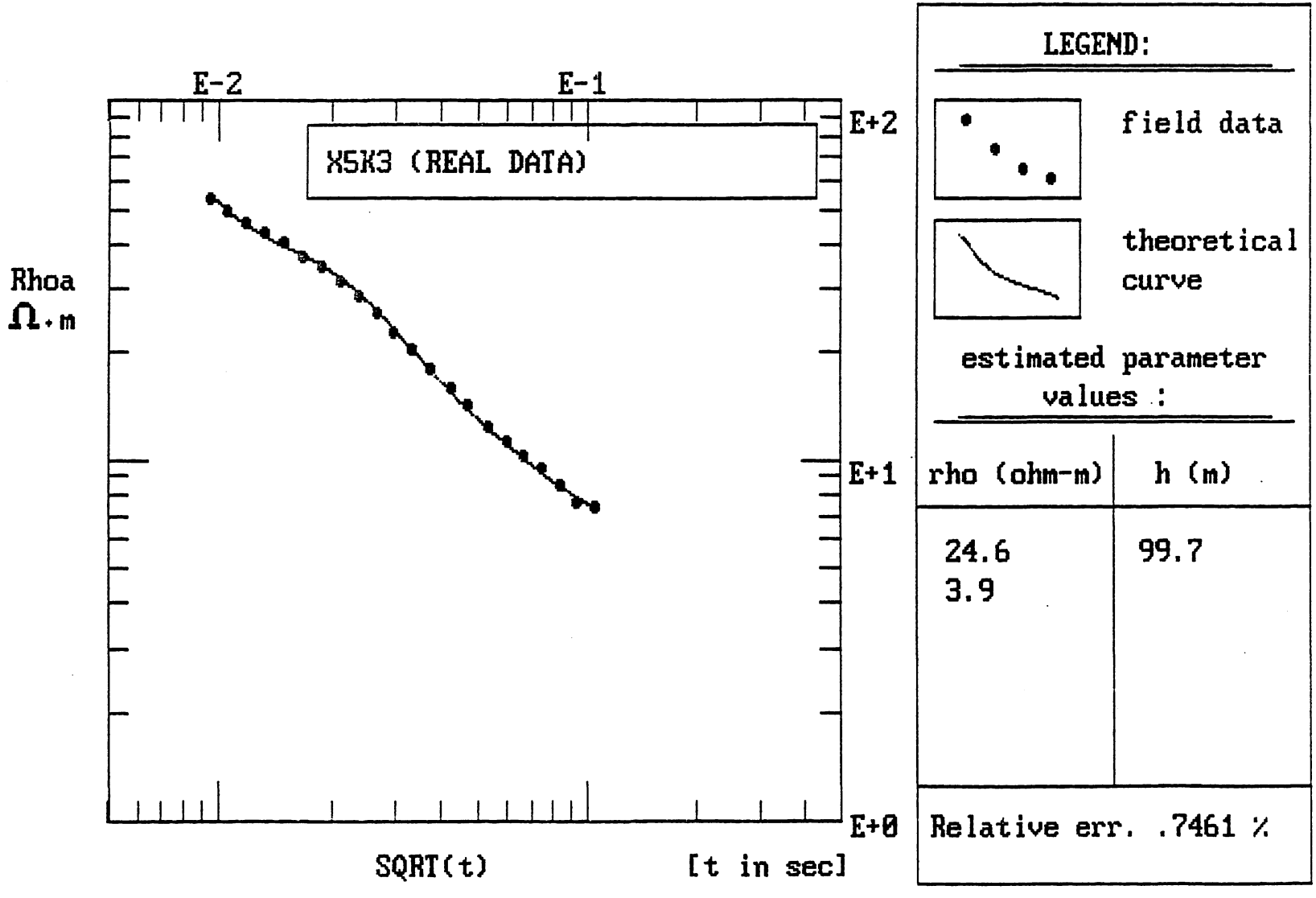

E-2
E-1
E+2
E+1
E+0
X5K3 (REAL DATA)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values :
rho (ohm-m)
h (m)
24.6
3.9
99.7
Relative err. .7461 %

Figure 50

6. NITZANIM AREA: FRESHWATER/SEAWATER INTERFACE DETECTION

6.1 Background

Estimation of the maximum penetration of the seawater (toe of the interface) into the coastal plain aquifer of Israel is most important for aquifer management. The monitoring of the depth of the interface requires a dense network of observation wells technically suited to this purpose. Unfortunately, the present well network is very poor and only a few observation wells satisfy the technical requirements.

In order to overcome the above difficulties, it was suggested that the depth to the interface be determined using direct surface geophysical measurements. An attempt was made to determine whether the TDEM method can satisfactorily replace the conventional direct measurements.

6.2 Hydrogeological Background

The Nitzanim area (Fig. 51) was chosen for the investigation owing to the existence of long term observation wells and the simplicity of the geological setup.

The geological sequence consists of calcareous sandstone intercalated by some clay horizons and underlain by the Saqiye marl beds. The sequence is divided into two main units, namely the upper subaquifer, AB, and the lower subaquifer, C (Fig. 52). The upper subaquifer is more affected by seawater intrusion and was, therefore, the main target for investigation.

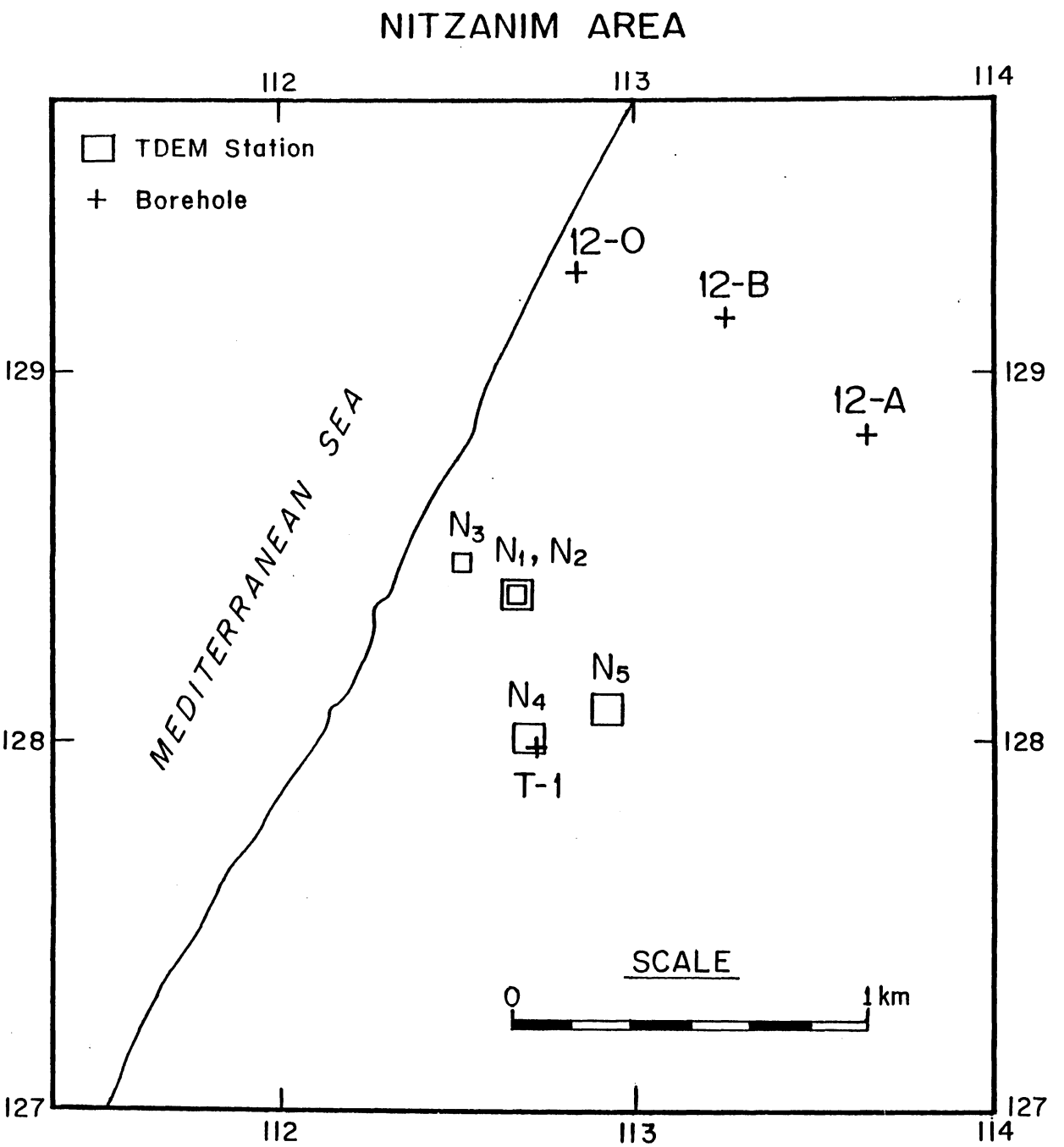
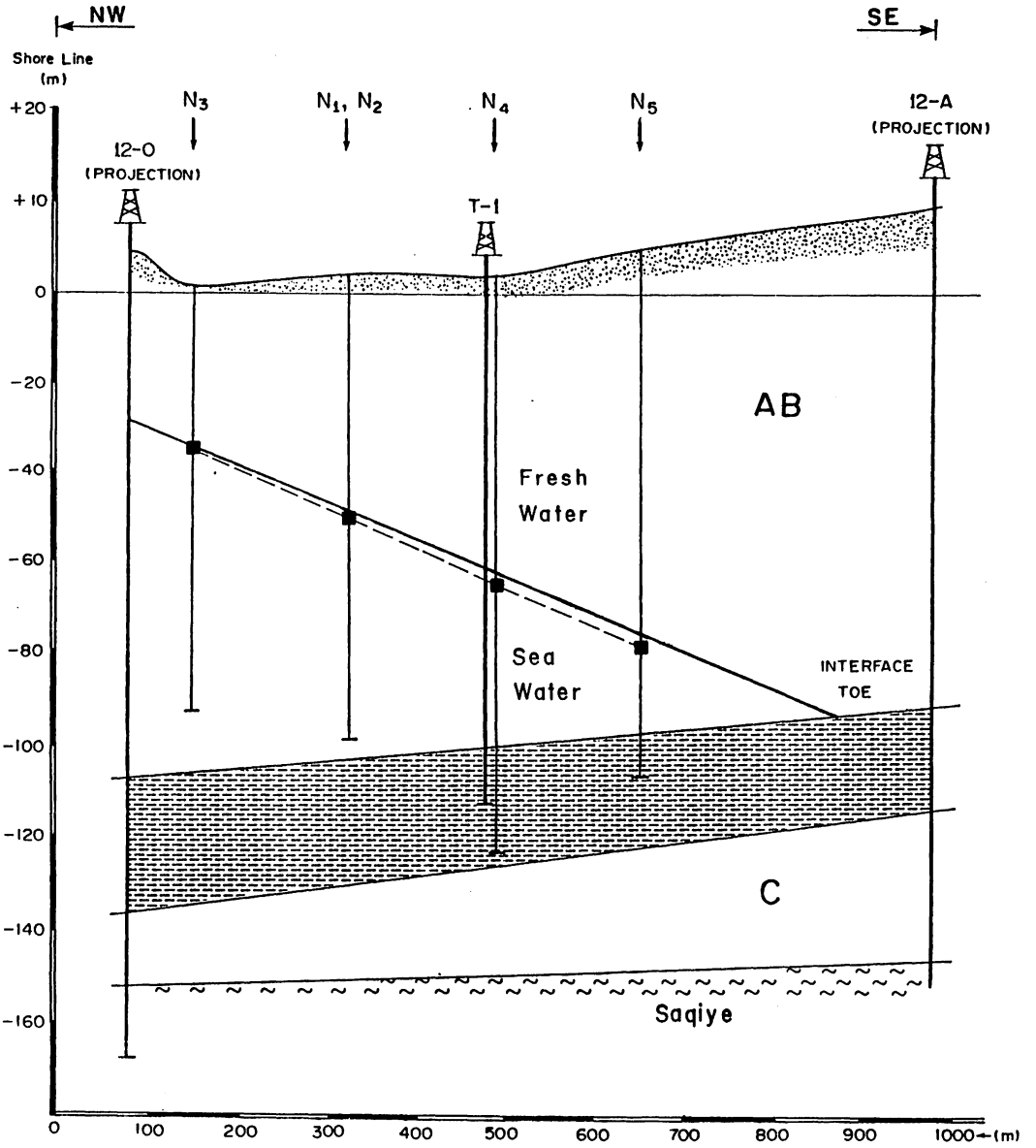

NITZANIM AREA
112
113
114
TDEM Station
Borehole
12-O
12-B
12-A
129
128
127
MEDITERRANEAN SEA
N3
N1, N2
N5
N4
T-1
SCALE
0
1 km

Figure 51

NITZANIM : GEOLOGICAL CROSS-SECTION



LEGEND

	SANDS, DUNES	N_{1-5}	TDEM STATIONS
	SANDSTONE	T-1	WELL NUMBER
	CLAYS	———	INTERFACE 17 500 mgl. T.D.S.
	CLAYS & MARLS	—■—	TOP OF THE LOW RESISTIVITY LAYER

Figure 52

6.3 Procedure and Results

Four sites (Fig. 51) were selected for TDEM soundings:

a) Sounding N1,N2 (two measurements at the same site using different transmitter loops of 100mx100m and 50mx50m respectively) was located approximately 900m to the south of the observation well, 12-0.

b) Sounding N3 was located approximately 800m to the south of the observation well, 12-0.

c) Sounding N4 was located a few meters from the observation well Ashdod T-1.

d) Sounding N5 was located 200m NE of the observation well Ashdod T-1 and approximately 1000m SW of the observation well, 12-A.

The approximate maximum penetration depths of the above soundings vary between 90 to 130m. The results of the automatic inversion of the measured TDEM data (resistivities and depths), together with other relevant parameters, are represented in Table 2.

TABLE 2

TDEM Station	Coordinates	Distance from sea (m)	Elevation ASL (m)	Depth to top of very low resistivity layer (m)	Resistivity of very low resistivity layer (Ohm.m)
N1,N2	11265/12840	320	+4	55	1.5
N3	11249/12849	150	+1	36	1.7
N4	11266/12802	480	+4	70	1.8
N5	11288/12808	650	+10	89	1.4

Interpretation of the TDEM data distinguished between three main geoelectrical units: two of these were clearly detected while the lower one could be defined by approximation only. The upper unit has relatively high resistivity values in the range of 20 to 50 ohm.m. The intermediate horizon is characterized by significantly lower resistivity values within the range of 1.4 to 1.8 ohm•m. Unfortunately the lower unit was not clearly defined due to the insufficient penetration depth of the measurements.

The boundary between the upper and second resistivity units is regarded as a sharp contrast between a freshwater body and the seawater intrusion. The reasons for this are:

- The boundary is detected within a roughly homogeneous lithological sequence and, thus, reflects a salinity contrast.
- The absolute resistivity values beneath the interface are in the range of 1.4 to 1.8 ohm•m, typical to resistivity of seawater intrusion in coastal aquifers (Sengpiel, 1986).

A comparison of the TDEM results with measurements from the observation wells showed a very good coincidence between them (Fig. 52). For example, in station N4 next to the Ashdod T-1 well, the depth to the top of a low resistivity layer is 70m (Table 2 and Figs. 52 and 55); whereas the depth to the narrow transition zone of the seawater intrusion (17,500 mgl TDS), is 67m. Furthermore, a salinity of 35,000 mgl TDS, which approximately matches seawater, was found at a depth of 70m.

6.4 Conclusions

The main conclusions drawn from the above are as follows:

a) A geoelectric unit with a resistivity of about 1.5 ohm•m cannot be attributed to any known lithology and represents only seawater intrusion. The top of this unit lies within the transition zone between the freshwater body and the seawater intrusion.

b) The discrepancy between the TDEM results and the observation well measurements ranges between Ø-3 meters (Ø-4% respectively), depending on the interface definition (either 35,ØØØ or 17,5ØØ mgl TDS) used.

These preliminary positive results would allow us to recommend further employment of the TDEM method in order to define its feasibility for the above objectives. It is therefore proposed to continue the survey near available observation wells in order to ensure greater statistical significance of the results and to increase the penetration depth of the TDEM measurements in order to reach the base of the aquifer.

The results of this survey may indicate whether the proposed method could either replace or assist the conventional methods of detection of seawater intrusion into the coastal aquifer.

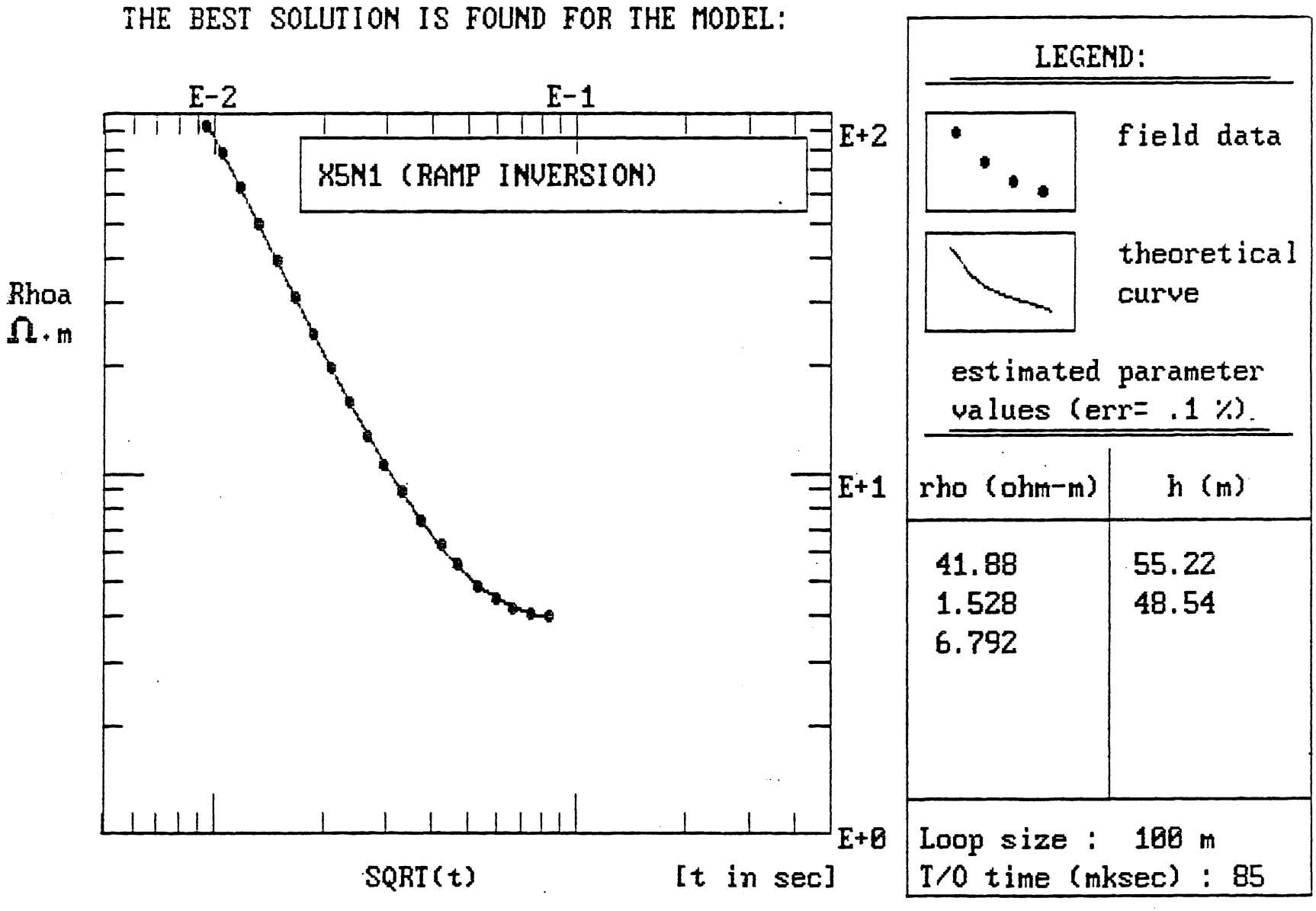

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+2
E+1
E+0
X5N1 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .1 %)
rho (ohm-m)
h (m)
41.88
1.528
6.792
55.22
48.54
Loop size : 100 m
T/O time (mksec) : 85

Figure 53

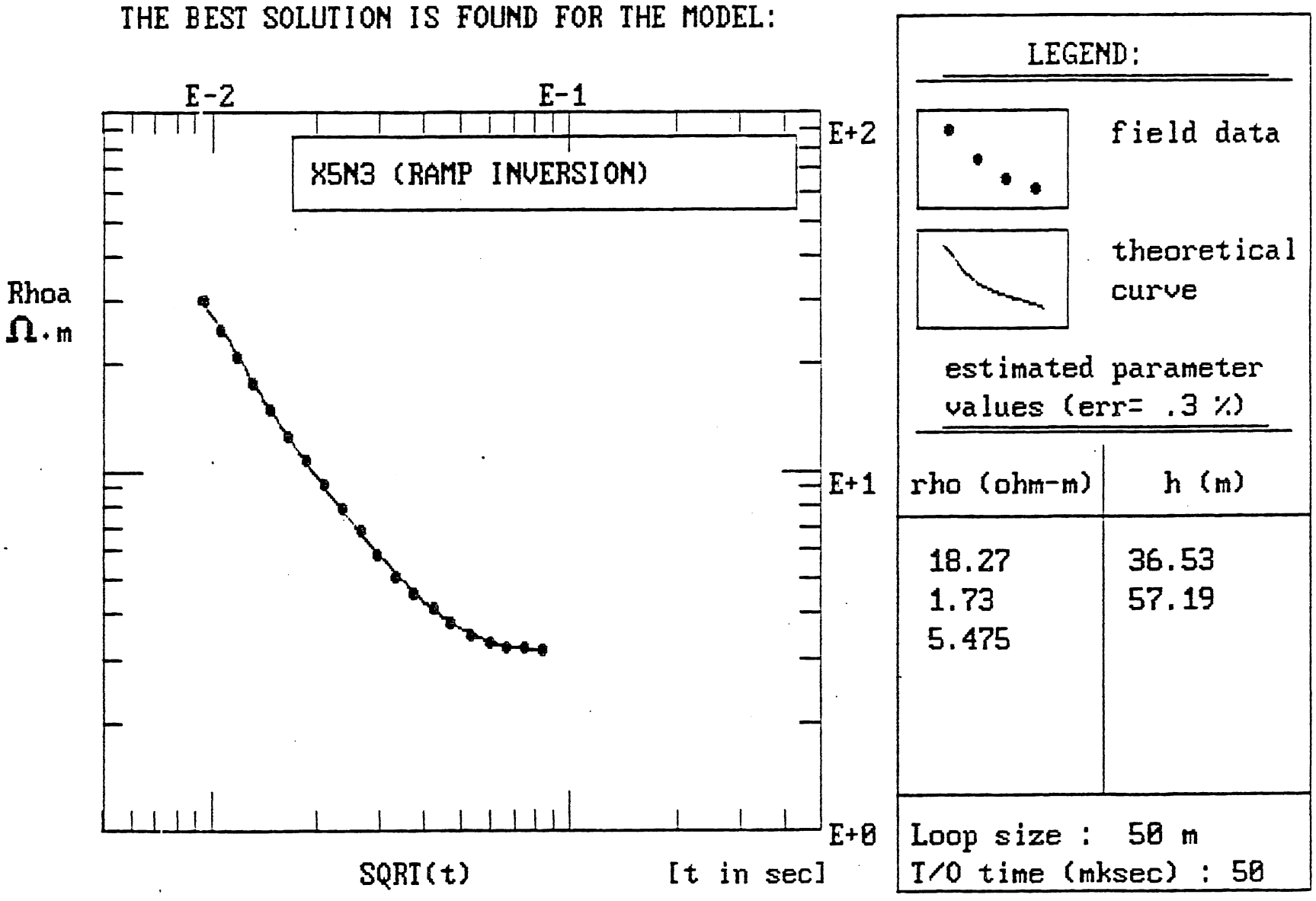

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+2
X5N3 (RAMP INVERSION)
Rhoa
Ω.m
E+1
E+0
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .3 %)
rho (ohm-m)
h (m)
18.27
1.73
5.475
36.53
57.19
Loop size : 50 m
I/O time (mksec) : 50

Figure 54

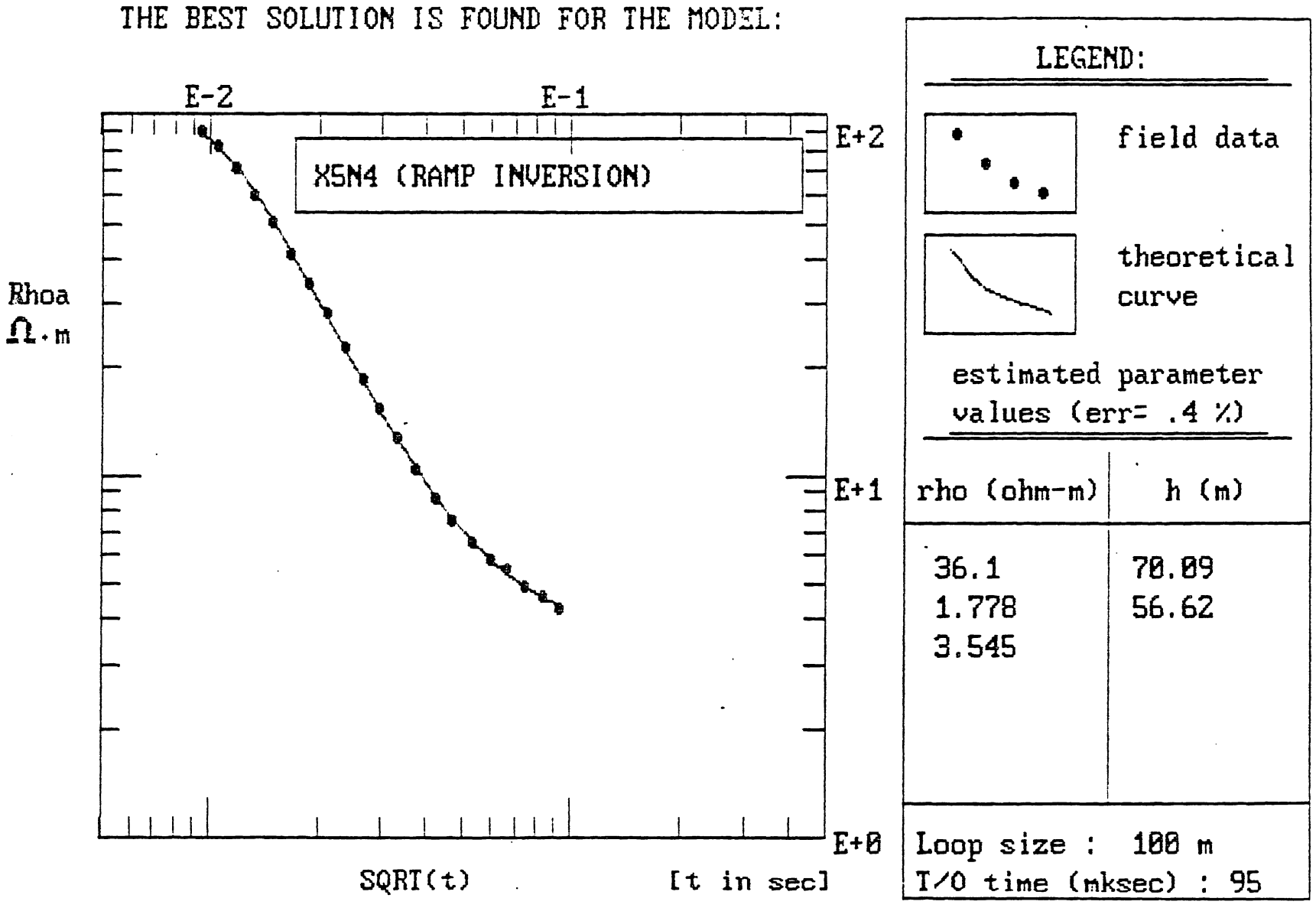

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
E+2
E+1
E+0
X5N4 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= .4 %)
rho (ohm-m)
h (m)
36.1
1.778
3.545
70.09
56.62
Loop size : 100 m
T/O time (mksec) : 95

Figure 55

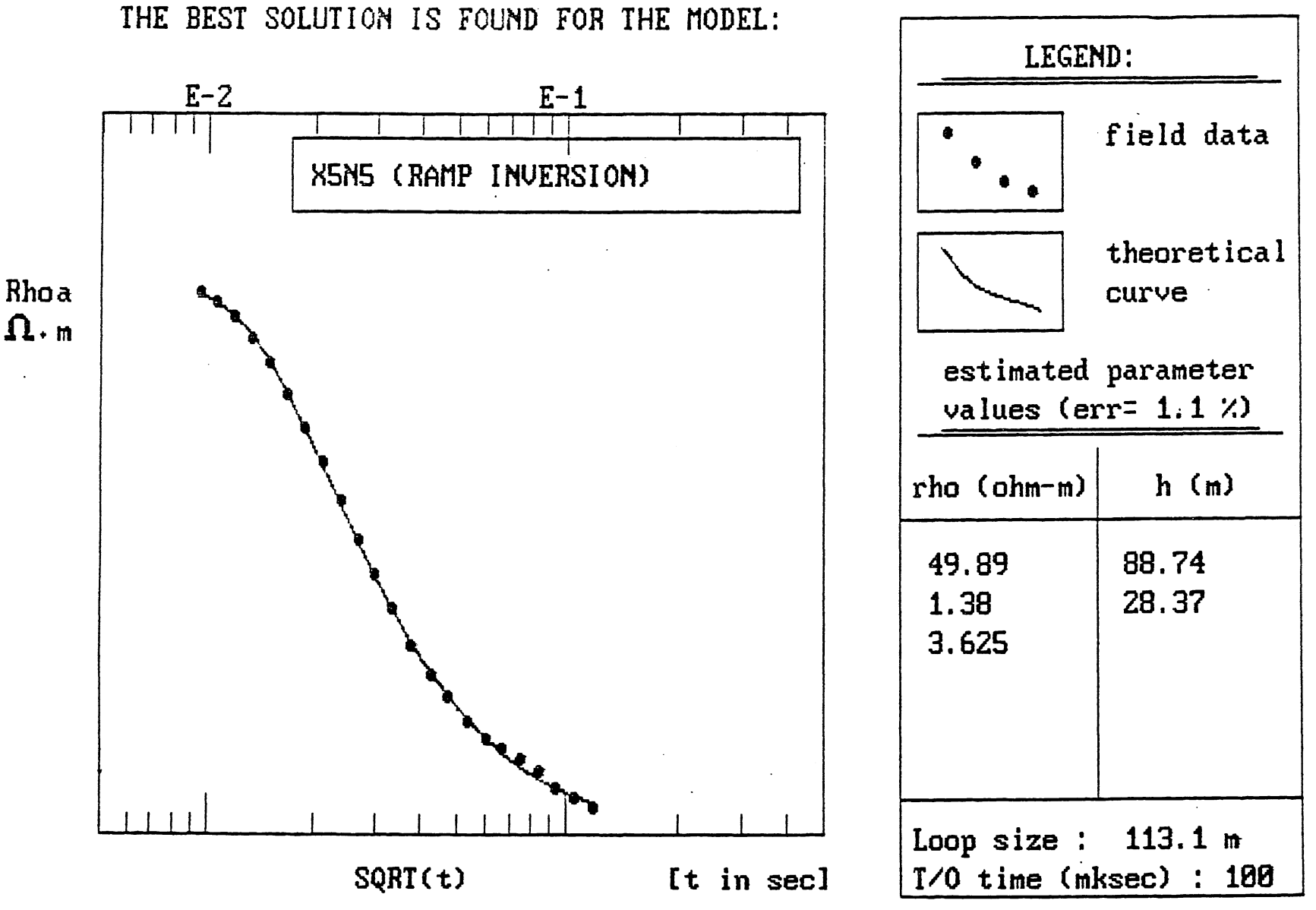

THE BEST SOLUTION IS FOUND FOR THE MODEL:
E-2
E-1
X5N5 (RAMP INVERSION)
Rhoa
Ω.m
SQRT(t)
[t in sec]
LEGEND:
field data
theoretical curve
estimated parameter values (err= 1.1 %)
rho (ohm-m)
h (m)
49.89
1.38
3.625
88.74
28.37
Loop size : 113.1 m
T/O time (mksec) : 100

Figure 56

SUMMARY AND RECOMMENDATIONS

Throughout the present study, 39 of the 40 soundings were of good quality permitting geophysical interpretation. Only one measurement was distorted by the influence of a nearby power line thus preventing any interpretation. As far as the geological interpretation is concerned, a comparison with boreholes showed that the correlation between geoelectrical and geological boundaries was not always straightforward. The general conclusions concerning the applicability of the TDEM method using the Geonics EM42 and EM37 systems for each specific geological target are detailed below:

- The most appropriate targets seem to be seawater intrusions into the shallow (up to depths of a few hundred meters) coastal and karstic aquifers connected to the sea. This seems to be feasilbe due to their very consistent and specific range of resistivities (around 1.5 ohm.m). According to relevant literature, this range is typical for many coastal aquifers around the world.

  Owing to the very high accuracy of the TDEM measurements and interpretation, the depth to the freshwater/seawater interface can be determined with almost the same degree of accuracy as in observation wells. The latter conclusion, however, requires verification through some additional tests in order to be statistically significant.

- A boundary between clastic and carbonate sediments and the quartz-syenite intrusions in southern Israel is detected very confidently by the TDEM measurements. It is true, however, that the accuracy of the TDEM results cannot yet be controlled due to the lack of boreholes in the studied area.
- The contact between the Neogene basaltic sequence and the Neogene to Quaternary clastic rocks in northern Israel can be clearly detected by the TDEM measurements. The required condition is that the alternations of basalts and clastic rock units should be sufficiently thick (more than 50m). It is not possible to distinguish the clastic rocks/basaltic contact from the clastic rocks/limestone (or dolomite) contact since the resistivity contrast across the contact is approximately the same in both cases. On the other hand, there is no clear resistivity contrast between the volcanic and limestone (dolomite) sequences. Thus, the geological interpretation of the geoelectric boundaries obtained is not always straightforward.
- Very deep targets, such as the Precambrian basement and/or hypothetical magmatic intrusions in southern Israel could not be detected by the TDEM measurements. If the only reason for this is that penetration depth was insufficient, then further tests of an alternative transmitter/receiver configuration and/or measuring equipment is be recommended.

A consideration of the overall results of this feasibility study allows us to conclude:

1. The TDEM method has been tested in a large variety of geological environments and as such resulted in usable data, albeit not always leading to an unequivocable geological interpretation.
2. Considering the variety of targets tackled, the effectiveness of this sounding method to detect geological phenomena up to depths of several hundred meters, has been demonstrated.
3. The geology of Israel ensures a variety of attractive targets which can be explored with the TDEM method. It is expected that the effectiveness of the geological interpretation of the data will increase significantly as experience is gained with the application to a wide range of problems even before future technological development of the method.
4. Future application of the TDEM method as a subsurface mapping tool should be conducted by teams composed of geophysicists and geologists. A relatively large amount of soundings should be allowed in order to refine the method and facilitate the construction of three dimensional geophysical and geological models.

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ACKNOWLEDGEMENTS
==========

This project has been successfully executed with the support of the Geological Survey of Israel and the Soreq Nuclear Research Center.

Mr. M. Blohm (ERTEC, Denver, Colorado) skillfully managed the instrumentation during all phases of the field work. The following persons from the Geological Survey of Israel and the Soreq Nuclear Research Center participated in various phases of the field work: S. Ashkenazi, Y. Assael, Y. Bassan, C. Dallal, H. Dingot, A. Madar, H. Hemo, G. Liphshitz, R. Madmon, L. Michaelescu, Y. Mizrahi, Y. Peled, Z. Peleg, E. Ram, Y. Raphael, G. Shaliv, D. Sidi, A. Strull, U. Vulkan and Y. Yaffe.

The manuscript was edited and typed by Mrs. D. Artzi. The most complicated sections of the graphical work were drafted by Miss I. Chelinsky. Some of the location maps were prepared by Mr. R. Assael.

Thanks are due to all of them.